



저작자표시-비영리-변경금지 2.0 대한민국

이용자는 아래의 조건을 따르는 경우에 한하여 자유롭게

- 이 저작물을 복제, 배포, 전송, 전시, 공연 및 방송할 수 있습니다.

다음과 같은 조건을 따라야 합니다:



저작자표시. 귀하는 원저작자를 표시하여야 합니다.



비영리. 귀하는 이 저작물을 영리 목적으로 이용할 수 없습니다.



변경금지. 귀하는 이 저작물을 개작, 변형 또는 가공할 수 없습니다.

- 귀하는, 이 저작물의 재이용이나 배포의 경우, 이 저작물에 적용된 이용허락조건을 명확하게 나타내어야 합니다.
- 저작권자로부터 별도의 허가를 받으면 이러한 조건들은 적용되지 않습니다.

저작권법에 따른 이용자의 권리는 위의 내용에 의하여 영향을 받지 않습니다.

이것은 [이용허락규약\(Legal Code\)](#)을 이해하기 쉽게 요약한 것입니다.

[Disclaimer](#)

Thesis for the Degree of Ph.D.

Deep Learning-Based Safety Analysis for High-Altitude Construction Workers: PPE Detection and Posture Recognition Model Development

School of Architectural, Civil, Environmental, and Energy Engineering,
Major in Architectural Engineering
The Graduate School

Yirong Mi

June 2025

The Graduate School
Kyungpook National University

Deep Learning-Based Safety Analysis for High-Altitude Construction Workers: PPE Detection and Posture Recognition Model Development

Yirong Mi

School of Architectural, Civil, Environmental, and Energy Engineering,
Major in Architectural Engineering
The Graduate School

Supervised by Professor Dong-Eun Lee

Approved as a qualified thesis of Yirong Mi
for the degree of Ph.D. by the Evaluation Committee

June 2025

Chairman _____ (인)

_____ (인)

_____ (인)

_____ (인)

_____ (인)

The Graduate School Council
Kyungpook National University

Table of Contents

Chapter 1. Introduction	1
1.1 Research background	1
1.2 Research objectives	11
Chapter 2. Review of related works	14
2.1 Research related to construction worker safety	14
2.1.1 Construction worker personal protective equipment	14
2.1.2 Construction worker action & posture	20
2.2 Research related to deep learning	25
2.2.1 Object detection model	25
2.2.2 Action recognition model	33
2.3 Research on YOLO family models	47
Chapter 3. Materials and dataset development	57
3.1 Data acquisition of construction worker dataset	57
3.2 Personal protective equipment dataset	60
3.2.1 Dataset organization and preprocessing	60
3.2.2 Dataset annotation	63
3.2.3 Data augmentation	64
3.3 Construction worker action dataset	71
3.3.1 Dataset organization and preprocessing	71
3.3.2 Data annotation	72
3.3.3 Data augmentation	75
Chapter 4. Development of the proposed framework	79
4.1 Method	79
4.2 Object detection model	82
4.2.1 Model optimization	86
4.2.2 Loss function	97
4.3 Action recognition model	100

4.3.1 Model improvement	103
4.4 Recognition of hazardous postures in construction workers	111
4.4.1 Joint angle calculation	112
4.4.2 Scale selection and score calculation	116
Chapter 5. Experiments and analysis	120
5.1 Experimental environment	120
5.1.1 Hardware environment	120
5.1.2 Virtual environment	121
5.2 Experimental analysis	123
5.2.1 Object detection model	123
5.2.1.1 Performance metrics	123
5.2.1.2 Training and validation	128
5.2.1.3 Visualization	141
5.2.1.4 Model optimization analysis	143
5.2.2 Keypoint detection model	151
5.2.2.1 Performance metrics	151
5.2.2.2 Training and validation	155
5.2.2.3 Visualization	165
5.2.2.4 Ablation study	167
5.3 Joint angle analysis and risk assessment	169
5.3.1 Calculation of joint angles	170
5.3.2 Assessment of worker risk levels	175
Chapter 6. Conclusion and future work	180
6.1 Conclusion	180
6.2 Future work	185
■ References	187
■ Abstract	207
■ Appendix A: Dataset Overview	209

Table

Table 1. Common object detection models for construction applications	33
Table 2. Activity recognition models in the construction industry	46
Table 3. Major models and improvements in the YOLO family	48
Table 4. Primary recording equipment in the high-altitude worksite dataset	58
Table 5. Data preprocessing and augmentation of the personal protective equipment dataset	70
Table 6. Composition of the augmented and original personal protective equipment dataset	70
Table 7. Data preprocessing and augmentation of action dataset	77
Table 8. Composition of the augmented and original of action dataset	78
Table 9. PPE_YOLO model architecture	96
Table 10. Hardware environment	121
Table 11. Virtual Environment	122
Table 12. Model performance evaluation formulas	128
Table 13. Hyperparameters	131
Table 14. Performance metrics of the YOLOv5 model	148
Table 15. Performance metrics of the YOLOv11 model	149
Table 16. Performance metrics of the PPE_YOLO model	150
Table 17. Model performance evaluation metrics	152
Table 18. Hyperparameters	156
Table 19. Ablation test results	167
Table 20. Joint angle calculation and visualization	173

Figure

Figure 1. Distribution of construction accident types in China	3
Figure 2. Distribution of fatal occupational injuries in the U.S. construction industry	4
Figure 3. Distribution of fatal occupational injuries in the South Korean construction industry	6
Figure 4. Flow of research	13
Figure 5. Personal protective equipment dataset examples: (a) Aerial platform operations; (b) Scaffolding work; (c) Ladder tasks	60
Figure 6. Example of construction workers' PPE annotation: (a) Scaffolding work; (b) Ladder tasks; (c) Aerial platform operations	66
Figure 7. Skeletal structure design for keypoint annotation of workers' postures	73
Figure 8. Example of construction worker action annotation: (a) Overhead; (b) Lifting	75
Figure 9. Research flow chart	81
Figure 10. Network architecture of YOLOv5	84
Figure 11. Channel attention module	87
Figure 12. Spatial attention module	88
Figure 13. Convolutional block attention module	88
Figure 14. CBAM and C3CBAM modules	89
Figure 15. Sampling positions	90
Figure 16. Deformable convolutional network module	91
Figure 17. Network architecture of C3_DCN	92
Figure 18. BiFPN module: (a) FPN, (b) PANet, and (c) BiFPN	94
Figure 19. Network architecture of YOLOv8	103
Figure 20. BottleneckCSP module	105
Figure 21. (a)Traditional convolution; (b)GhostConv module ($\emptyset$: Cheap operation)	106
Figure 22. Customer_YOLOv8 network architecture	110
Figure 23. Rapid upper limb assessment	117

Figure 24. Rapid entire body assessment	119
Figure 25. Confusion matrix	123
Figure 26. Model loss curve: (a) Box loss; (b) Object (Obj) loss; (c) Classification (Cls) loss	135
Figure 27. Model precision and recall curve	138
Figure 28. Mean average precision (mAP) curve	140
Figure 29. Visualization Results: (a) Ladder Work, (b) Scaffold Work, (c) Elevated Platform Work	143
Figure 30. Confusion matrix of the (a) YOLOv5 and (b) PPE_YOLO models	147
Figure 31. Model loss curve: (a) Box loss; (b) Pose loss	159
Figure 32. Model precision and recall curve	161
Figure 33. Model mean average precision (mAP) curve	163
Figure 34. Results visualization: (a) Working overhead; (b) Lifting	166
Figure 35. Risk level assessment process	178

Chapter 1. Introduction

1.1 Research background

Along with many opportunities, the construction industry's rapid growth has also presented several difficulties (Zhang et al., 2022)¹⁾. As one of the most hazardous industries in the world (Berglund et al., 2021)²⁾, construction has long been a global safety concern. This concern is due to the high risks associated with extensive construction activities, often leading to frequent accidents, casualties, and property losses (Guo et al., 2022)³⁾. Although it employs only 7% of the world's workforce, the construction sector is disproportionately responsible for 30–40% of all recorded fatalities (Islam et al., 2019)⁴⁾. This high-risk nature is particularly evident in rapidly developing economies. Over the past 30 years, real estate investment, infrastructure expansion, and rapid urbanization have played a significant role in the remarkable rise of China's construction industry (Li & Li, 2022)⁵⁾. However, this rapid expansion has also exposed numerous safety concerns, including poor working conditions, high labor turnover, and inadequate worker training.

Due to these ongoing safety concerns, China's construction sector has had the highest death rate of any industrial sector since 2012 (Zhang et al., 2019)⁶⁾. Statistics indicate that from 2011 to 2017, 4,766 accidents occurred in China's construction industry, averaging 1.87 deaths per day. As of November 30, 712 accidents occurred in 2018, leading to 798 fatalities. By 2019, the number of accidents for the year had risen to 734, resulting in 904 deaths—an increase of 5.3% in

accidents and 7.6% in fatalities compared to the previous year (Lu et al., 2023)⁷⁾. In the first half of 2022 alone, 1,222 accidents were reported, causing 1,303 deaths (<https://www.mem.gov.cn>). These figures underscore the persistent safety challenges in the construction sector, despite the implementation of various safety management measures. A particularly alarming example is the 3,843 fatal accidents recorded in China's construction industry in 2017, further reinforcing its status as one of the most hazardous sectors in the country.

As shown in Figure 1, the number of accidents in China significantly increased between 2014 and 2016, followed by a sharp decline after 2017. The period from 2014 to 2016 was marked by significant economic growth in China, characterized by large-scale infrastructure projects and the rapid expansion of the real estate sector. This rapid development, while contributing to the country's economic prosperity, also led to increased construction activity, which in turn heightened the risk of accidents on construction sites. Additionally, the accidents were categorized into six types. Among these, the most frequent type of accident was falls from height, accounting for 43.7%. Falls are particularly concerning due to their high fatality rate, emphasizing the need for improved fall protection measures. Collapses, accounting for 25.6%, highlighted the risks of structural failure, calling for stronger safety standards and monitoring. Struck-by incidents accounted for 10%, underscoring the importance of site organization and protective barriers. Lifting injuries (5.5%) and electrocutions (4.6%) further emphasize the need for better equipment safety and worker training to minimize hazards. The remaining 10.6% were categorized as other

accidents, indicating the wide range of potential risks in construction (Xia et al., 2024)⁸⁾.

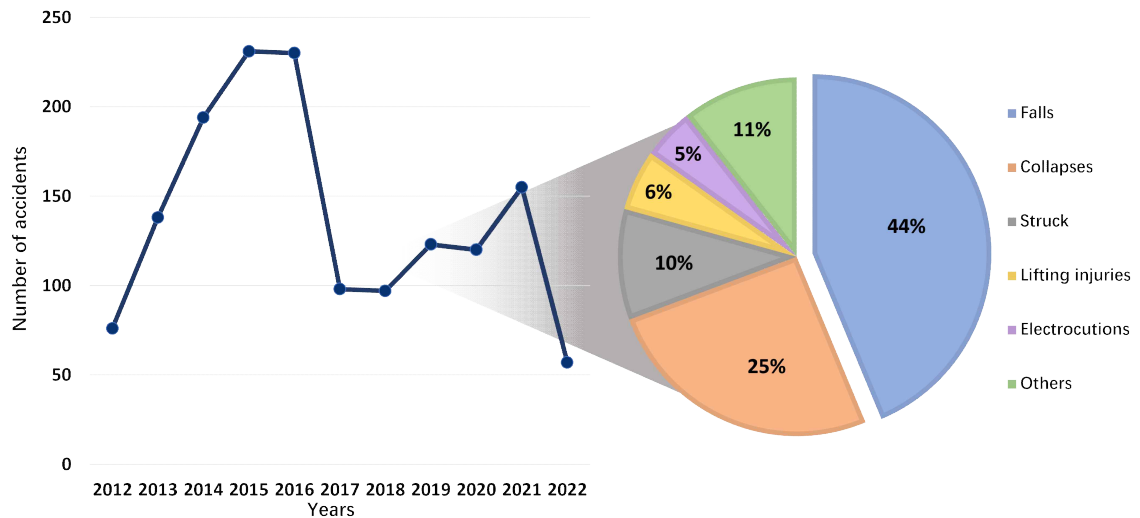


Figure 1. Statistics on the number and types of construction accidents in China (2012–2022).

China is not the only country with a serious occupational safety problem in the building sector. Taking the United States as an example, statistics from the Occupational Safety and Health Administration (OSHA) (Statistics, 2020)¹⁰⁾ indicate that in 2016, there were 991 construction worker fatalities, with a fatal injury rate of 10.1%, ranking third among all industries. In 2014, construction-related fatalities accounted for 18% of all occupational deaths in the United States, highlighting the high-risk nature of the industry. Moreover, 5,333 occurrences involving construction workers were reported in 2019, equivalent to 3.5 events per 100,000 full-time workers. This is a rate of more than 100 incidents each week, or an average of 15 fatalities daily (Statistics, 2016)¹¹⁾.

In the U.S. construction business, falls have been the most common cause of fatalities and injuries among these accidents (Hu et al., 2011)¹²⁾. In 2010, falls accounted for 33.3% of all workplace fatalities. By 2014, they caused 349 deaths, representing 39.9% of total construction-related fatalities (OSHA, 2015)¹³⁾. As shown in Figure 2, between 2011 and 2021, approximately 10,000 construction workers in the United States lost their lives due to work-related injuries. Of these fatalities, 65.5% were associated with four leading causes: falls, being struck by objects, being caught in or between equipment or structures, and electrocutions. Falls constituted the most prevalent cause, accounting for 53.5% of the fatalities among the four primary incident categories. Furthermore, falls alone were responsible for 35.0% of all fatal injuries in the construction sector during this period (Yoon et al., 2024)⁹⁾.

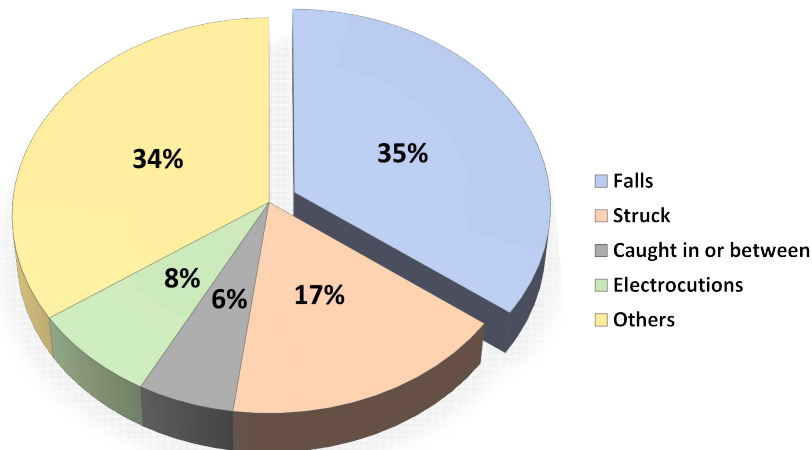


Figure 2. Types of construction accidents in the United States (2011–2021).

These statistics highlight the fact that, despite the United States' relatively advanced occupational safety management systems, construction remains one of the most hazardous industries.

In South Korea, where construction is a major employer and a high-risk industry, this worldwide trend is also noticeable. Workers' Compensation data from 2020 showed that the construction industry had an injury rate of 1.17%, which was more than twice as high as the industry average of 0.57%. The construction industry is the leading area of workplace fatalities, accounting for 51.9% of all occupational deaths (882 fatalities). The fatality rate per 10,000 workers was 2.48, compared to the overall industry average of 1.09 (Khosravi et al., 2014)¹⁴⁾. According to occupational safety statistics from 2022, the construction industry in South Korea recorded the highest number of work-related fatalities, with 402 deaths, accounting for 46.0% of the total. This figure was significantly higher than those in other sectors, such as manufacturing (184 deaths, 21.1%), the service industry (150 deaths, 17.2%), and other industries (104 deaths, 11.9%). As shown in Figure 3, the number of fatalities in South Korea by type of accident from 2018 to 2022 reveals that falls were the leading cause of death in the construction sector. A detailed analysis indicates that falls were responsible for 215 deaths, accounting for 53.5% of all construction-related fatalities. Other major causes included collisions (48 deaths, 11.9%), being struck by objects (22 deaths, 5.5%), being crushed (22 deaths, 5.5%), and various other types of accidents categorized as miscellaneous (21 deaths, 5.2%)(Yoon et al., 2024)⁹⁾.

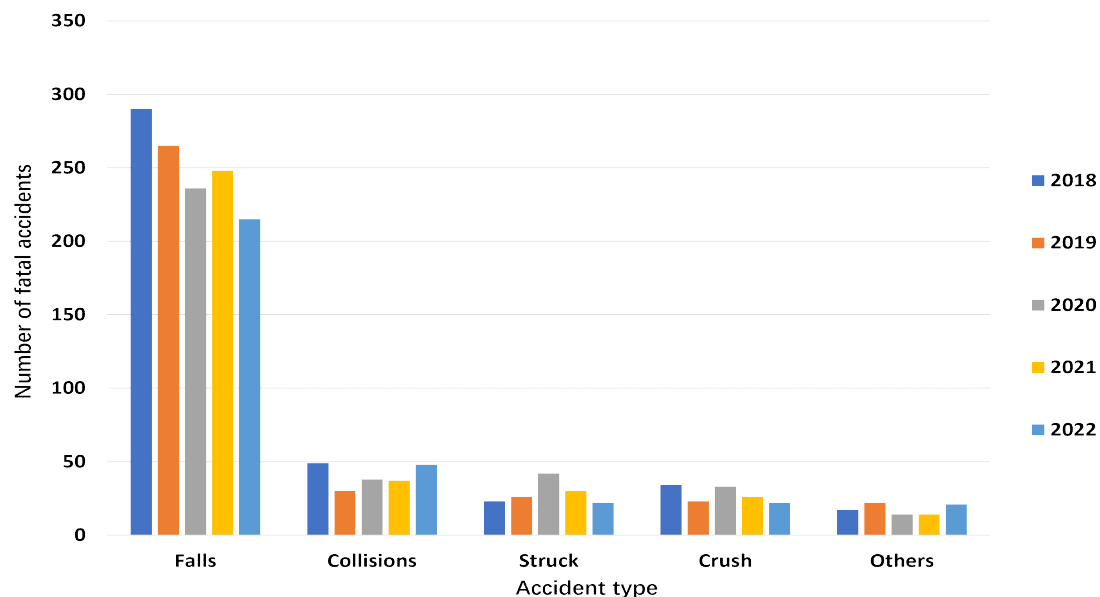


Figure 3. Number of fatalities in South Korea by type of accident (2018–2022).

High-risk activities, like working at heights, lead to severe accidents. Despite strengthened safety management and awareness efforts in South Korea, the high rate of construction accidents persists, highlighting the need for further safety improvements.

According to the Health and Safety Executive (HSE), the fatal injury rate has stayed at 1.64 per 100,000 workers over the past five years, with falls from height (47%) and collapsing objects (12%) being the leading causes. From 2017 to 2018, 58,000 workplace injuries were reported across industries, and in construction alone, injuries and illnesses led to an annual loss of about 2.4 million working days between 2015 and 2018 (HSE, 2019)¹⁵⁾. Despite strict regulations, construction accidents remain frequent due to complex sites and high-risk tasks like working at heights and using heavy machinery.

Similarly, Malaysia's construction industry has witnessed a sharp rise in reported incidents. In 2016 alone, 7,338 accidents were reported, a 69.47% increase from 4,330 incidents in 2011. The sector remains the leading industry of fatal workplace accidents, with a mortality rate of 14.57 per 10,000 workers (Zid et al., 2020)¹⁶⁾. Moreover, Russia, a critical player in the global construction market, faces equally significant occupational safety and health challenges. Despite employing just 7% of workers, construction accounts for 30% to 40% of global workplace deaths, according to the International Labour Organization. Russian workers frequently face risks from heavy machinery, working at heights, and repetitive tasks, with around 30% suffering from musculoskeletal issues due to physical strain and poor posture.

Recently, computer vision technology has gained significant attention in the architecture, engineering, and construction industry, enabling computers to analyze digital images and videos, extract relevant information, and understand incidents and objects in a scenario.

For instance, a study (Wu et al., 2021)¹⁷⁾ proposed a computer vision and ontology-based framework for identifying site hazards and recommending mitigation strategies, aiding managers in developing cognitively informed safety plans. Another study (Moohialdin et al., 2021)¹⁸⁾ developed a TensorFlow-based computer vision system using a two-layer decision algorithm for real-time PPE and posture detection, showing strong potential to improve on-site safety management. In (Fang et al., 2020)¹⁹⁾, a construction safety framework was proposed that uses data fusion and digital technology to improve safety management by automatically detecting dangerous behaviors and

hazardous conditions through deep learning and computer vision. Similarly, a study (Balakreshnan et al., 2020)²⁰⁾ demonstrated the use of artificial intelligence (AI) and machine vision to detect PPE, particularly safety glasses, in real-time, promoting compliance and preventing accidents. Transfer learning also showed significant benefits in construction safety management. Researchers (Lee & Lee, 2023)²¹⁾ proposed a transfer learning-based safety management system that trains object detection models with synthetic data to prevent accidents. The system includes models for detecting workers, assessing fall risks at heights, and verifying the use of safety vests and helmets. Luo et al. (Luo et al., 2020)²²⁾ introduced a framework using computer vision and deep learning to predict equipment component keypoints from surveillance images or videos, providing accurate and efficient posture estimation to support safe equipment operation.

Deep learning provides powerful technological support for data analysis in computer vision, overcoming the limitations of traditional manual observation and recording of unsafe behaviors.

For instance, Kolar et al. (Kolar et al., 2018)²³⁾ developed a convolutional neural network (CNN)-based safety barrier detection model that used synthetic data augmentation and Visual Geometry Group 16 (VGG-16) architecture for feature extraction, achieving 96.5% detection accuracy, demonstrating the effectiveness of CNN and synthetic data in safety monitoring. In the domain of construction worker safety, Fang et al. (Fang et al., 2018)²⁴⁾ proposed a high-precision, fast, and widely applicable Faster Region-based CNN (Faster R-CNN) method for detecting unsafe worker behaviors. The

method, tested on over 100,000 frames from 25 construction sites, showed high accuracy and recall, significantly improving safety inspections and supervision. Wu et al. (Wu et al., 2019)²⁵⁾ proposed a helmet detection system based on a single-shot multibox detector (SSD) integrated with a reverse progressive attention module. Trained on a custom dataset, the model effectively detected small-scale helmets under diverse conditions, supporting real-time on-site monitoring. Further research on safety equipment detection (Nath et al., 2020)²⁶⁾ proposed three AI-based methods for verifying PPE compliance. The first detects workers, helmets, and vests using a machine learning model. The second uses a single CNN framework for simultaneous detection of workers and PPE, while the third employs CNN classifiers like VGG-16, residual network 50 (ResNet-50), and Xception for detection and classification. Beyond PPE detection, Jeelani et al. (Jeelani et al., 2021)²⁷⁾ developed a framework for hazard detection and real-time worker tracking using wearable cameras. The system tracks worker locations, issues warnings near hazards, and helps prevent musculoskeletal disorders caused by poor postures. Finally, Rantwi-Afari et al. (Antwi-Afari et al., 2022)²⁸⁾ introduced a method using wearable insole sensors and deep learning networks to identify poor working postures that lead to musculoskeletal disorders. The system monitored five postures, using plantar pressure data to train recurrent neural networks (RNNs) like long short-term memory (LSTM), bidirectional LSTM (Bi-LSTM), and gated recurrent units (GRUs), achieving accurate posture detection.

In summary, deep learning and computer vision are transforming

construction safety management by improving PPE detection, monitoring hazardous behaviors, and analyzing work postures, thereby enhancing safety and promoting a more sustainable work environment.

Despite significant advancements in construction site safety management, several challenges remain:

- **Dataset limitations:** Most studies have relied on small-scale or laboratory-based datasets with limited sample sizes and little diversity. These constraints hinder the generalization of research findings to real-world construction environments, affecting the reliability and practical effectiveness of safety management technology. Expanding and varying datasets is crucial to improving the robustness and suitability of these methods.

- **Monotony of posture estimation methods:** Many studies have primarily applied sensor-based approaches to estimate worker postures and joint angles, with relatively few attempts to infer posture directly from images or videos. In addition to making data gathering and model training more complicated, an over-reliance on sensor data raises the possibility of interfering with the natural movements of employees, affecting operational efficiency and overall workplace ergonomics. Advancing image-based posture estimation methods could mitigate these challenges and provide a more practical alternative.

- **Lack of diversity in risk assessment methods:** The current research has predominantly employed single-method approaches for risk assessment, failing to capture and evaluate worker risk levels accurately across varying contexts and environments. Worker posture, environmental conditions, and task complexity are often overlooked,

resulting in assessments that may be insufficient to address the dynamic and multifaceted risks in construction work. A more integrated and context-aware approach to risk assessment is needed to enhance accuracy and effectiveness.

Fragmented research focus: Although numerous studies have contributed to technological advancements in worker safety, most have focused on isolated aspects, such as helmet detection or safety vest recognition. These single-task detection models lack comprehensiveness and struggle to provide holistic safety solutions for workers operating in complex and dynamic construction environments. Integrating multiple detection tasks into a unified system is crucial for achieving more effective and intelligent safety monitoring.

1.2 Research objectives

Given the high proportion of accidents related to working at heights in the construction industry worldwide, this study proposes an improved deep learning model and real datasets from construction sites to address critical challenges in the safety detection of construction workers at heights. The primary goal is to improve the precision and dependability of safety detection while preserving model simplicity, ensuring that the proposed approach remains efficient and feasible for real-world construction applications.

To achieve this improvement, we employ datasets from the AI-Hub platform (<https://www.aihub.or.kr/>), providing comprehensive records of real-world construction scenarios. These datasets capture detailed information on the behaviors, postures, and surroundings of

construction workers working at heights during tasks. Regarding model selection and optimization, this study employs the You Only Look Once (YOLO) model family for multitask detection, including PPE detection and skeleton keypoint-based posture recognition of construction workers. This study introduces several crucial optimizations to the YOLO framework to enhance the detection capabilities of the model:

- **Module replacement:** Critical modules in the YOLO architecture affecting detection accuracy are replaced with more efficient feature extraction layers and fusion mechanisms, improving feature representation and detection precision.
- **Attention mechanism integration:** The model incorporates attention techniques to enhance object detection accuracy by enabling the selection of crucial feature areas associated with workers and nearby objects.
- **Ablation experiments:** Systematic ablation studies are conducted to assess the individual contributions of various modules and mechanisms. The optimal configuration is identified to ensure improved performance while controlling computational costs.

This thesis is organized as follows. Chapter 1 introduces the research background, objectives, and methodology. Chapter 2 reviews existing studies on construction worker hazard detection. Chapters 3 and 4 describe the proposed hazard detection and analysis method using an improved YOLO framework, integrating attention mechanisms and multitask detection techniques, along with model construction and optimization strategies. Chapter 5 presents experimental analysis and results, validating the effectiveness of the approach. Finally, Chapter 6

summarizes the findings, discusses limitations, and suggests future research directions (Figure 4).

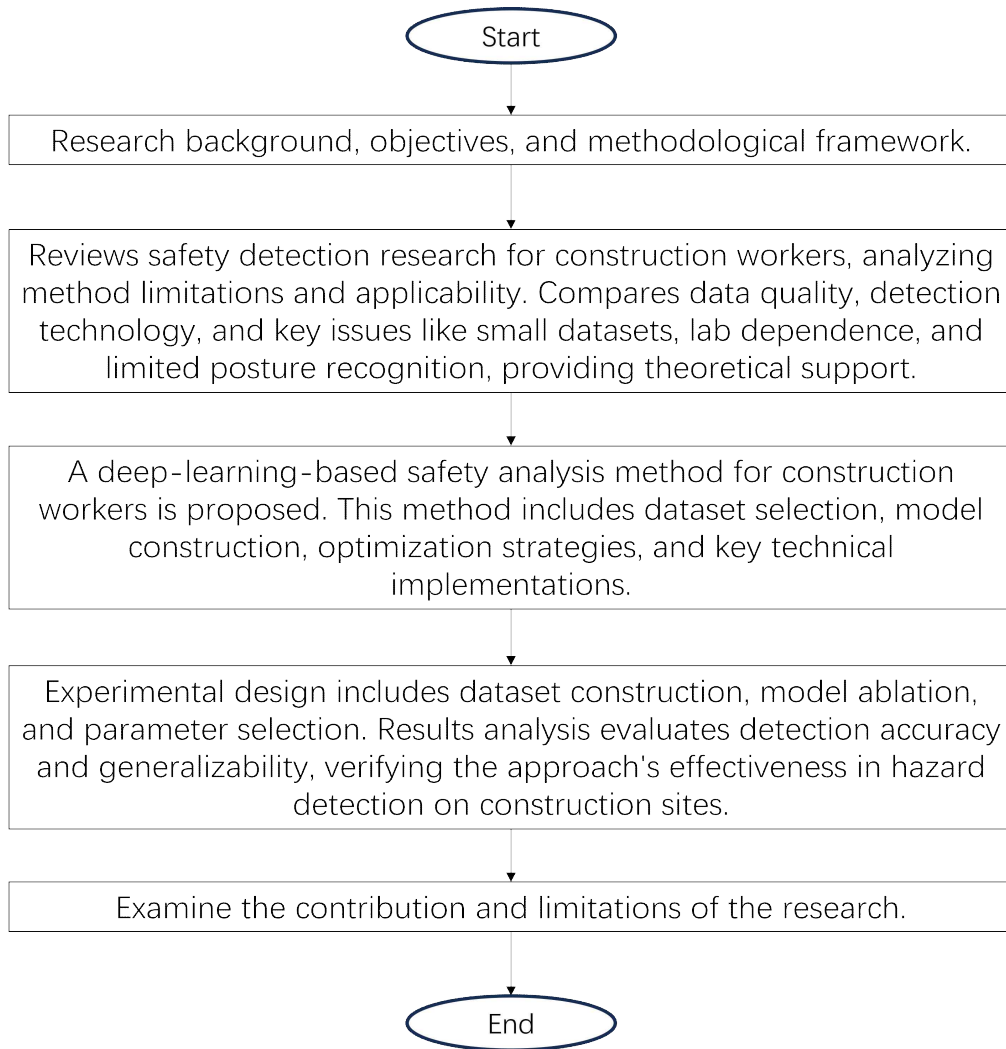


Figure 4. Flow of research.

Chapter 2. Review of related work

2.1 Research related to construction worker safety

2.1.1 Construction worker personal protective equipment

Currently, automatic PPE compliance monitoring technology is predominantly classified into sensor and vision-based approaches. The sensor-based method uses devices to collect real-time physiological and environmental data from the wearer, assessing the effectiveness and compliance of PPE. In contrast, the vision-based approach tracks PPE use in real time and verifies appropriate wear using cameras to capture images, which are processed using image processing and computer vision algorithms.

Regarding sensor-based monitoring, several recent studies have focused on PPE compliance on construction sites.

Kelm et al. (Kelm et al., 2013)²⁹⁾ provide an overview of technologies enabling automated PPE monitoring, discussing human identification methods, radio-frequency identification (RFID) tagging, control strategies, and the pros and cons of passive RFID. They also explore technology for training and compliance verification, data processing for safety benchmarking, and how these tools can integrate into existing workflows and safety management systems. Barro-Torres et al. (Barro-Torres et al., 2012)³⁰⁾ proposed a system for real-time PPE monitoring using two networks: body area and mesh networks. RFID technology detects PPE, while Zigbee enables efficient data transmission. This integration ensures real-time tracking and improves safety management. Literature (Naticchia et al., 2013)³¹⁾ explores a

similar approach, where each piece of PPE is equipped with a short-range responder, and a wireless system verifies workers' compliance by detecting the type of PPE worn. However, the actual implementation of sensor-based approaches may be limited due to the high cost of procuring, installing, and maintaining complex sensor networks. Zhang et al. (Zhang et al., 2019)³²⁾ proposed a system that uses sensors, mobile communication, and cloud technology for real-time monitoring and localization of nonhuman work on construction sites. Safety helmets with infrared sensors and RFID triggers enable precise detection and localization. A web tool and smartphone app manage alerts, while a cloud server ensures efficient data storage. The system enhances construction safety with high accuracy and reliability. Kim et al. (Kim et al., 2021)³³⁾ proposed a method using wearable devices to detect helmet-wearing status on construction sites. The device integrates an accelerometer, photoplethysmogram sensor, long-range network module, and GPS, all controlled by a microcontroller. The receiving node connects to a smartphone via IoT to display results. This technique automatically alerts management when a helmet is worn incorrectly by analyzing photoplethysmogram sensor data from the chin strap. Kim et al. (Kim et al., 2018)³⁴⁾ developed a system integrating a three-axis accelerometer into a safety helmet to detect proper helmet usage. The system analyzes sensor data and uses the RandomTree algorithm for classification. Experimental results showed high accuracy—96.89%, 96.90%, and 98.26% for correctly worn, improperly worn, and no helmets, respectively. The system improves safety management by

providing real-time helmet status monitoring, enhancing worker safety awareness, and offering managers timely feedback to reduce accident risks. Márquez-Sánchez et al. (Márquez-Sánchez et al., 2021)³⁵⁾ proposed a PPE system with three wearable devices: a helmet, wristband, and belt. These devices collect data processed via edge computing and AI to monitor worker safety and health. The helmet tracks head posture and environment, the wristband monitors physiological status and activity, and the belt observes movement and load. The system analyzes data in real time to detect hazards like falls or fatigue and sends alerts to workers and management for prompt intervention, enhancing safety on construction sites. Nath et al. (Nath et al., 2020)³⁶⁾ designed a PPE-tool pairing system to improve construction safety. The system ensures workers wear the required PPE before using hazardous tools by employing wear detection technology. Each tool is equipped with a vibration motor, buzzer, and Wi-Fi module, while wearable sensors and batteries monitor PPE usage. If a worker fails to wear the proper gear, the system issues alerts and transmits data to an online platform for real-time monitoring, enabling quick safety interventions and reducing accident risks. To mitigate falls from heights, Choo et al. (Choo et al., 2023)³⁷⁾ developed an automated monitoring system to mitigate falls from heights using wearable sensors. The system tracks worker altitude and operational status in real time, using a barometer, accelerometer, and gyroscope to estimate altitude accurately. Machine learning algorithms analyze sensor data to assess the fastening state of safety hooks. This allows for quick identification of improperly fastened hooks, enabling

safety supervisors to address hazards promptly and improve overall safety during high-altitude operations, reducing the risk of accidents. Pisu et al. (Pisu et al., 2024)³⁸⁾ proposed a new operator local area network system to enhance PPE monitoring by improving portability, noninvasiveness, and privacy. Using machine learning, the system analyzes the received signal strength indicator (RSSI) between PPE items to detect improper usage. Instead of using RSSI as a distance proxy, it helps assess proper PPE wear. The system uses post-processing algorithms to enhance the accuracy of a high-frequency support vector machine model, enabling real-time monitoring and alerts. Experimental results show an 80% reduction in false alarms, improving workplace safety and safety management efficiency.

Vision-based approaches use cameras instead of sensors to take pictures or videos of construction sites and examine them to confirm PPE compliance. By analyzing image data, vision-based methods detect PPE usage and capture site-wide dynamics, assisting managers in assessing safety risks and optimizing protective measures (Seo et al., 2015)³⁹⁾.

Du et al. (Du et al., 2011)⁴⁰⁾ proposed an innovative helmet detection method using video sequences, combining Haar-like face features, motion information, and color analysis to improve construction site safety. The system first detects faces using Haar-like features, then applies motion detection and skin color analysis to reduce false positives. This approach enables real-time detection of workers' facial features and helmet-wearing status, offering an effective solution for

enhancing safety on construction sites. Shrestha et al. (Shrestha et al., 2015)⁴¹⁾ developed a software tool in Microsoft Visual Studio 2012 for real-time helmet detection using closed-circuit television (CCTV) footage. The system segments video at 32 fps, detects faces using Haar-like features, and identifies helmets based on color (red) and shape (semicircular). Edge detection algorithms recognize contours, and if the number of helmets matches the number of faces, a “safe work” notification is issued. Otherwise, alerts are sent to office monitors, workers' mobile devices, and on-site speakers, improving the timeliness and accuracy of construction site safety management. Park et al. (Park et al., 2015)⁴²⁾ proposed a vision-based method for automatic helmet-wearing detection. The approach involves detecting human figures and helmets in video frames and matching helmets to workers based on geometric and spatial relationships. Park and Brilakis (Park & Brilakis, 2012)⁴³⁾ introduced an automatic method for detecting construction workers in video frames. The approach includes background subtraction to isolate moving objects, Histogram of Oriented Gradients (HOG) for worker detection, and color histograms to classify individuals as workers or nonworkers. The system aims to maximize detection accuracy while minimizing time delays, ensuring effective monitoring of workers' safety vest usage. Fang et al. (Fang et al., 2018)⁴⁴⁾ developed a computer vision-based method for detecting workers not wearing safety belts during high-altitude operations. The system combines two deep learning models: Faster R-CNN for worker detection and CNN for safety belt classification. The results show high accuracy in detecting workers without safety

belts, with Faster R-CNN improving annotation focus and reducing computation time. Al-Azani et al. (Al-Azani et al., 2024)⁴⁵⁾ developed a real-time PPE detection system using CCTV data and compared three models: CenterNet, DAB-Deformable-DETR, and YOLOv7. The YOLOv7 model performed best, with 92.36% precision on familiar data and 66.70% on new environments. This system helps managers detect and address PPE noncompliance in real time, enhancing workplace safety. Riaz et al. (Riaz et al., 2023)⁴⁶⁾ introduced PPE Swin, a method based on Swin U-Net and self-attention for automatic PPE detection in construction environments. The model integrates global and local feature extraction and was trained on a new, diverse dataset. PPE Swin achieved 97% detection accuracy, outperforming existing methods, offering an advanced AI-driven solution to improve PPE compliance and construction site safety management. Velasco and Marasigan (Velasco & Marasigan, 2024)⁴⁷⁾ developed an AI-based system for ensuring proper PPE usage among construction workers. The system includes a web-based real-time monitoring application, achieving 92% accuracy in PPE compliance detection. It allows managers to monitor worker safety and intervene promptly. Gil and Lee (Gil & Lee, 2024)⁴⁸⁾ introduced a zero-shot PPE monitoring model that doesn't require a training dataset. The system uses mm-Pose to detect human keypoints and ClipCap technology to generate captions for cropped body-part images, providing semantic support for compliance evaluation. It then uses an MPNet-based sentence bidirectional encoder representations from transformers (SBERT) model to assess compliance with OSHA PPE standards. This

method outperforms traditional object detection, achieving an F1 score of 82.6% and 53% accuracy, demonstrating its flexibility and efficiency in monitoring PPE compliance without large labeled datasets. Kumar et al. (Kumar et al., 2024)⁴⁹⁾ developed a drone-based PPE monitoring system using TensorFlow and Faster R-CNN for real-time helmet compliance detection. The system ensures robust PPE recognition in varying conditions and enables immediate feedback, reducing injury risks through rapid intervention. With a high accuracy of 93.1%, a mean average precision of 58.45%, and a real-time processing rate of 27 fps, the system automates safety inspections, improving monitoring efficiency and reducing supervisors' workloads. Nguyen et al. (Nguyen et al., 2024)⁵⁰⁾ introduced a stochastic hybrid optimization YOLOv5 (SHO-YOLOv5) model to improve PPE detection. By optimizing the YOLOv5 network with a refined scale and loss function, the SHO algorithm enhances both detection accuracy and computational efficiency. This model enables automated PPE compliance monitoring, improving real-time safety assessments and reducing workplace accident risks.

2.1.2 Construction worker action & posture

Many researchers have noted that adopting an improper posture during work is a leading cause of safety accidents among construction workers (Zhang et al., 2020)⁵¹⁾. Behavior-based safety has gained significant attention in safety management research to mitigate this problem. Traditional behavior-based safety methods predominantly rely on manual observation and on-site supervision, where safety managers

monitor workers' postures and behaviors to identify unsafe actions. However, such approaches suffer from high cost, low efficiency, and limited scalability, making them inadequate for large-scale construction projects (Han & Lee, 2013)⁵²⁾. Computer vision technology has become a viable option for automated behavior monitoring in response to these problems. Vision-based behavior detection has advanced, enabling automatic identification of risky postures and behaviors (Han et al., 2014)⁵³⁾.

Han et al. (Han et al., 2012)⁵⁴⁾ proposed a motion capture method based on vision to identify dangerous behaviors on construction sites. The method uses prior models of unsafe behaviors, developed through experimental analysis, as references for detecting similar actions in videos. By extracting 3D human skeletal models from on-site footage and comparing them to predefined unsafe behavior models, the system enables real-time action recognition. Han and Lee (Han & Lee, 2013)⁵⁵⁾ developed a vision-based framework for automated detection of unsafe behaviors in construction. The system follows four steps: identifying critical unsafe behaviors, collecting motion templates and video data, extracting 3D skeletal information from video, and detecting unsafe actions using motion templates and skeletal models. Focusing on ladder-climbing scenarios, the framework successfully identified hazardous behaviors like excessive lateral reaching. Han et al. (Han et al., 2014)⁵⁶⁾ developed a method for detecting unsafe actions through feature extraction using four key motion features, motion pattern simulation via average trajectory estimation, and spatiotemporal similarity-based classification. The method, validated

with red, green, blue, and depth (RGB-D) sensors for ladder-climbing scenarios, achieved a 99.5% classification accuracy in a controlled lab environment. Han et al. (Han et al., 2013)⁵⁷⁾ introduced a cost-effective depth sensor for capturing worker motion data to detect hazards. The framework involves preprocessing raw motion data into structured 3D spatial data for analysis and using template-based motion classification to identify unsafe behaviors. The system, tested on ladder-climbing actions like climbing backward and carrying objects, achieved a 90.91% detection accuracy, demonstrating its reliability and effectiveness in real-world applications. Guo et al. (Guo et al., 2018)⁵⁸⁾ developed a skeleton-based method for real-time recognition of unsafe behaviors, simplifying dynamic movements into static postures for efficient detection. The system consists of three modules: a database of hazardous action patterns, a real-time data collection module, and a behavior judgment module. Laboratory tests confirmed the method's feasibility, efficiency, and accuracy, highlighting its potential for identifying causes of unsafe behavior, such as poor management, inadequate safety measures, and suboptimal decision-making.

Recently, deep learning has been widely adopted to recognize construction worker behavior. With rapid advancements in neural network architectures, automated worker action recognition has significantly improved. A review of the research highlights the widespread adoption of several mainstream deep learning models. These models include the CNN for spatial feature extraction from images and videos, deep neural networks for high-dimensional feature representation learning, and RNNs (particularly LSTM networks) for

temporal sequence modeling, enabling the detection of dynamic posture changes and action patterns. These deep learning approaches have enhanced the accuracy and automation of behavior recognition, facilitating the transition from traditional manual safety monitoring to intelligent recognition systems (Fang et al., 2020)⁵⁹⁾.

Ding et al. (Ding et al., 2018)⁶⁰⁾ proposed a CNN-LSTM hybrid model for automated unsafe behavior recognition in construction. The system classifies actions as safe or unsafe by collecting real-world motion data and video footage, extracting visual features via CNN, and analyzing temporal sequences using LSTM. Experimental results showed high accuracy, validating the model's effectiveness for real-world safety monitoring. Luo et al. (Luo et al., 2019)⁶¹⁾ proposed a hierarchical statistical method for recognizing complex worker activities in remote monitoring videos. The approach combines a temporal segment network with a novel fusion strategy and a Dirichlet process-based hidden semi-Markov model for detailed action sequence analysis. It achieved 84% accuracy on a remote monitoring dataset, demonstrating its effectiveness for large-scale safety monitoring in industrial environments. Zhang et al. (Zhang et al., 2018)⁶²⁾ proposed a PPR method to improve vision-based ergonomic posture recognition in construction. This method employs a multistage CNN architecture to estimate 3D skeletal and joint information from 2D video frames. The method integrates a convolutional pose machine, 3D joint estimation, 3D joint position and angle extraction, and an end-to-end trained posture classifier. This method enables 3D ergonomic posture recognition with a single 2D camera, outperforming previous

approaches. Chu et al. (Chu et al., 2020)⁶³⁾ designed a monocular vision-based framework for biomechanical analysis consisting of tracking, 2D recognition, refinement, and 3D modeling. This method estimates 14 joint angles with an average error of 11° using video data from a single camera in an actual construction workshop. The framework identifies improper postures and actions of workers, providing strong support for safety management on construction sites. Chu et al. (Chu et al., 2020)⁶⁴⁾ proposed a two-stage method for accurately detecting construction workers in dynamic scenes, combining ResNet-152 for feature extraction and Faster R-CNN for detection. Tested on real construction site footage, the system achieved 94.3% accuracy, 96.03% precision, and 98.13% recall, demonstrating strong performance across varied postures and backgrounds. Kong et al. (Kong et al., 2018)⁶⁵⁾ proposed a computer vision-based 3D biomechanical framework to estimate construction workers' mechanical energy expenditure. By integrating smart insoles for foot pressure and acceleration data with a 3D posture prediction algorithm from 2D video, the system quantifies energy use across joints and tasks such as walking, lifting, and bending. The approach enables estimation of daily total energy expenditure, offering insights to optimize workloads and reduce worker fatigue. Yu et al. (Yu et al., 2019)⁶⁶⁾ developed an automatic ergonomic assessment tool for construction workers, combining 3D posture estimation from 2D images with rapid entire body assessment (REBA)-based risk scoring. The system accurately captures 3D joint positions and provides multilevel ergonomic evaluations with low intrusion, making it suitable for

dynamic site conditions. Field tests confirmed its feasibility and effectiveness compared to traditional methods. The joint-level visual ergonomic assessment for construction workers (JVEC) tool enables comprehensive ergonomic risk management by identifying high-risk tasks, detecting overloaded joints, and informing work-rest scheduling strategies. Yang et al. (Yang et al., 2023)⁶⁷⁾ proposed the spatiotemporal relation (STR)-Transformer, a deep learning model for recognizing construction worker behaviors from multi-view video streams. It integrates spatial and temporal features using transformers, 3D CNNs, and a relative position fusion module. Validated on the newly built Construction Meta Action (CMA) dataset with seven behavior classes, the model achieved 88.7% accuracy, surpassing baselines by four points. The STR-Transformer demonstrates strong potential for precise, automated safety monitoring on construction sites.

2.2 Research related to deep learning

2.2.1 Object detection model

Object detection in computer vision involves locating and categorizing items in an image by generating bounding boxes and classifying the categories. The process comprises two components: object localization and object classification. Localization determines object positions in an image, typically using bounding boxes, whereas classification identifies the detected object category. Deep learning developments have greatly enhanced object detection algorithms, allowing for broad applicability in various fields. However, the development of deep neural networks has

historically been limited by computational resources, dataset constraints, and theoretical bottlenecks. Notable methods include the deformable part model, selective search, Oxford-MKL, and NLPR-HOGLBP. These approaches, developed under limited computational power, rely on handcrafted features and relatively simple classification models. In contrast, modern object detection employs deep learning, particularly the CNN, to enhance accuracy and robustness significantly. Based on the detection process, modern algorithms are classified into two-stage and single-stage detectors. Two-stage detectors (e.g., region-based CNN features (R-CNN)) classify and improve bounding boxes for accuracy after first producing candidate regions. In contrast, single-stage detectors (e.g., YOLO) directly perform localization and classification across the image, offering faster processing speeds, which are ideal for real-time applications. These methods achieve efficient and precise object detection by integrating feature extraction, bounding-box prediction, and classification, even under challenging conditions, such as lighting variations, partial occlusions, or background clutter.

A. Two-stage object detection architecture

The R-CNN marked an early integration of deep learning into object detection. Inspired by AlexNet, Girshick et al. introduced deep CNNs, replacing traditional methods, such as the histogram of oriented gradients and scale-invariant feature transform (Girshick et al., 2015)⁶⁸). The R-CNN architecture, which utilizes region proposal algorithms like selective search, improved object detection accuracy by about 30% over traditional methods. However, it suffers from major

limitations: high computational cost due to processing $\sim 2,000$ proposals per image, fixed input size requiring region resizing to 227×227 (risking information loss), and inefficient feature extraction, as features are computed separately for each region without leveraging CNN feature sharing. Despite its accuracy gains, these inefficiencies hinder its scalability for large-scale applications.

Compared to the R-CNN, the Fast R-CNN offers significant advantages in speed, memory efficiency, and training performance. Girshick et al. introduced the region of interest (ROI) pooling layer, also known as single-level spatial pyramid pooling (SPP), to address the limitations of the R-CNN. In the Fast R-CNN (He et al., 2015)⁶⁹, a deep CNN computes the image feature map, onto which region proposals generated using the selective search algorithm are mapped. The ROI pooling layer transforms regions of varying sizes into fixed-size feature vectors, which are input into fully connected (FC) layers for further processing. Finally, a softmax function predicts object classes, and bounding-box regression refines the object localization. The Fast R-CNN employs a multitask loss function that jointly trains classification and bounding-box regression, allowing these tasks to share convolutional features and enabling end-to-end training. Compared to the R-CNN and the SPP network, the Fast R-CNN offers several advantages: improved accuracy, a more efficient training process due to the multitask loss function, and the ability to update all network layers during training, unlike the SPP network, where only FC layers are updated. Feature caching no longer requires disk storage, reducing memory consumption, and overall training and testing speeds

are significantly faster than those of the R-CNN and SPP network.

The region proposal network (RPN) was introduced in the Faster R-CNN (Ren et al., 2016)⁷⁰⁾ to increase object detection efficiency. This network improves the architecture by combining object identification and region proposal creation into a single, cohesive network, building on the Fast R-CNN. Unlike the Fast R-CNN, which relies on external region proposal algorithms (e.g., selective search), making the process computationally expensive and independent of the detection network, the Faster R-CNN incorporates the RPN, enabling end-to-end training and shared convolutional features and optimizing computational efficiency. By sharing convolutional features, the Faster R-CNN improves computational efficiency and eliminates dependence on external region proposal algorithms. The RPN applies convolutional features to generate more accurate candidate regions, enhancing object localization and spatial relationship modeling. The network dynamically generates adaptive candidate regions suitable for diverse image scenes. The end-to-end training of the Faster R-CNN enables the joint optimization of the entire network, reducing training time and complexity. Overall, the Faster R-CNN surpasses the Fast R-CNN in speed, accuracy, training efficiency, and storage requirements.

The Mask R-CNN (He et al., 2017)⁷¹⁾ adds a mask branch (e.g., segmentation) to the Faster R-CNN to enable simultaneous object identification and instance segmentation. The network's core improvements include pixel-level segmentation and the introduction of ROIAlign. The mask network employs a simplified, fully convolutional network to generate pixel-level segmentation masks for each candidate

region, allowing for precise object segmentation. To address the misalignment caused by quantization in ROI pooling, He et al. introduced the ROIAlign layer, which employs bilinear interpolation to align the ROI feature map accurately with the original image region, significantly improving mask prediction accuracy. The critical advantage of the Mask R-CNN is in its ability to perform object detection and instance segmentation concurrently, with ROIAlign enhancing the segmentation accuracy and detection performance. The experimental results indicate that the Mask R-CNN based on the ResNet-101 feature pyramid network (FPN) surpasses the Faster R-CNN with G-RMI in fine-grained segmentation tasks.

Lin et al. presented the FPN in 2017 as a novel approach to overcome the drawbacks of conventional object identification techniques (Lin et al., 2017)⁷²). Traditional detectors typically used high-level features for detection or performed independent detection at various feature layers, ineffectively combining classification and location information. Low-level features offer rich location information but poor semantics, whereas high-level features offer better semantics but less precise location data, making multiscale object detection challenging for traditional methods. The main improvements of the FPN include top-down feature enhancement (refining high-level features to become more suitable for multiscale object detection), lateral connections (combining enhanced high- and low-level features to retain location information while improving semantic representation), and multilevel object detection (performing detection on the fused feature pyramid, capturing objects at various scales). The feature

fusion method of the FPN significantly improves detection accuracy, particularly in multiscale object scenarios, and is better able to capture fine details, greatly enhancing detection precision.

B. Single-stage object detection architecture

The YOLO end-to-end model directly regresses object class probabilities and bounding boxes, eliminating region proposals (Redmon et al., 2016)⁷³⁾. This design significantly improves detection speed. The YOLO and R-CNN family represent two fundamentally different approaches to object detection. Some limitations of YOLO include its suboptimal performance in detecting dense, small objects because each grid cell is restricted to predicting only two bounding boxes and associating them with a single class. Moreover, YOLO has weak generalization for objects with varying aspect ratios because its ability to generalize is limited when objects of the same class appear with a new aspect ratio in test images. In addition, YOLO has a flawed loss function, affecting the detection performance improvement. Despite YOLO's significant advantages in real-time detection tasks, it does not perform as well as the R-CNN family in complex scenes and small-object detection.

To address YOLO's limitations in detecting multiscale objects, Liu et al. proposed the SSD, combining the strengths of the Faster R-CNN and YOLO to achieve efficient and high-precision object detection (Liu et al., 2016)⁷⁴⁾. The SSD uses the VGG16 as its backbone for feature extraction, substituting convolutional layers (Conv6 and Conv7) for the FC layers (FC6 and FC7). This method introduces four extra convolutional layers (Conv8 to Conv11) to enhance multiscale feature

representation. The core idea of the SSD is to extract features hierarchically, generating feature maps at various semantic levels from convolutional layers, which are employed for object classification and bounding-box regression. The SSD effectively adapts to multiscale object detection tasks using multiscale feature maps and anchor boxes, addressing YOLO's shortcomings when handling objects with significant scale variations. In terms of performance, the SSD outperforms the Faster R-CNN and YOLO. In addition, SSD512 has significantly better accuracy than the Faster R-CNN and is three times faster. Compared to YOLO, SSD300 runs at 59 fps, significantly improving detection quality. The advantage of the SSD is that it combines the efficiency of single-stage detection methods with the powerful detection capability of multiscale features, balancing accuracy and speed.

The deconvolutional SSD (DSSD) uses ResNet-101 as its backbone network and introduces deconvolutional modules and skip connections to improve the representation of low-level feature maps and enhance the expressive capability of low-level feature maps in SSD (He et al., 2016)⁷⁵). Compared to the SSD, the DSSD enhances the understanding of low-level features in complex scenes. The DSSD restores the spatial resolution of low-level feature maps using deconvolution modules, whereas the skip connections improve the fusion of low- and high-level features, resulting in better detection accuracy. Although the DSSD achieves significant improvements in accuracy, the addition of deconvolution and skip connection modules increases its computational complexity and model parameters, leading to higher computational and storage costs.

The YOLO, SSD, and DSSD methods each have unique strengths and weaknesses. For example, YOLO is fast and ideal for real-time tasks but struggles with small, dense objects. The SSD balances accuracy and speed with multiscale feature fusion and anchor mechanisms, making it versatile for various tasks. Finally, the DSSD improves low-level feature representation for better accuracy but requires more computation and storage.

C. Construction-related tasks

The object detection tasks in the construction field are highly complex because of the numerous details involved, diverse object scales, and complex geometries in buildings and facilities. These tasks are also influenced by weather, lighting, and changes in perspective. Thus, traditional object detection methods must be adjusted and optimized to meet the demands of the construction industry. By integrating advanced object detection architectures, such as YOLO, the Faster R-CNN, and the SSD, with the unique characteristics of the construction sector, detection accuracy and robustness can be enhanced, particularly in building facade detection and construction site monitoring. Object detection has already been applied in the automated inspection of construction projects, construction process monitoring, and structural health monitoring. These applications improve work efficiency and accuracy and reduce the cost of manual inspections, enhancing safety and quality control. Table 1 presents several standard object detection models for research.

Table 1. Common object detection models for construction applications.

Research	Detection Target	Year	Model	Content
(76)	Helmet	2018	Faster R-CNN	A method based on the Faster R-CNN for detecting nonhardhat-use of construction workers in far-field surveillance videos
(77)	PPE	2018	SSD	A method for automatically inspecting safety measures for workers at heights using deep learning-assisted computer vision technology
(78)	Excavators and dump trucks	2019	ResNet-101	A method for automatically identifying visually unique working activities of construction equipment from video sequence frames
(79)	Workers	2019	Mask R-CNN	A method for developing the Mask R-CNN to detect workers passing through concrete/steel scaffolds during construction accurately
(80)	PPE	2020	YOLO-v3	A method for determining the compliance of workers wearing PPE from real-time images/videos
(81)	Construction vehicles	2020	FSSD	An end-to-end network detection method based on deep learning and the CNN

2.2.2 Action recognition model

Human action recognition (HAR) focuses on automatically identifying and understanding human behavior, which is crucial for practical applications. In visual surveillance systems, HAR can identify potentially dangerous behaviors, enhancing public safety. Moreover, HAR methods can be categorized based on whether the data modality is visual or nonvisual. The visual modality includes RGB image

sequences, skeleton data, depth images, and infrared images, among others. These data directly represent human actions, offering rich spatial and semantic information. In contrast, nonvisual modalities include audio, accelerometer data, radar, and Wi-Fi signals. Although not directly visual, these modalities offer distinct advantages in privacy-sensitive environments or conditions with low visibility and restricted lighting. For instance, audio data are well-suited for identifying sound-related actions from time-series data. Accelerometer data are helpful for fine-grained behavior recognition, such as gait or gesture detection. Radar data can penetrate obstacles for behavior detection through walls, and Wi-Fi signals enable contactless behavior modeling by capturing changes in signal reflections (Sun et al., 2020)⁸²⁾.

The RGB framework is widely used in visual surveillance, autonomous navigation, and motion analysis due to its accessibility and ability to capture rich visual information. However, RGB-based action recognition faces challenges such as background changes, viewpoint variations, human scale differences, and lighting fluctuations. Additionally, the large data volumes of RGB videos increase computational complexity in spatiotemporal modeling. Deep learning techniques have significantly advanced to address these challenges, with three primary approaches: dual-stream 2D CNN, RNN, and 3D CNN-based methods. These methods focus on efficiently extracting and applying spatiotemporal information to improve action recognition accuracy.

Girdhar et al. (Girdhar et al., 2017)⁸³⁾ introduced a novel video

representation method for action classification by aggregating local convolutional features across the spatiotemporal domain. This method combines a dual-stream network with a learnable spatiotemporal feature aggregation mechanism, resulting in an end-to-end trainable architecture for video classification. Key innovations include joint pooling across spatial and temporal dimensions to capture complex motion dynamics and separately aggregating appearance and motion stream features to preserve their unique characteristics, thus enhancing classification performance. Experimental results on datasets like HMDB51, UCF101, and Charades showed a 13% relative performance improvement over traditional dual-stream architectures, outperforming other state-of-the-art methods. Diba et al. (Diba et al., 2017)⁸⁴⁾ introduced Temporal Linear Encoding (TLE), a CNN layer that captures appearance and motion features across video sequences. TLE aggregates spatiotemporal information into a compact representation, improving action classification. It learns discriminative features while maintaining efficiency and adaptability, and outperforms state-of-the-art methods on datasets like HMDB51 and UCF101, demonstrating its effectiveness in capturing complex spatiotemporal dynamics. Feichtenhofer et al. (Feichtenhofer et al., 2017)⁸⁵⁾ designed a generic CNN architecture for video action recognition based on spatiotemporal feature multiplicative interactions. The model incorporates a motion gating mechanism to fuse appearance and motion path features in a dual-stream network, optimizing the system via end-to-end training. The motion gating mechanism enhances action recognition by integrating spatiotemporal features. Zong et al. (Zong et

al., 2021)⁸⁶⁾ introduced the MSM-ResNets, a multistream multiplicative residual network for video action recognition. It consists of three interdependent streams: appearance, motion, and motion saliency. The model enhances action recognition by capturing spatiotemporal features and applying interactions between these streams. The appearance stream extracts static features, the motion stream captures dynamic patterns, and the motion saliency stream highlights significant motion while suppressing background noise. Two multiplicative connection mechanisms facilitate efficient feature interaction, improving robustness and accuracy by focusing on crucial motion information while reducing background interference. Wang et al. (Wang et al., 2017)⁸⁷⁾ proposed an RGB-D video representation method for efficient action recognition, using bidirectional rank pooling to encode depth sequences into structured dynamic images at multiple levels. Unlike traditional methods that apply separate CNNs to each body part or joint, this approach constructs dynamic image pairs at various granularity levels, preserving spatiotemporal characteristics and improving feature representations. It enhances action dynamics capture across body parts, joints, and temporal scales, while being computationally efficient and memory-friendly, making it suitable for resource-constrained environments and real-world applications. Xiao et al. (Xiao et al., 2019)⁸⁸⁾ introduced a depth-based video behavior recognition method using dynamic images, which projects depth video data into multiple viewpoints and extracts dynamic images to capture 3D features. The key innovation is the multiview fusion strategy, which enhances spatiotemporal information and improves behavior recognition accuracy.

Their CNN model uses shared convolutional layers across different viewpoints and distinct fully connected (FC) layers, improving multiview feature fusion while reducing computational overhead. This design also mitigates the gradient vanishing problem in deep networks, enabling more stable gradient propagation. The method optimizes multiview dynamic image processing, offering a computationally efficient solution for real-time behavior recognition. Zhang et al. (Zhang et al., 2016)⁸⁹⁾ proposed using motion vectors instead of optical flow to accelerate video action recognition. Motion vectors, extracted directly from compressed video streams, offer significant computational efficiency but suffer from lower recognition performance due to noise and lack of fine structural details. To address this, they introduced a strategy that transfers knowledge from optical flow to motion vectors, leveraging their intrinsic correlation. This approach enhances motion vector performance, boosting recognition accuracy and mitigating their limitations, while still achieving substantial computational acceleration for video action recognition. In contrast, Piergiovanni and Ryoo (Piergiovanni et al., 2019)⁹⁰⁾ introduced the representation flow layer, a convolutional layer inspired by optical flow algorithms, to learn motion representations in videos and improve action recognition. Unlike traditional optical flow methods, this fully differentiable layer allows joint learning with other CNN parameters through end-to-end optimization, enhancing computational efficiency while capturing dynamic information. The representation flow layer models spatiotemporal dynamics flexibly and, when stacked, improves the network's ability to capture long-term dependencies for recognizing

complex actions. Experimental results show that this approach outperforms traditional methods in robustness and accuracy for action recognition.

The RNN are well-suited for processing time-series data due to their recurrent connections, which integrate current inputs with previous hidden states, capturing temporal dependencies. This makes RNN effective for sequential tasks like speech recognition, natural language processing, and video analysis. LSTM networks, a sophisticated gated RNN architecture, address long-term dependencies by solving the gradient vanishing and explosion problems common in traditional RNN. LSTM's gating mechanism selectively retains and updates information over long sequences, greatly improving performance in tasks involving long-term sequential data, such as video analysis. Donahue et al. (Donahue et al., 2015)⁹¹⁾ proposed the Long-term Recurrent Convolutional Network (LRCN), an end-to-end trainable architecture combining a 2D CNN for frame-level feature extraction and an LSTM for modeling temporal dependencies. This integration enables effective video action recognition by jointly capturing spatial and temporal dynamics. Experimental results show that LRCN outperforms models treating spatial and temporal components separately, thanks to its ability to learn both detailed visual features and long-range temporal patterns. Majd and Safabakhsh (Majd & Safabakhsh, 2020)⁹²⁾ proposed C²LSTM, an improved LSTM variant that integrates convolutional operations to better capture spatial features, motion dynamics, and temporal dependencies in video data. Built upon this unit, they developed a deep neural network for HAR,

enabling end-to-end learning of spatial, motion, and temporal representations. Evaluated on UCF101 and HMDB51, C²LSTM demonstrated superior performance by enhancing traditional LSTM's temporal modeling with motion-aware features, leading to notable improvements in action recognition accuracy. He et al. (He et al., 2021)⁹³⁾ introduced the Dense Bi-LSTM (DB-LSTM) architecture, which captures both forward and backward temporal dependencies using stacked Bi-LSTM units with dense skip connections. These connections enhance information flow and improve long-term dependency modeling. DB-LSTM also incorporates a modality fusion module to effectively combine appearance and motion features, leading to a richer spatiotemporal representation. Evaluated on HMDB51 and UCF101, the model outperformed several state-of-the-art methods, demonstrating the effectiveness of combining bidirectional temporal modeling, dense connectivity, and multimodal fusion for robust action recognition. Zhao and Jin (Zhao & Jin, 2020)⁹⁴⁾ proposed a “dual depth” architecture for action recognition, enhancing spatiotemporal modeling through deep spatial and temporal pathways. An improved p-nonlocal operation captures long-range dependencies, while CNNs extract spatial features. A keyless attention mechanism with Bi-LSTM enables efficient temporal learning, and a multiepoch fusion strategy improves accuracy. Tested on HMDB51 and Hollywood2, the model outperformed traditional CNN and RNN approaches. Sudhakaran et al. (Sudhakaran et al., 2019)⁹⁵⁾ introduced the Long Short-Term Attention (LSTA) mechanism to enhance action recognition by focusing on spatially important regions while maintaining consistent attention over

time. Integrated into a dual-stream architecture, LSTA simultaneously processes appearance and motion information, enabling effective spatiotemporal feature extraction. This design allows the model to track dynamic changes across frames, improving recognition accuracy through refined attention to critical video regions. Wu et al. (Wu et al., 2019)⁹⁶⁾ proposed AdaFrame, a framework for efficient video recognition that adaptively selects the most informative frames based on task relevance. By using an LSTM with a global memory mechanism, AdaFrame captures temporal dependencies and guides future frame selection. It employs a policy gradient method to learn a frame selection strategy that maximizes expected utility. During inference, this strategy enables adaptive reasoning, significantly reducing computation while maintaining high accuracy. Liu et al. (Liu et al., 2020)⁹⁷⁾ proposed a multistream residual network, STS-ResNet, and an enhanced variant, STS-ALSTM, for action recognition. The model combines spatial (RGB), temporal (optical flow), and spatiotemporal saliency (STS) streams to emphasize dynamic regions and suppress background noise. STS-ALSTM further incorporates an attention-aware LSTM to capture long-term temporal dependencies and adaptively focus on important frames. This design improves recognition accuracy and robustness, especially in complex scenes with cluttered backgrounds and dynamic actions. Kim et al. (Kim et al., 2018)⁹⁸⁾ introduced a method for recognizing group activities in surveillance videos using discriminative group context features (DGCFs). This approach combines individual trajectory data with key subevent characteristics to capture interindividual relationships and dynamic

changes in group behavior. The DGCFs, paired with a gated recurrent unit (GRU) model, capture long-term temporal dependencies, improving group activity recognition. Trajectory data augmentation enhances generalization and reduces overfitting. This method effectively distinguishes group actions by focusing on interactions within subgroups, improving recognition accuracy and robustness in complex scenarios.

Many researchers have recently expanded 2D CNN to 3D structures to collect spatial and temporal context information, both critical for HAR from videos. The ability of the 3D CNN to model the spatial characteristics of individual frames and temporal dynamics between successive frames concurrently is greatly improved by incorporating the temporal dimension into conventional 2D convolutional networks. Unlike the traditional 2D CNN, which focuses on spatial information in each frame, the 3D CNN can capture spatial and temporal dependencies, making them exceptionally effective at identifying spatiotemporal patterns in video sequences.

Ji et al. (Ji et al., 2012)⁹⁹⁾ proposed a novel 3D CNN model for HAR that performs 3D convolutions across both spatial and temporal dimensions. The model uses multiple channels, each conducting independent convolutions and subsampling to extract spatial and temporal features. This approach captures local spatiotemporal patterns while reducing computational complexity. By integrating data from all channels, the model creates a comprehensive video feature representation, improving both accuracy and efficiency in action recognition. Li et al. (Li et al., 2020)¹⁰⁰⁾ introduced a spatiotemporal

deformable convolutional network (ConvNet) with an attention mechanism for action recognition. This module captures long-range spatiotemporal dependencies by learning adaptive offsets, allowing it to model irregular action patterns more effectively. The attention-based deformable module adjusts the receptive field dynamically, enhancing the model's ability to capture complex spatiotemporal structures. Experimental results on datasets like UCF-101 and HMDB-51 show that this approach significantly outperforms existing methods, proving its effectiveness in capturing detailed action information. Yang et al. (Yang et al., 2020)¹⁰¹⁾ proposed a universal temporal pyramid network (TPN) that integrates with 2D and 3D backbones for enhanced video action recognition. The TPN captures multiscale temporal information through feature sources and combines it into a hierarchical structure, improving the model's ability to recognize actions with varying rhythms and speeds. Easily integrated into existing CNN architectures, the TPN demonstrated consistent performance improvements, including a 2% boost on the Kinetics-400 validation set with 3D ResNet-50. Varol et al. (Varol et al., 2021)¹⁰²⁾ introduced an approach to enhance action recognition by using synthetic videos for data augmentation. Their method, SURREACT, generates diverse action data by simulating different viewpoints, body shapes, clothing variations, and action-related augmentations, such as nonuniform frame sampling and interpolation. This synthetic data generation improves model robustness by increasing action diversity. The method showed significant performance gains on human action multiview datasets like NTU RGB+D and UESTC, as well as the Kinetics dataset, enhancing

recognition accuracy, especially with limited training data. Fayyaz et al. (Fayyaz et al., 2021)¹⁰³⁾ introduced a distinguishable similarity-guided sampling module to enhance 3D CNN architectures. This module adapts the resolution of temporal features by dynamically grouping similar frames, allowing the resolution to adjust based on the characteristics of each video clip. This adaptive mechanism reduces computational redundancy by varying feature extraction precision. When integrated into 3D CNNs, it reduces computational costs by around 50% while maintaining or improving action recognition accuracy. Li et al. (Li et al., 2021)¹⁰⁴⁾ introduced the channel tensorization network (CT-Net), which optimizes feature processing by decomposing the channel dimension into subdimensions, reducing computational complexity. The CT-Net enhances feature interactions by expanding them into a 3D receptive field and includes a tensor excitation mechanism that applies attention across spatial, temporal, and channel dimensions, improving classification accuracy and performance in complex tasks.

In nonvisual modalities, audio signals are frequently provided alongside video. Because the visual and audio streams are synchronized, activities can be identified using audio data, reducing computational costs and the need for manual labeling. This approach enables the model to capitalize on the temporal alignment of audio and visual cues, enhancing its ability to recognize actions more competently. By integrating both modalities, the system can focus on more relevant portions of the video, improving accuracy while minimizing processing.

Liang and Thomaz (Liang & Thomaz, 2019)¹⁰⁵⁾ introduced an audio-based activity recognition framework that extracts features from sound clips using the pretrained VGGish model. These features capture environmental sounds, speech, and other auditory cues, which are crucial for recognition, especially when visual data is incomplete or noisy. The audio features are processed through a deep classification network, significantly improving activity recognition performance by incorporating multimodal information. Xiao et al. (Xiao et al., 2020)¹⁰⁶⁾ introduced the innovative audiovisual slowfast (AVSlowFast) network, which integrates slow and fast visual pathways with a fast audio pathway to jointly model audiovisual information. It combines multiple feature levels, with audio playing a key role, and uses the DropPathway technique to improve robustness by randomly dropping the audio pathway during training. Khan et al. (Khan et al., 2010)¹⁰⁷⁾ proposed an activity recognition system using kernel discriminant analysis to extract nonlinear features that enhance class differentiation by maximizing interclass variance and minimizing intraclass variance. This method improves classification performance by capturing activity differences more accurately. The system was validated using smartphone data collected from five body positions, offering flexibility over traditional methods. Unlike conventional systems that require specific body placement, smartphones with accelerometers can be placed in any pocket, allowing continuous monitoring over extended periods. Ignatov (Ignatov, 2018)¹⁰⁸⁾ developed a CNN-based method for user-independent HAR that enhances statistical features from accelerometer time-series data. This approach requires minimal feature

engineering and preprocessing, allowing for short recognition intervals (as brief as 1 second). The algorithm's shallow architecture makes it suitable for real-time operation on mobile devices with low computational demands. Shah and Fioranelli (Shah & Fioranelli, 2019)¹⁰⁹⁾ developed a radar-based system for human activity and fall detection. The system converts micro-Doppler features into images, using transfer learning and classifiers like support vector machine and k-nearest neighbors. Deep learning algorithms such as GoogleNet improve feature extraction, enabling the radar to outperform traditional methods in detecting motion and falls. Zhou et al. (Zhou et al., 2020)¹¹⁰⁾ proposed a dual-stream CNN for action classification, combining spectrograms and echo signals. The model was trained on a simulated Motion Capture (MoCap) dataset, where the echo signals were augmented with Gaussian white noise to simulate real-world conditions. The dual-stream approach, using both spectrograms and echo signals, achieved over 90% accuracy, outperforming traditional single-stream networks in action recognition. Chen et al. (Chen et al., 2018)¹¹¹⁾ proposed an attention-based Bi-LSTM (ABLSTM) for passive human activity recognition (HAR) using Wi-Fi channel state information. The method utilizes Bi-LSTM networks to capture sequential patterns from both forward and reverse directions. The attention mechanism dynamically adjusts the relevance of features and time steps, improving activity detection accuracy. ABLSTM enhances contextual information during sequence learning, leading to superior performance in action recognition compared to existing methods.

HAR in construction is challenging due to diverse actions,

spatial-temporal variations, and environmental factors. Integrating advanced models like 3D CNNs, LSTMs, or transformers improves accuracy for tasks such as process monitoring, worker behavior identification, and safety inspections. HAR enhances efficiency, reduces costs, and improves safety and quality control. Table 2 presents studies on typical activity recognition models for construction workers.

Table 2. Activity recognition models in the construction industry.

Research	Year	Model	Content
(112)	2022	GAN+ Resnet	A combined skeleton encoding, GAN inpainting, and ResNet classification model was used to address occlusion issues in construction worker behavior recognition, achieving an accuracy of 86.62%
(113)	2023	YOWO	A motion detection system assesses productivity and pinpoints inefficient actions in column formwork tasks at the worker level
(114)	2023	LSTM	LSTM-based motion detection technology based on a 3D skeleton assists in automatically checking whether safety belts are fastened correctly on the job site
(115)	2024	GCN+ TCN	Six 3D skeletal-based characteristics in the spatio-temporal multifeature network help track and record construction worker motions to identify their behavior
(116)	2024	CNN-LSTM	The improved CNN-LSTM deep learning architecture with three optimizations enhances the accuracy of traditional CNN-LSTM models applied to human activity recognition
(117)	2023	OpenPose	An OpenPose-based framework extracts high-altitude work-posture patterns, combining posture recognition with a Gaussian model for instability detection and assessment

2.3 Research on YOLO family models

YOLO models, widely used in construction for their real-time performance and efficiency, divide images into grids and perform regression-based predictions for object detection. This makes YOLO ideal for applications such as object recognition, equipment and material management, personnel monitoring, and safety enforcement. Construction sites present challenges like lighting variations, occlusions, and viewpoint changes, but YOLO maintains high inference speeds with low computational costs, making it suitable for real-time monitoring. By providing precise classification and localization, YOLO optimizes resource allocation and improves operational efficiency. Additionally, YOLO helps track worker movements and ensure compliance with PPE requirements, reducing safety risks. Overall, YOLO's ability to process data quickly, its computational efficiency, and its adaptability make it an effective tool for intelligent monitoring and safety management in dynamic construction environments (Terven et al., 2023)¹¹⁸⁾. Table 3 summarizes the versions of the YOLO family, highlighting the improvements introduced in each version.

Table 3. Major models and improvements in the YOLO family.

Version	Improvements
YOLOv1 (119)	• Unified convolutional network: Simultaneously performs object detection and classification. • Global inference: Performs object detection for the entire image at once. • Grid division: Creates grids of the image, each predicting the class and position of an object. • Bounding-box prediction: Predicts confidence scores of bounding boxes. • Real-time detection: Enables real-time object detection. • Low-resolution input: Uses 224×224 resolution images to accelerate processing.
YOLOv2 (120)	• Batch normalization: Improves convergence and reduces overfitting. • High-resolution classifier: Fine-tuned on 448×448 resolution, improving high-resolution performance. • Fully convolutional architecture: Removes dense layers and adopts a fully convolutional structure. • Anchor box prediction: Anchor boxes predict bounding-box coordinates and classes. • Dimensional clustering: Selects appropriate anchor boxes via k-means clustering to improve prediction accuracy. • Direct position prediction: Directly predicts bounding-box coordinates for each grid cell. • Finer features: Removes pooling layers to retain fine-grained features. • Multiscale training: Randomly adjusts input size to enhance model robustness.

YOLOv3 (121)	• Bounding-box prediction: Assigns an anchor box to each object and uses logistic regression to predict object confidence scores based on four coordinates (tx, ty, tw, and th). • Class prediction: Uses binary cross-entropy for multilabel classification, enabling multiple labels per box. • New backbone network: Uses a 53-layer feature extractor with residual connections. • Spatial pyramid pooling (SPP): Improved SPP block increases the receptive field, boosting performance. • Multiscale prediction: Predicts boxes at three scales for better detection accuracy. • Prior bounding boxes: Uses k-means clustering to determine anchor boxes, with three prior boxes used at three scales.
YOLOv4 (122)	• Enhanced architecture: Uses cross-stage partial connection (CSP) Darknet53-PANet-SPP architecture, combining the CSP network (CSPNet), Mish activation, SPP, PANet, and spatial attention module (SAM). • CSPs: Reduce computation while maintaining accuracy. • Modified PANet: Improves feature connection methods to enhance feature fusion. • Bag-of-specials: Uses backbone architectures (e.g., ResNeXt50, EfficientNet-B3, and Darknet-53) to boost performance. • Bag-of-freebies: Improves training and regularization using mosaic augmentation, DropBlock, label smoothing, or other such methods. • Complete intersection over union (CIoU) loss and cross-batch normalization: Adds CIoU loss and cross-batch normalization to optimize detection performance. • Self-adversarial training: Enhances model robustness against interference using adversarial attacks. • Genetic algorithm for hyperparameter optimization: Optimizes hyperparameters using genetic algorithms and cosine annealing schedulers to enhance training.

YOLOv5 (123)	• Backbone network: Uses a modified CSPDarknet53 with stem, SPPF layers, and upsampling layers to improve feature extraction and computational speed. • Feature processing: By pooling features at various scales, batch normalization, and sigmoid linear unit (SiLU) activation following convolution, the SPPF layer accelerates processing. • Neck and head: The neck uses SPPF and improved CSP-PAN, with the head similar to YOLOv3. • Augmentation features: Includes mosaic, copy-paste, random affine, MixUp, HSV augmentation, and random horizontal flipping. • Grid sensitivity: Enhances grid sensitivity for improved gradient stability. • Multiple versions: Provides YOLOv5n, s, m, l, x versions to meet hardware and application needs. • Open source and maintenance: YOLOv5 is open source and supports mobile versions with training and deployment integration. • Performance: YOLOv5x achieves 50.7% AP on the Common Objects in Context (COCO) dataset, with up to 200 fps inference speed and 55.8% AP with larger input sizes.
YOLOv6 (124)	• New backbone network: Uses EfficientRep, based on RepVGG, for higher parallelism. • Neck enhancement: Uses PAN and enhances with RepBlocks or CSPStackRep blocks for larger models. • Decoupled head: Developed an efficient decoupled head, applied after YOLOx. • Task-aligned learning: Uses task-aligned learning for label assignment in TOOD. • Classification and regression loss: Uses VariFocal classification loss and skew intersection over union (SIoU)/generalized IoU regression loss. • Self-refinement strategy: Implements self-refinement strategies for regression and classification. • Quantization scheme: Achieves faster detector performance via RepOptimizer and channel distillation.

YOLOv7 (125)	• E-ELAN: Enhances network learning ability by shuffling and merging cardinalities without disrupting the original gradient paths. • Model scaling: YOLOv7 adopts a connected model scaling strategy, adjusting block depth and width to maintain optimal model structure and avoid reduced hardware utilization.
YOLOv8 (126)	• Cross-level feature fusion module: Enhances detection accuracy by combining contextual information with high-level features. • Anchorless model and decoupled head: Improves accuracy by independently handling objectness, classification, and regression tasks. • Loss functions: Improves small-object detection with binary cross-entropy for classification and CIoU and DFL for bounding-box loss. • Semantic segmentation: Provides the YOLOv8-Seg model, combining the cross-level feature fusion module and segmentation head for object detection and semantic segmentation. • High performance: Version 8x achieves 53.9% AP on the MS COCO dataset with 280 fps on an Nvidia A100 with TensorRT, outperforming YOLOv5. • Usability: Runs through CLI and supports installation as a PIP package, offering training and deployment integrations.
YOLO-NAS (127)	• Quantization-aware module: Combines QSP and QCI with 8-bit quantization reparameterization to reduce accuracy loss in post-training quantization. • Automated architecture design: Uses Deci's AutoNAC technology for automated architecture design. • Hybrid quantization: Model parts are selectively quantized, balancing accuracy and latency. • Pretraining scheme: Offers pretraining solutions with automatic data labeling, self-refinement, and large dataset pretraining.

YOLOv10 (128)	• No nonmaximum suppression (NMS) training: Eliminates the need for NMS via consistent dual assignment, reducing inference delay. • Overall model design: Optimizes every element, including the spatial-channel decoupled downsampling, lightweight classification heads, and tiered guidance block designs, to increase accuracy and efficiency. • Enhanced model capability: Enhances performance without increasing computational costs by combining partial self-attention modules with large kernel convolutions.
YOLOv11 (129)	• Enhanced feature extraction: Uses new modules (e.g., C3k2 and CBS) to enhance feature extraction efficiency and accuracy. • Lightweight design: Reduces the parameter count by 22% compared to YOLOv8m, offering faster inference. • Broad task support: Supports computer vision tasks. • Efficient deployment: Can be deployed on edge devices, cloud platforms, and Nvidia graphics processing unit (GPU) systems, offering cross-environment adaptability.

Worker safety is a critical concern in construction settings. The YOLO series has demonstrated significant potential in personnel safety monitoring by enabling real-time detection and tracking of worker positions. This capability is beneficial on large construction sites, where monitoring worker movements enhances safety oversight and helps prevent accidents. The YOLO models can identify potential safety hazards, such as the absence of hard hats, inadequate protective clothing, or unsafe practices. The automated detection in YOLO technology can promptly issue alerts, notifying workers and management to implement necessary safety measures and mitigating

risks throughout the construction process.

Park et al. (Park et al., 2023)¹³⁰⁾ introduced SOC-YOLO, a real-time operator detection model based on YOLOv5, with four optimization strategies to improve accuracy and efficiency. Trained on the MOCS dataset and tested on the SODA dataset, SOC-YOLO outperformed traditional methods, showing enhanced accuracy and robustness. The model's attention mechanism, which extracts features across both channel and spatial dimensions, enables better generalization to diverse conditions. SOC-YOLO also offers practical safety benefits, such as real-time detection and tracking of workers during heavy machinery operations, helping avoid collisions. Li et al. (Li et al., 2023)¹³¹⁾ proposed a method combining spatiotemporal graph convolutional networks and YOLO to identify unsafe interactions on construction sites. The approach detects unsafe interactions between workers, machinery, and materials, aiding safety management. Key findings include the effectiveness of YOLO, OpenPose, and spatiotemporal networks, though action recognition's sensitivity to context highlights the need for the right method. The study's tasks, based on field observations, were practical, but using convenience sampling may have affected results. Wang et al. (Wang et al., 2020)¹³²⁾ proposed an improved YOLOv3-based method for real-time detection of safety helmets and protective clothing, aiming to enhance PPE monitoring on construction sites. Key improvements include: optimizing the residual convolutional structure for better feature extraction, using large-scale feature maps for multiscale prediction to detect small or complex targets, and refining anchor box clustering to boost detection precision

and stability. Experimental results show the model significantly improves accuracy while maintaining high speed, offering a practical and reliable solution for construction safety supervision. Kim et al. (Kim & Xiong, 2025)¹³³⁾ enhanced the YOLOv8 model for real-time detection of multiple PPE types on construction sites to reduce risks like falls. They built a comprehensive dataset covering helmets, harnesses, lanyards, and their absence. Model improvements include coordinate attention for better localization, GhostConv for reduced computation, transfer learning for small-sample performance, and merged nonmaximum suppression for optimized detection. Mahmud et al. (Mahmud et al., 2023)¹³⁴⁾ applied the YOLOv8 algorithm to an automatic safety helmet detection system, aiming to improve accuracy and robustness in complex environments. Using a dataset of 4,200 images—4,000 from Kaggle and 200 custom images with challenging weather conditions—the YOLOv8m-based model achieved 92% accuracy. The results demonstrate strong real-time detection performance under diverse lighting and weather conditions, making it well-suited for dynamic construction settings. Gu et al. (Gu et al., 2021)¹³⁵⁾ combined an improved OpenPose model with YOLOv4 to detect safety helmets across complex worker postures. OpenPose provides accurate pose estimation, enabling posture-based head localization—using a three-point method for upright poses and skin color detection for side-lying ones. YOLOv4 then checks for helmet use based on head-helmet overlap. The method achieves higher accuracy than previous approaches by improving head region localization. To improve the precision and resilience of safety helmet

identification, Wu et al. (Wu et al., 2019)¹³⁶⁾ proposed an enhanced YOLOv3 model by integrating DenseNet into its backbone, creating a YOLO-Densebackbone CNN. This design improves feature extraction while maintaining detection speed. Experimental results show the model handles challenges like low resolution, helmet damage, and occlusion, achieving a 2.44% accuracy improvement over standard YOLOv3. Liu et al. (Liu et al., 2024)¹³⁷⁾ developed a safety harness detection method for high-altitude workers using drone imagery and an enhanced YOLOv5 model. Key improvements include integrating CSPNet into the Darknet53 backbone, merging the focus layer, replacing the SPP block for better feature fusion, and applying CSPNet in PANet for improved multiscale processing. The improved model achieved 99.2% accuracy, outperforming Fast R-CNN, SSD, YOLOX-m, and YOLOv7, offering a highly effective solution for automated safety monitoring at elevated work sites. Song (Song, 2022)¹³⁸⁾ proposed an improved YOLOv3-based detection method, replacing the Res8 module with a RepVGG skip squeeze excitation module to boost speed and accuracy. The Res2 module replaces CBL×5 to address gradient vanishing and enhance feature reuse. Additional improvements include higher input resolution, a four-scale prediction mechanism for small targets, and the CIoU loss function for better localization. The method increased precision by 3.9%, recall by 5.2%, and mAP by 4.7% over the original YOLOv3. Ngoc-Thoan et al. (Ngoc-Thoan et al., 2024)¹³⁹⁾ improved the YOLOv5 model for automatic detection of PPE on construction sites to enhance safety management. Compared to manual inspection, the automated approach improves efficiency and data

handling. The model was trained on 11,978 images from real sites and web sources, covering six common types of PPE. Wang et al. (Wang et al., 2020)¹⁴⁰⁾ proposed CSYOLOv3, an improved YOLOv3-based algorithm for detecting safety helmets on complex construction sites. It integrates CSPNet and an enhanced CSPDarknet53 for better feature extraction, along with an upgraded SPP for improved multiscale prediction. Trained on a newly labeled dataset, CSYOLOv3 achieved over 90% accuracy and exceeded 20 fps. Li et al. (Li et al., 2024)¹⁴¹⁾ introduced YOLO-PL, a lightweight safety helmet detection algorithm based on YOLOv4. The YOLO-P optimization improves small-object detection and anchor point allocation, while an enhanced PANet boosts accuracy by integrating high- and low-level features. YOLO-PL replaces the SPP module with dilated convolution CSP and X res unit, and improves the model with L-VoVN, max-pooling, and swish activation. YOLO-PL outperforms YOLOv5 and YOLOv7 in helmet detection, making it ideal for real-time, resource-limited devices. Zhang and Shen (Zhang & Shen, 2024)¹⁴²⁾ introduced YOLO Dark Segment (DS), based on YOLOv8, for instance segmentation in low-light construction sites. The model incorporates a low-light enhancement network and a refined training strategy to handle challenges like equipment deformation and blurred edges. YOLO-DS achieved a mAP of 55.5% on MOCS, improving 2.8% over YOLOv8, and 56.5% on DarkMOCS, outperforming YOLOv8 by 4.5%, showcasing its effectiveness in low-light conditions.

Chapter 3. Materials and dataset development

3.1 Data acquisition of construction worker dataset

The High-altitude Worksite Real-time Video Dataset was sourced from the AI-Hub platform (The dataset details are provided in Appendix A.). The dataset was developed to facilitate creating services to prevent safety incidents during high-altitude operations. The primary goals of the dataset are to monitor unsafe behaviors, identify potential safety hazards, and detect problems related to equipment usage, enhancing safety in high-risk work environments. The dataset was constructed by realistically simulating high-altitude work scenarios with a range of high-risk operational settings. The dataset includes recordings of typical work behaviors and reenactments of potentially hazardous actions. The data collection involved the following high-precision devices to enable multiangle and comprehensive monitoring of high-altitude operations:

Head-mounted cameras (NexSys and LinkFlow): These cameras provide a first-person perspective of the workers, capturing detailed information about daily operations and the surrounding environment. This perspective is beneficial for observing activities closely related to workers, such as specific actions during operations, instances of unsafe behavior, and environmental risks.

Portable action cameras (GoPro9): These versatile devices can be quickly repositioned to suit diverse work scenarios. They are effective in recording dynamic and transient events, making them ideal for capturing complex or exceptional conditions, such as hazardous

situations arising from rapid movements at high-altitude construction sites.

Mobile CCTV systems: These systems offer long-term and continuous monitoring of large-scale worksite environments, providing comprehensive coverage of high-altitude operations. They produce stable data suitable for analyzing long-term behavior patterns, equipment conditions, and overall worksite environments.

The dataset achieves extensive, multiangle scene coverage through the collaborative use of these devices, reflecting the complexity of behavioral patterns and variety of potential risks in high-altitude work environments. Table 4 presents an overview of the primary recording equipment used in the dataset.

Table 4. Primary recording equipment in the high-altitude worksite dataset.

Equipment Name	Model/ Type	Purpose	Resolution/ Features	Remarks
Head-mounted camera	NexSys, LinkFlow	Captures the worker's head-mounted perspective, recording behaviors and surroundings during operations	1920×1080	Suitable for high-altitude operations, providing a first-person view
Action camera	GoPro9	Portable camera to record worker activities and work environments	1920×1080	Highly mobile, ideal for dynamic recording
Mobile CCTV	No specific model	Continuous monitoring of fixed scenes, recording on-site operations	1920×1080	Suitable for real-time monitoring of larger areas

During data collection, all videos were recorded in full high-definition resolution (1920×1080) in MP4 format, using x264 and xvid encoding standards to ensure clarity, storage ease, and processing efficiency. The total recorded duration exceeded 330 h, encompassing 19 construction sites in typical high-altitude work environments in Seoul, Incheon, Suwon, Hwaseong, Jicheon, and other areas. Each site was set up to simulate common high-altitude work scenarios, including scaffolding operations, ladder use, and aerial platform tasks. These scenarios represented the typical characteristics of real high-altitude work and incorporated simulated hazardous situations. All participants were required to wear standard-compliant PPE, including safety helmets, harnesses, and gloves, to ensure worker safety during data collection. This approach ensured safety during the data collection process and preserved the authenticity and validity of the data. Data collection occurred during regular work hours at each site (typically 8:00 a.m. to 6:00 p.m.) to capture typical behaviors in real-world work scenarios.

The images related to high-altitude operations were selected from the High-altitude Worksite Real-time Video Dataset, aligning with the study objectives, to construct the PPE dataset to analyze workers' PPE use. High-resolution and clear images were prioritized during the selection process to ensure image quality and the effectiveness of the study. Figure 5 depicts the sample images from the dataset, which feature three typical work scenarios: aerial platform operations, scaffolding work, and ladder tasks.



(a) Aerial platform operations

(b) Scaffolding work

(c) Ladder tasks

Figure 5. Personal protective equipment dataset examples: (a) Aerial platform operations; (b) Scaffolding work; (c) Ladder tasks.

3.2 Personal protective equipment dataset

3.2.1 Dataset organization and preprocessing

During dataset construction, cleaning and standardization procedures were implemented to ensure consistency and high quality. These measures optimized the dataset structure and content, providing a solid foundation for model training and validation. First, all images were thoroughly reviewed and manually inspected to address potential

duplication problems. Duplicate or highly similar samples were identified and removed via screening, ensuring that each dataset image had unique characteristics and value. Special attention was focused on discrepancies arising from changes in shooting angles, lighting conditions, and resolution adjustments. Strict criteria were applied to eliminate images with minor variations due to these factors, maintaining the quality and diversity of the dataset. The exclusion of duplicate images reduced data redundancy and prevented overfitting caused by repetitive data, substantially enhancing the generalizability and training efficiency.

Second, all cleaned images were standardized to a resolution of 1920×1080 pixels to ensure uniformity across the dataset. This standardization step eliminated potential errors in model training caused by inconsistent image dimensions. A consistent resolution also improved the ability of the model to capture fine details and enhanced its processing capacity, particularly for accurately identifying critical features in the images. All images were converted to the JPG format to ensure data compatibility and storage efficiency. This format offers a favorable balance of visual quality, file size, and widespread compatibility, providing significant advantages in data storage and processing.

These cleaning and standardization procedures created a refined and optimized dataset. The dataset's uniqueness and consistency improved the efficiency of the data analysis and model development and enhanced the reliability and applicability of the research findings. This dataset provided a robust foundation for model training and validation.

The finalized dataset was divided as follows:

Aerial platform operations: 800 high-quality images present typical worker behaviors during operations on aerial platforms;

Scaffolding work: 600 high-quality images reflect worker behavior patterns during scaffolding tasks; and

Ladder tasks: 600 high-quality images cover typical worker actions during ladder operations.

Based on these selected samples, a high-quality personal protective equipment detection dataset containing 2,000 images was constructed.

The dataset was split using the 80:10:10 ratio, allocating 80% for training, 10% for validation, and 10% for testing.

According to the Ultralytics YOLO official documentation, allocating approximately 70–80% of a dataset for training is a common practice in object detection tasks. This is particularly important for single-stage detectors like YOLO, where using as much training data as possible helps the model learn object appearance variations and contextual cues. Allocating 80% of the data for training ensures the model has enough examples to learn the structural features of targets, improving robustness to variations in scale, shape, and background, as well as enhancing convergence speed and detection accuracy. Existing research has demonstrated that, when using YOLO models for object detection, a higher proportion of training data generally correlates with improved mAP scores, thereby confirming the critical impact of abundant training samples on model performance (Apeināns, 2024)¹⁴³.

Additionally, reserving 10% of the data as a validation set enables dynamic performance evaluation during training, supports

hyperparameter tuning, and helps prevent overfitting. Another 10% is typically used as a test set for final performance assessment, ensuring the model's ability to generalize to real-world scenarios.

In summary, the 80:10:10 dataset split strategy is widely adopted in YOLO-based and other deep learning detection tasks. It effectively balances training efficiency with evaluation comprehensiveness and is considered a best practice in aligning performance with practical feasibility. This configuration has been consistently recognized across YOLO application studies as a reliable approach for balancing efficient training with rigorous evaluation. (Rahimi et al., 2024)¹⁴⁴.

3.2.2 Dataset annotation

Object detection aims to identify target categories and accurately locate objects using bounding boxes. The precision of these boxes is vital for model training and inference, as it affects both object localization and classification accuracy. High-quality labeled data is essential, considering factors like scale, pose, and occlusion.

This study employs the Roboflow platform (<https://roboflow.com/>), a widely adopted tool in computer vision projects, to perform annotations efficiently. Roboflow supports several functions, including uploading, annotating, augmenting, and converting image data formats. During annotation, Roboflow supports several annotation formats, such as bounding boxes, segmentation masks, and keypoint annotations, making it suitable for object detection and localization tasks. The platform enables manual annotations via intuitive drawing tools and provides features to ensure annotation accuracy and consistency. These

features include automated annotation assistance, data augmentation options, and annotation review functionalities. Once the annotations are completed, Roboflow can automatically convert the annotated data into several standard formats (e.g., YOLO, TensorFlow, and Common Objects in Context [COCO]), ensuring seamless compatibility with machine learning frameworks. Furthermore, the data augmentation features of Roboflow, including rotation, scaling, cropping, and flipping, contribute to the dataset diversity and improve model robustness. This approach produces more varied samples, enhancing the learning capacity of the model and lowering the overfitting possibility, making it especially beneficial when working with tiny or unbalanced datasets.

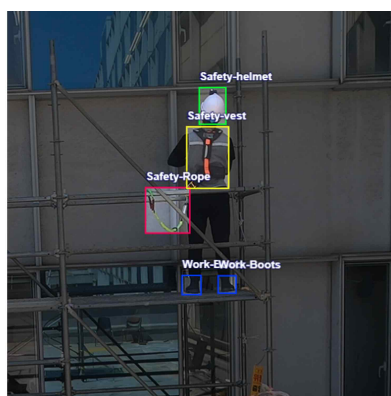
This study employs object detection annotation to label the PPE worn by construction workers, including helmets, safety vests, fall-arrest ropes, safety boots, and gloves. Using the automation tools from Roboflow, each PPE item was marked with a bounding box, and the corresponding labels were assigned to ensure accurate identification and precise localization of each item in the images. Figure 6 presents the annotation results, where bounding boxes highlight the PPE items worn by the workers. Each box contains category labels, ensuring dataset consistency.

3.2.3 Data augmentation

Data augmentation can significantly increase the size, diversity, and complexity of the dataset without additional data collection by applying a series of transformations to images, effectively enhancing the learning capability of the model. Some transformations, such as

rotation, scaling, translation, flipping, cropping, and color adjustments, are applied to the original images, simulating myriad real-world scenarios, viewing angles, and environmental factors. This process generates a diverse set of image samples, increasing the data diversity and exposing the model to a broader range of potential visual variations during training. Data augmentation alleviates overfitting problems caused by insufficient data and prevents the model from over-relying on local details in the training data by artificially synthesizing new training samples. This method encourages the model to learn more generalizable features, improving its ability to generalize to unseen data.

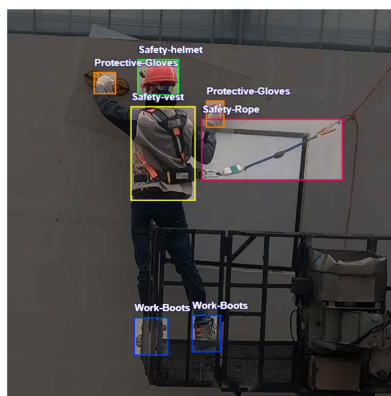
Data augmentation is crucial for small datasets, enhancing the adaptability and resilience of the model to environmental changes. For instance, the accuracy of visual models can be affected by several variables, such as object occlusion, viewpoint alterations, and lighting variations. The model can be trained under various lighting conditions using data augmentation techniques and can learn to handle challenges arising from perspective changes or partial occlusions, improving its performance in real-world applications. This approach is especially crucial in construction site safety monitoring and PPE detection tasks, where the diversity of environmental factors requires the model to be highly adaptable. Data augmentation is an effective solution to address these challenges.



(a) Scaffolding work



(b) Ladder tasks



(c) Aerial platform operations

Figure 6. Example of construction workers' PPE annotation: (a) Scaffolding work; (b) Ladder tasks; (c) Aerial platform operations.

Considering the need for diverse images in PPE detection tasks, particularly across diverse work environments, data augmentation is a crucial strategy for enhancing the robustness of the model. For each training sample, three augmented versions were generated, ensuring that the model encountered a wide range of images and improving training effectiveness. The data augmentation techniques applied in this study are discussed below.

Horizontal and vertical flip: In the context of construction work environments, varying shooting angles can result in target objects being presented from various perspectives. Thus, horizontal and vertical flip operations were applied to enhance the ability of the model to adapt to diverse viewpoints. Horizontal flipping generates left-right mirrored images that are useful for simulating the appearance of PPE from various directions, such as helmets or safety vests viewed from the side. Vertical flipping increases data diversity, particularly for unique work environments (e.g., aerial tasks or shots from above or below), strengthening the ability to detect workers and equipment. Overall, these flipping operations offer a rich set of input scenarios, considerably improving the multiview detection capabilities of the model.

90° rotation: Images were rotated by 90° in clock- and counterclockwise directions to simulate various angular perspectives in the construction environment. This operation is beneficial for scenes involving elevated platforms, scaffolding, and ladders, where the camera angle or the worker's perspective may vary. The 90° rotation ensures that the model retains efficient recognition capabilities when

facing objects from different angles, enhancing its adaptability and robustness in multiangle scenarios.

Random rotation: A random rotation operation, within a range of -15° to $+15^{\circ}$, introduces asymmetric angular changes to the dataset. This adjustment simulates real-world scenarios where workers wear PPE during dynamic tasks or assume various postures. For example, safety vests may slightly tilt during movement, or a helmet may shift when a worker bends. The model can learn a broader range of features owing to random rotation, enhancing its performance in challenging real-world scenarios and making it more appropriate for nonstandard work conditions.

Hue adjustment: Random hue adjustments within a range of -15° to $+15^{\circ}$ were applied to account for varying lighting conditions that affect image colors. This operation alters the overall image tone, simulating common lighting conditions on construction sites, where a combination of natural and artificial lighting can cause PPE colors to appear differently. Hue adjustment ensures that the model maintains stable detection performance across diverse lighting conditions, improving its ability to recognize PPE objects in diverse environments.

Saturation adjustment: The image saturation was randomly adjusted between -25% and $+25\%$ to simulate the effects of environmental lighting and imaging devices on the vibrancy of colors. For instance, daylight might enhance saturation, whereas indoor or shaded lighting could reduce it. By adjusting saturation, the model learns to recognize features under different color conditions, improving its ability to detect brightly colored PPE items, such as safety vests or helmets, and

enhancing its robustness across scenarios.

Brightness adjustment: The image brightness was randomly adjusted within a range of -20% to $+20\%$ to simulate changes in lighting environments on construction sites. Bright sunlight could overexpose an image, whereas cloudy weather or nighttime conditions might cause images to appear dimmer. Brightness adjustment improves the ability of the model to adapt to extreme lighting variations, ensuring reliable PPE detection in a wide range of conditions, from low light to bright environments.

Noise addition: A small amount of noise (up to 0.1%) was randomly added to the images to simulate potential real-world image quality problems, such as camera sensor defects, low resolution, or signal interference during image transmission. Adding noise helps the model learn to detect targets in lower-quality images, improving its robustness. This noise addition is helpful in construction site monitoring scenarios, ensuring that the model remains reliable even when faced with blurry or lower-quality images. Table 5 summarizes the operations and parameter settings for PPE in the data preprocessing and augmentation processes.

Table 5. Data preprocessing and augmentation of the personal protective equipment dataset.

Preprocessing	
Resize	640×640 pixels
Auto-Adjust Contrast	Using Contrast Stretching
Augmentation	
Flip	Horizontal, Vertical
90° Rotate	Clockwise, Counter-Clockwise
Rotation	-15° to $+15^\circ$
Hue	-15° to $+15^\circ$
Saturation	-25% to $+25\%$
Brightness	-20% to $+20\%$
Noise	Up to 0.1% of pixels

These data augmentation techniques increased the original dataset from 2,000 to 5,200 images, substantially expanding the sample diversity. The validation and testing sets constitute a smaller proportion of the total dataset; however, the number of images in each set remains sufficient to support model tuning and evaluation. Hence, data augmentation was not applied to the validation and testing sets to avoid introducing potential biases from the augmentation operations, ensuring the objectivity and consistency of the evaluation results. Table 6 summarizes the composition of the augmented and original datasets.

Table 6. Composition of the augmented and original personal protective equipment dataset.

Dataset	Original Images	Augmented Images
Training	1600	4800
Validation	200	200
Test	200	200
Total	2000	5200

3.3 Construction worker action dataset

3.3.1 Dataset organization and preprocessing

The dataset preparation process for construction worker actions involved cleaning and standardization steps to ensure data consistency and high quality. Duplicate or highly similar samples were identified and removed via rigorous screening, ensuring that each dataset image displayed unique features and research value. Special attention was focused on slight variations arising from shooting angles, lighting conditions, resolution adjustments, or other factors. Strict deletion criteria were applied to remove images affected by these factors, guaranteeing the dataset quality and diversity. The removal of duplicate images reduced data redundancy and mitigated potential overfitting associated with repetitive data, improving the generalization and training efficiency. After cleaning, all images were standardized to a resolution of 1080×1080 pixels to ensure consistency across the dataset. This standardization eliminated potential training errors caused by inconsistent image sizes and enhanced the ability of the model to capture critical image details, particularly when processing critical features. All images were converted to JPG format to improve data compatibility and storage efficiency.

The dataset construction process focused on two crucial worker tasks: high-altitude operations and lifting/moving tasks. These actions are among the most common and hazardous on construction sites. High-altitude operations are hazardous due to the risk of falls, improper posture, and external environmental factors. Lifting and

moving tasks are physically demanding due to the strain of carrying heavy objects, platform instability, and increased operational complexity. Both actions often involve frequent collaboration of critical joints, representing core construction tasks and concentrated safety hazards. The dataset comprises the following task images:

- Overhead tasks (600 images) focused on displaying typical behaviors of workers operating on ladders and
- Lifting and moving tasks (600 images) depicting worker movements on elevated platforms.

This study uses an 80:10:10 split for training, validation, and testing sets to ensure effective training and comprehensive evaluation.

3.3.2 Data annotation

Keypoint annotation is a widely employed image annotation method in deep learning. Its core objective is to accurately identify specific keypoints in images, such as the joints of the human body or important object structures. This technology provides valuable insight into the posture, shape, and motion patterns of the target by precisely locating the keypoints, and it is commonly applied in human pose estimation, motion detection, and gesture recognition tasks.

This stage focuses on the safety monitoring and dynamic behavior analysis of workers in construction site scenarios, aiming to explore their movement characteristics and potential safety risks via keypoint annotation technology. This research aims to detect workers' uncomfortable postures, awkward movements, and potentially hazardous stances, incorporating joint angle analysis to achieve precise motion

recognition and risk assessment. For instance, the model can identify uncomfortable postures leading to musculoskeletal strain or workplace injuries, such as excessive bending or kneeling, by analyzing the relative positions of the elbow and knee. A skeletal structure with 16 keypoints was designed, including the head, neck, shoulders, elbows, wrists, hands, hips, knees, and feet (Figure 7).

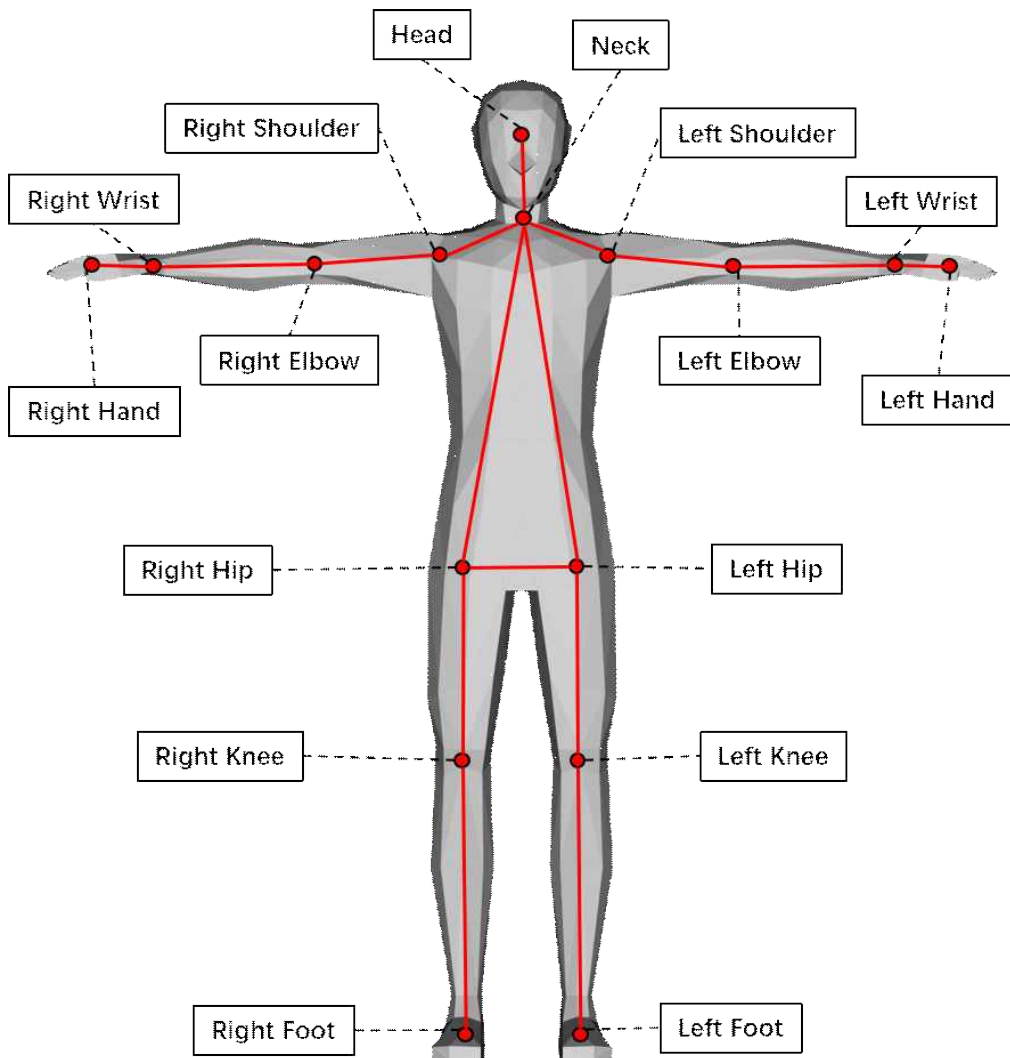


Figure 7. Skeletal structure design for keypoint annotation of workers' postures.

This structural design aligns with basic human physiological characteristics and considers the practical work needs of workers on construction sites. For instance, the keypoint layout of the elbows and knees aids in analyzing worker postures during tasks (e.g., bending, carrying, ladder climbing, or other common work scenarios). The annotation of the shoulders and hips can be employed to assess the worker's standing stability or load-bearing status, whereas the foot keypoints support the analysis of the standing posture and gait. The annotation work was conducted using the keypoint annotation tool provided by Roboflow. The annotation process followed several steps (Figure 8 presents the completed action dataset sample):

1. Target box selection and object identification: The first step involves selecting a target box around the worker in the image, clearly defining the scope of the annotated object. The box selection process ensures complete coverage of the worker's joint positions while excluding background interference and other nontarget objects.

2. Keypoint location and adjustment: Within the selected box, each keypoint is annotated individually, ensuring precise placement on the predefined skeletal structure nodes.

3. Handling occlusion and missing keypoints: Keypoints not visible due to the worker's posture or a limited viewing angle are marked as "occluded." If a keypoint is outside the image boundary or does not exist in the specific pose (e.g., a limb at a particular position is not visible in the image), that keypoint is omitted. This approach effectively prevents the negative affect of incorrect annotations on model training.

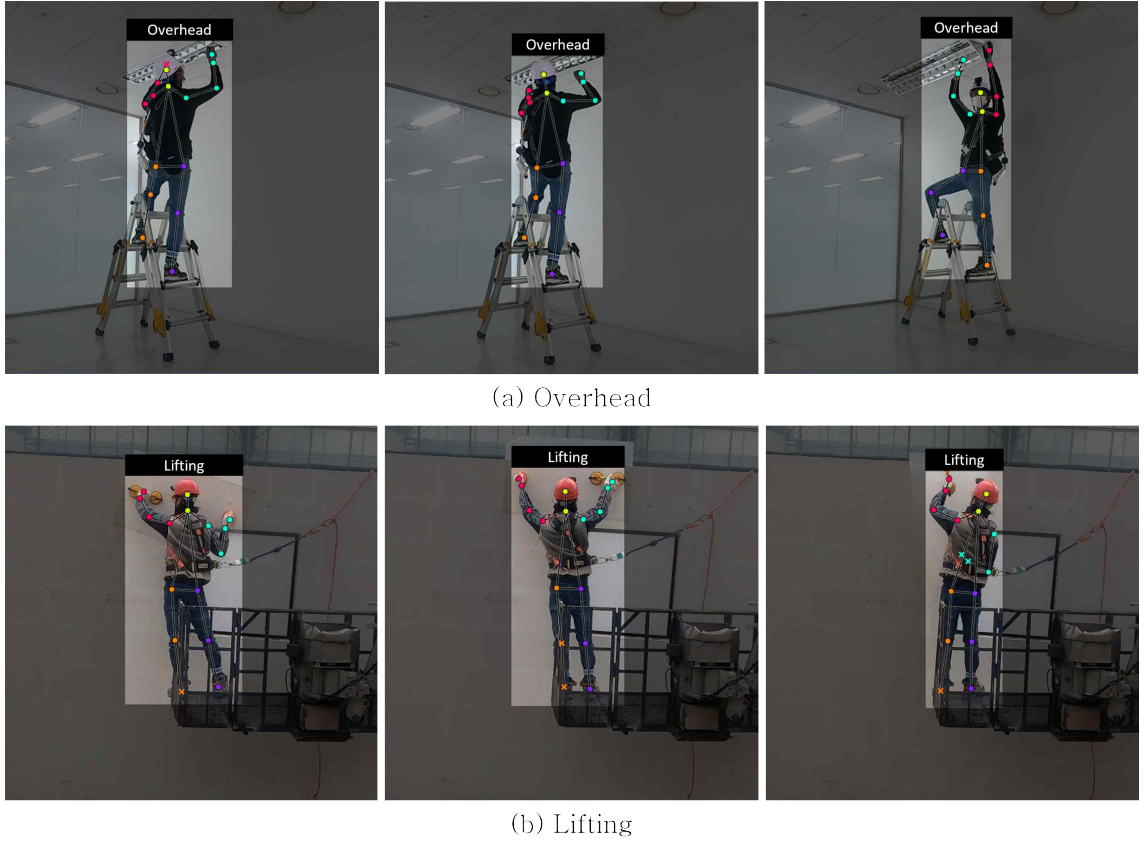


Figure 8. Example of construction worker action annotation: (a) Overhead; (b) Lifting.

3.3.3 Data augmentation

In the data preprocessing stage, all original images were resized to a uniform resolution of 640×640 pixels to ensure consistent dimensions across the dataset. This standardization enhanced the efficiency of model training and minimized potential errors arising from inconsistent input sizes.

An automatic contrast stretching technique was applied during preprocessing to optimize image quality. This technique redistributes the range of brightness values in the image, improving detail

representation and enhancing clarity, making the images easier to observe. This adjustment optimized the overall visual quality of the images, ensuring accurate detection and analysis of keypoints and texture details in the following steps.

Additionally, data augmentation strategies were designed and implemented to improve the dataset diversity and enhance the adaptability of the model to complex scenes. Special attention was focused on the unique characteristics of human keypoint recognition and the practical requirements of construction site environments. In this process, image flipping was avoided. Although horizontal or vertical flipping is commonly employed in traditional computer vision tasks to improve the invariance of the model to direction, flipping could lead to confusion in keypoint positions for human keypoint recognition tasks. Flipping could cause discrepancies between annotations and actual body poses, particularly if the positions of the left and right shoulders or arms were swapped. Further, flipping operations might introduce unnatural or unrealistic poses, reducing the compatibility of the augmented data with real-world scenarios. This study does not apply flipping augmentation to ensure data consistency and authenticity.

Finally, the hue, saturation, brightness, and exposure values of the images were adjusted to improve the simulation of the varying lighting conditions in construction sites. This approach generates augmented versions with diverse lighting scenarios, helping the model maintain a practical posture analysis and action detection capability across a range of environmental conditions. The specific adjustment parameters

are -15° to $+15^\circ$ for hue, -25% to $+25\%$ for saturation, -15% to $+15\%$ for brightness, and -10% to $+10\%$ for exposure.

These augmentation operations simulate ideal conditions with adequate lighting and generate images under low light or overexposed conditions, reflecting diverse shooting environments typical of real construction site scenarios. The adjusted images preserve the critical features of the original data while enhancing the robustness of the model to complex lighting variations. For instance, images created in low-light conditions help train the model to recognize worker actions in dimly lit environments, whereas high-exposure images improve the performance of the model under direct, intense lighting.

Thus, this study expands the original dataset via a diversified data augmentation strategy. For each original image, five augmented versions were generated using the mentioned augmentation techniques, resulting in 5,040 images. This substantial expansion extensively enhanced the diversity of the dataset while mitigating the risk of overfitting associated with limited data. Table 7 summarizes the specific operations and parameter settings in the data preprocessing and augmentation steps of this study.

Table 7. Data preprocessing and augmentation of action dataset.

Preprocessing	
Resize	640×640 pixels
Auto-Adjust Contrast	Contrast Extending
Augmentation	
Hue	-15° to $+15^\circ$
Saturation	-25% to $+25\%$
Brightness	-15% to $+15\%$
Exposure	-10° to $+10^\circ$

This study applied the conventional 80:10:10 split ratio to separate the dataset into training, validation, and testing sets to train and validate the deep learning model efficiently. Initially, 960 images were included in the training set; however, 2,880 images were added. This measure significantly increased the training data diversity, helping the model exhibit better robustness when facing variations in angles, lighting, occlusions, and poses, improving its generalizability and enhancing its ability to recognize dangerous, uncomfortable, and awkward postures in construction site scenarios. No data augmentation was applied in the partitioning of the validation and testing sets, ensuring that their diversity remains intact and unaffected by augmented data and guaranteeing the objectivity and accuracy of the model evaluation. The samples in the validation and testing sets were not augmented; hence, they effectively test model performance on unseen data, avoiding biases caused by augmented data. Table 8 lists the structure of the split dataset and its corresponding proportions.

Table 8. Composition of the augmented and original of action dataset.

Dataset	Original Images	Augmented Images	Augmented Ratio
Training	960	2880	92%
Validation	120	120	4%
Test	120	120	4%
Total	1200	3120	100%

Chapter 4. Development of the proposed framework

4.1 Method

This study proposes an approach to enhance safety management efficiency on construction sites focusing on PPE detection and the assessment of worker safety during operations. A system was introduced to achieve this (Figure 9), comprising two primary modules: PPE detection and action safety recognition modules. The integrated operation of these two modules provides a comprehensive evaluation of construction worker safety during their tasks.

The PPE detection module was designed to identify the PPE worn by workers accurately, ensuring compliance with construction safety regulations and promptly detecting any instances of noncompliance, such as the absence of helmets or gloves. The action safety recognition module extracts and analyzes workers' skeletal and joint information using deep learning techniques to assess the safety of their actions in work scenarios. This module identifies potentially hazardous postures and evaluates the associated risks by integrating keypoint detection technology with posture evaluation algorithms.

The proposed framework comprises the following six stages to achieve these objectives:

1. Construction site dataset acquisition: Data were collected from workers in construction environments to ensure the dataset is representative and diverse. This data includes information from worker groups and work conditions.

2. Data collection and cleaning: Raw data was gathered and

processed to remove inconsistencies, ensuring high-quality inputs for further analysis.

3.Data annotation: Professional annotation tools were employed to label the data precisely, including PPE types (e.g., helmets and safety shoes) and human keypoints (e.g., joint positions and skeletal structure).

4.Image preprocessing: The annotated images underwent preprocessing, such as image augmentation, resolution adjustment, and noise reduction, to enhance data quality and improve model robustness.

5.Object detection and action recognition model development: A PPE detection model was constructed and optimized, and the existing action recognition models were refined to address the unique challenges of construction work scenarios. Model enhancements improve detection accuracy and robustness.

6.Analysis and evaluation: Statistical analyses were performed on the detection results to assess the recognition accuracy of various PPE types. Keypoint detection results were employed to evaluate worker postures, identifying potentially hazardous actions based on predefined risk standards.

Worker data from real high-altitude construction work, including various job types and complex real-world backgrounds.



- **Data collection and cleaning**

Manually clean the dataset to eliminate similar and problematic data to ensure the high quality of the dataset.

- **Data annotation (PPE)**

Labels include PPE types (e.g., helmets, safety shoes).



- **Data annotation (key points)**

Human key points (e.g., joint positions, skeletal structure) are annotated.



- Image augmentation (PPE)

Flip, Rotation, Color Adjustment (Hue, Saturation, Brightness), Noise Addition

- **Image augmentation (key points)**

Hue, Saturation,
Brightness, Exposure

- **Object detection model**

A PPE detection model is constructed and optimized.

[illegible]

- **Action recognition model**

Action recognition models are improved for better accuracy and robustness in construction.

- **Analysis and evaluation**

- A. Statistical analyses assess the recognition accuracy of different PPE types.
- B. Analyze the accuracy and robustness of the key point detection model.
- C. Analyze the dangerousness of construction workers' actions.

Class ¹	Images ²	Instances ²	P ³	R ³	mAP50 ⁴	mAP50-95 ⁴
all ⁵	200 ⁶	1160 ⁶	0.980 ⁷	0.970 ⁷	0.980 ⁷	0.772 ⁷
Protective Gloves ⁸	200 ⁶	302 ⁶	0.922 ⁷	0.858 ⁷	0.925 ⁷	0.564 ⁷
Safety-Rope ⁹	200 ⁶	139 ⁶	1.000 ⁷	1.000 ⁷	0.995 ⁷	0.889 ⁷
Safety-helmet ¹⁰	200 ⁶	200 ⁶	1.000 ⁷	1.000 ⁷	0.995 ⁷	0.821 ⁷
Safety-vest ¹¹	200 ⁶	138 ⁶	0.986 ⁷	1.000 ⁷	0.989 ⁷	0.869 ⁷
Work-Boots ¹²	200 ⁶	381 ⁶	0.992 ⁷	0.992 ⁷	0.995 ⁷	0.728 ⁷

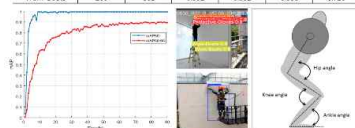


Figure 9. Research flow chart.

4.2 Object detection model

Jocher introduced YOLOv5 in 2020, a single-stage object detection model with four versions (s, m, l, and x) based on network depth and feature map coefficients. Among these, YOLOv5s displays the fastest computational speed, making it suitable for embedded devices with constraints on storage and computational resources. In this study, the YOLOv5s model was modified to enhance PPE detection accuracy to create the PPE-YOLO model.

Figure 10 presents the input layer, backbone network, neck network, and three detecting heads comprising the four parts of the YOLOv5 structure. Multiscale feature maps were extracted by the backbone network and integrated by the neck network, and the detection heads conducted object location and categorization. The YOLOv5 model was built on the advancements of YOLO versions 1 through 4, transitioning from the Darknet framework to PyTorch and improving flexibility and compatibility. Similar to other single-stage object detection models (e.g., SSD, YOLO, and RetinaNet), YOLOv5 formulates object detection as a regression problem, simultaneously predicting object locations and class probabilities by computing bounding-box coordinates. Continuous improvements have enabled YOLOv5 to achieve significant performance gains on benchmark datasets, such as Pascal Visual Object Class (VOC) and COCO.

The backbone network of YOLOv5 is based on CSPDarknet53, incorporating CSPNet into the traditional Darknet architecture. The CSPNet enhances the learning efficiency of the model by reducing

redundant gradient information during network optimization. This approach optimizes feature extraction, fuses gradient changes for improved accuracy, and reduces network complexity and model parameters. The backbone network integrates an SPP module, expanding the receptive field with pooling windows of varying sizes to extract critical features.

In the neck network, YOLOv5 employs the PANet, an improved version of the FPN. The PANet facilitates multiscale feature fusion by combining feature layers via top-down and bottom-up pathways, enhancing the propagation of low-level features. This architecture improves object localization, particularly for small-object detection tasks.

The detection head in YOLOv5 follows a structure similar to YOLOv3 and YOLOv4, using three resolution feature maps for multiscale object detection. This design efficiently identifies objects of various sizes, ranging from small to large. The image first undergoes feature extraction using CSPDarknet53, followed by multiscale feature fusion via PANet, and finally, the YOLO layer generates the detection results (Jocher et al., 2021)¹⁴⁵.

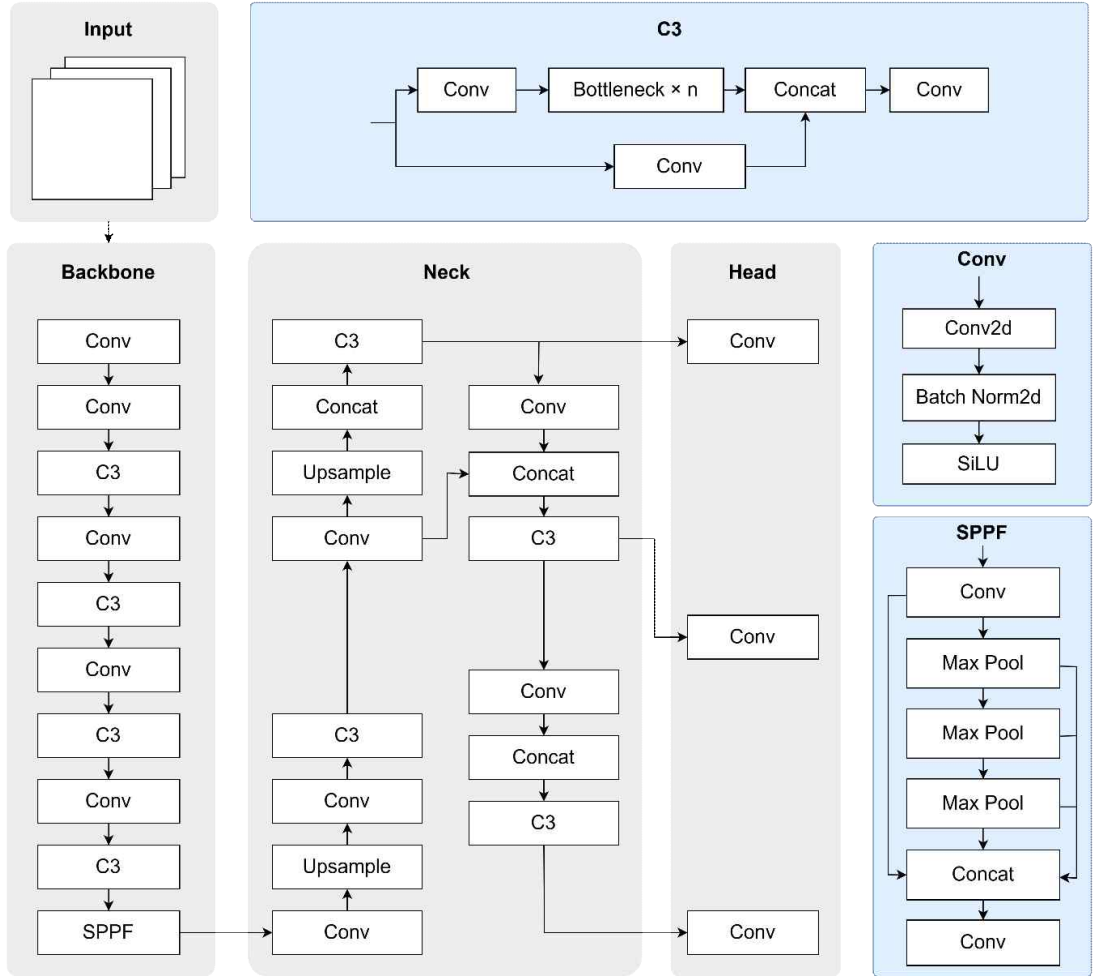


Figure 10. Network architecture of YOLOv5.

The dynamic and diverse nature of the environment makes it challenging to recognize PPE in construction site settings. These difficulties call for creating more reliable and effective detection techniques. Compared to conventional object detection tasks, PPE detection must address three primary difficulties:

Small objects: On construction sites, PPE items (e.g., helmets, goggles, and gloves) are typically small and may appear with low

resolution in images. These targets are often distant from the camera, limiting the captured feature information. The low resolution and reduced signal-to-noise ratio of small objects make them prone to being obscured by complex backgrounds or other irrelevant targets. The traditional CNN is susceptible to missing these details when extracting low-level features, leading to false negatives (FNs) or false positives (FPs).

Complex background: The construction site environment is filled with potential sources of interference, including scaffolding, machinery, and construction materials. These distractors often share visual features or colors similar to PPE items, complicating object detection. The complexity of the background increases the likelihood of false detections and makes object localization and recognition more challenging. In dynamic scenes, the interaction between the construction worker's movements and the background may cause significant occlusion, complicating the separation of the target from the surrounding environment.

Worker's dynamic deformation: Construction workers frequently perform myriad complex actions, such as climbing ladders, bending, or waving, resulting in significant changes to the spatial features and positional information of the targets. Moreover, PPE items may deform, shift, or become occluded due to these movements, increasing the difficulty of detection. For traditional detection methods, such deformations can cause critical feature loss or incorrect identification, adversely affecting the robustness and accuracy of the model.

To address these challenges, we introduced the C3 convolutional

block attention module (C3CBAM), C3 deformable convolutional network (C3-DCN), and bidirectional FPN (BiFPN) modules. These enhancements aim to improve the performance and robustness of PPE detection by more effectively handling small objects, complex backgrounds, and dynamic worker deformations.

4.2.1 Model optimization

A. C3CBAM module

The CBAM (Xin & Song, 2024)¹⁴⁶⁾, a sophisticated attention mechanism, improves the feature extraction capabilities of deep convolutional networks. This mechanism comprises two crucial components: the channel attention module (CAM), refining feature representation by emphasizing the most informative channels, and the spatial attention module (SAM), improving localization by focusing on critical spatial regions. The CAM dynamically adjusts the response weights of channels to emphasize significant feature channels, whereas the SAM adaptively adjusts the response weights across spatial positions to highlight critical spatial regions.

The CAM aims to enhance the feature representation of each channel. Global max-pooling and global average pooling are applied to each channel of the input feature map, computing its maximum and average feature values to achieve this enhancement. These operations generate two vectors that capture the global maximum and average features across all channels.

A shared FC layer processes these feature vectors and learns attention weights for every channel. Thus, the network may

dynamically determine the channels most pertinent to the current task and rank them. The global maximum and average feature vectors are from the final attention weight vector.

A sigmoid activation function ensures the weights remain within the range of 0 to 1. The computed channel attention weights are applied elementwise to the original feature map, generating a channel-weighted feature map highlighting the most significant channels while diminishing the influence of less relevant ones. Figure 11 illustrates the structure of the CAM.

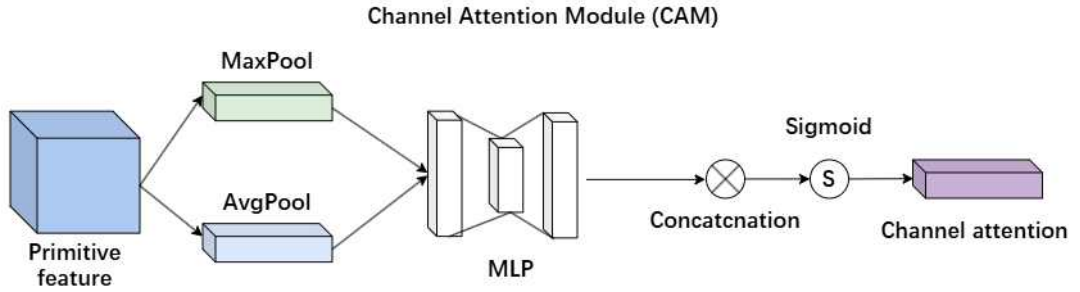


Figure 11. Channel attention module.

The SAM is designed to emphasize the significance of spatial locations in an image, which is accomplished using max-pooling and average pooling along the channel dimension of the input feature map, extracting features at various contextual scales. A multiscale contextual feature map is created by concatenating pooled features along the channel dimension.

The concatenated feature map passes through a convolutional layer to generate spatial attention weights, constrained by the sigmoid activation to the range from 0 to 1. The feature intensities at each

spatial position are adaptively adjusted by multiplying the resultant weights elementwise by the original feature map. This process highlights important regions and suppresses less relevant areas. Figure 12 presents the SAM.

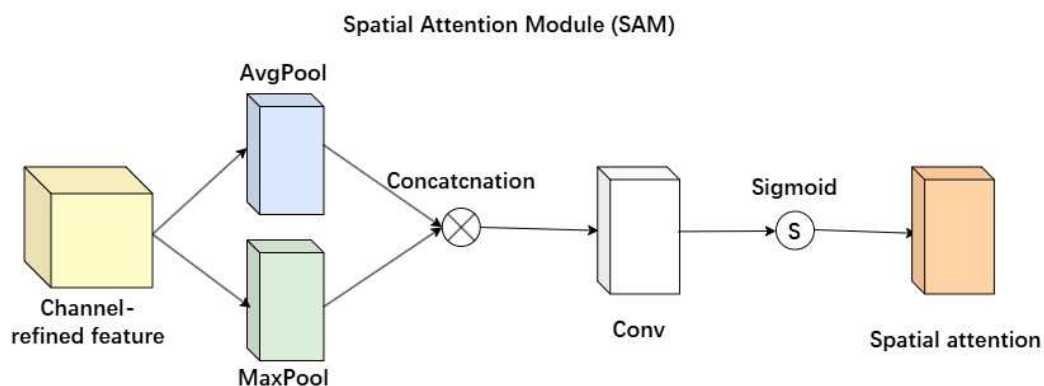


Figure 12. Spatial attention module.

The final attention-enhanced features are produced using CBAM via the elementwise multiplication of the SAM and CAM outputs. The later network layers receive these improved features, filtering noise and extraneous details while maintaining essential information. Figure 13 depicts the CBAM.

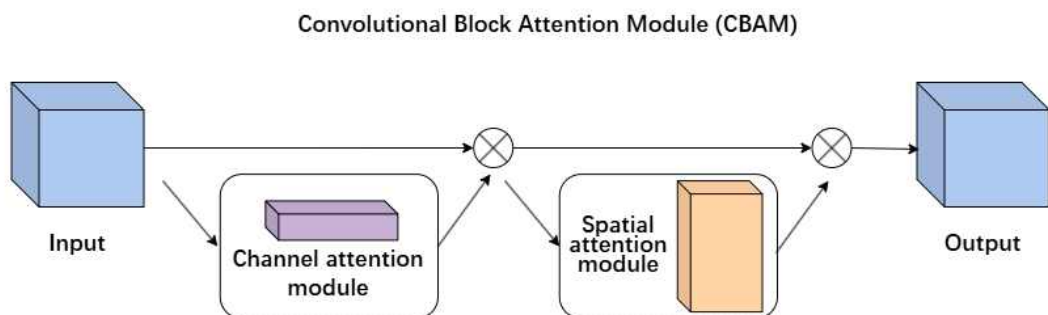


Figure 13. Convolutional block attention module.

The C3 module comprises three convolutional layers, bottleneck modules, and a concatenation module. Integrating the CBAM module into C3 forms the C3CBAM module, enhancing feature extraction precision while maintaining efficiency and rendering the model effective for small objects, complex backgrounds, and dynamic targets. Figure 14 illustrates the overall structures of the CBAM and C3CBAM modules.

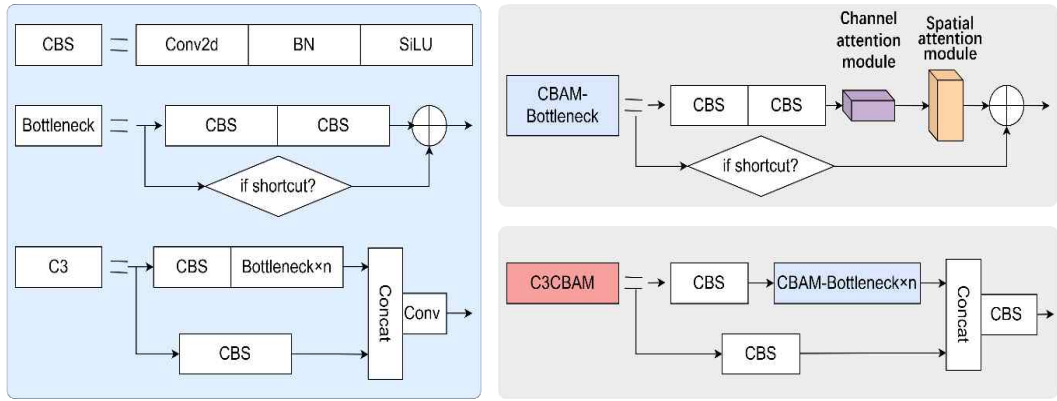


Figure 14. CBAM and C3CBAM modules.

B. C3_DCN module

The DCN is an enhancement of the traditional convolutional operations designed to improve feature extraction efficiency by introducing dynamic offsets to each sampling point in the convolutional kernel (Peng et al., 2023)¹⁴⁷⁾. This mechanism enables the kernel to adjust its sampling positions adaptively based on the target shape variations, capturing features more effectively. The DCN incorporates an offset learning branch, predicting the offset values of the sampling points via a parallel convolutional branch. This approach overcomes the limitations of traditional convolution kernels that sample from fixed

rectangular regions by allowing the sampling positions to align more closely with the actual target shape. As Figure 15 illustrates, blue points represent the original sampling positions of a 3×3 convolutional kernel, whereas green points indicate the adjusted positions after applying the learned offsets.

This mechanism significantly enhances the robustness of the model in complex scenes and when handling target deformations, allowing it to focus more precisely on the target area despite challenging backgrounds. Unlike randomly generated offsets, the offsets in deformable convolution are learned, ensuring that they do not introduce irrelevant noise. Instead, they dynamically adjust to better align with the target shape, improving the ability to extract meaningful features and enhance detection performance in intricate environments.

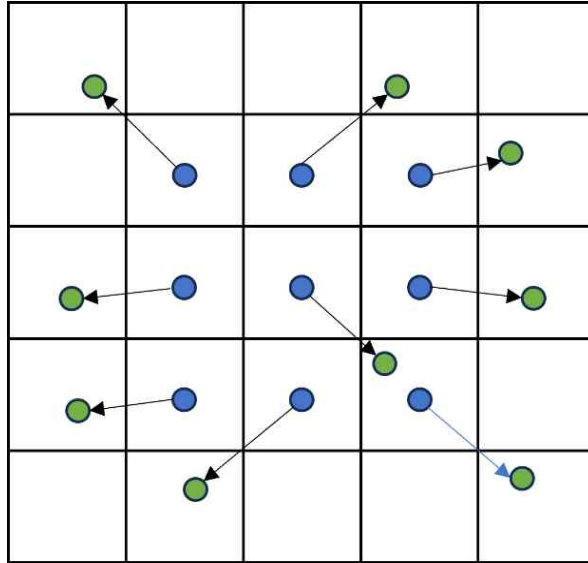


Figure 15. Sampling positions.

Figure 16 presents the DCN. Unlike standard convolution, deformable convolution introduces an additional convolutional layer (the green convolutional layer in Figure 16) explicitly designed to predict offsets. This layer is responsible for learning the 2D offsets corresponding to each sampling point in the input feature map. These offsets are typically noninteger values; therefore, bilinear interpolation is employed during computation to retrieve the appropriate values from the input feature map, ensuring an accurate convolutional operation. This mechanism enhances the adaptability of the model to the target deformations; however, it also introduces additional computational overhead, affecting the inference speed.

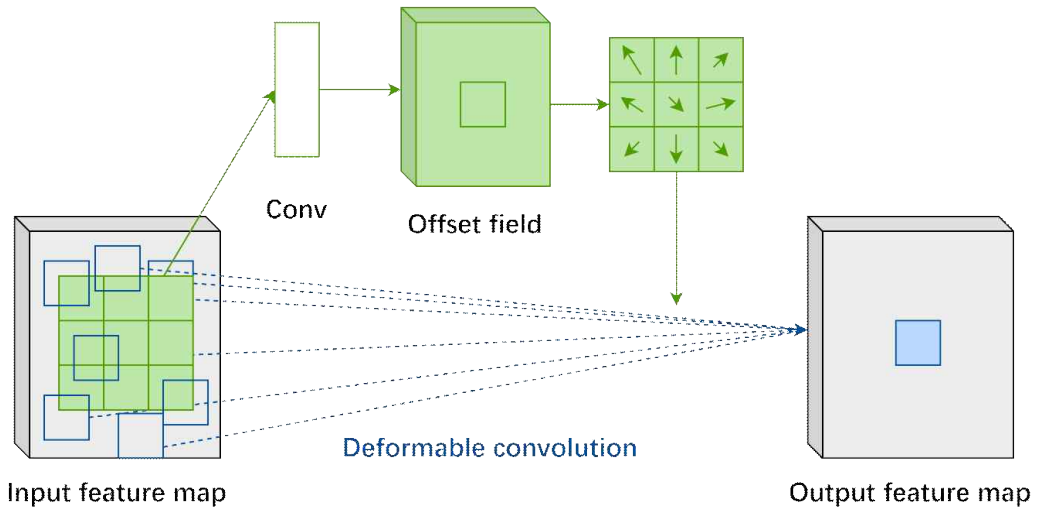


Figure 16. Deformable convolutional network module.

The C3 module, primarily responsible for extracting large target features in the backbone network of the object detection model, is optimized to enhance model accuracy while preserving real-time performance. The original standard convolutional layer (Conv) is

replaced with a deformable convolutional layer (the DCN), as illustrated in Figure 17, to construct the new module, the C3-DCN. This modification enables efficient feature extraction, improving the adaptability of the network to deformed targets while balancing detection accuracy and real-time speed.

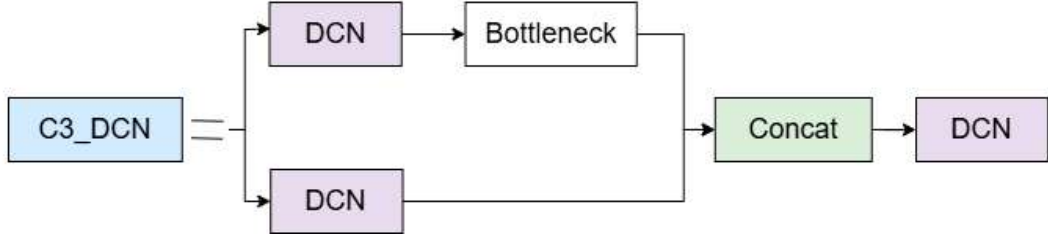


Figure 17. Network architecture of C3_DCN.

C. BiFPN module

The BiFPN is an improved multiscale feature fusion network designed to enhance efficiency and accuracy (Tan, Pang, & Le, 2020)¹⁴⁸. The BiFPN is a core component of the EfficientDet model, built on the EfficientNet backbone. Prior to BiFPN, two widely used feature pyramid architectures were the FPN and the PANet, both aiming to improve multiscale feature representation.

By adding a top-down pathway that makes it easier to fuse features at various levels, the FPN improves multiscale feature representation. The network can take advantage of high-resolution, low-semantic characteristics and low-resolution, high-semantic features owing to this design. However, the FPN employs a unidirectional feature fusion strategy, relying solely on the top-down pathway, limiting the information flow between feature levels.

The PANet introduces an additional bottom-up pathway, reinforcing

the representation of lower-level features to enhance the effectiveness of the FPN. Moreover, the PANet strengthens the interaction between multiscale features via path aggregation, improving the ability to capture fine-grained details and global semantic information.

Although PANet improves the FPN by introducing a bottom-up pathway and enhancing multiscale feature aggregation, it still has limitations. For example, its unidirectional pathways may not fully exploit the potential of feature fusion, causing suboptimal information flow across scales. The BiFPN introduces a more efficient and precise multiscale feature fusion strategy to address these challenges, incorporating the following vital improvements:

- Bidirectional feature fusion enables bidirectional information flow by introducing weighted feature fusion across multiple scales, improving feature interaction efficiency.
- Repeated fusion blocks enhance feature extraction by allowing iterative interactions between high- and low-level features, enabling deeper contextual understanding.
- Weighted feature fusion employs learnable weights to balance feature contributions at different scales adaptively, improving detection accuracy and robustness.

By integrating these enhancements, the BiFPN achieves effective feature fusion across multiple scales while maintaining—or even reducing—the overall computational overhead. Figure 18 presents the structures of the FPN, PANet, and BiFPN. Among these, P3, P4, P5, P6, and P7 represent multiscale feature maps corresponding to feature layers at various depths in the backbone network. Despite their

exceptionally high resolution, P1 and P2 primarily capture low-level features, such as edges and textures. However, this results in increased noise sensitivity and high computational overhead, making it less suitable for deep feature extraction. In contrast, P3 balances spatial details and semantic information, offering optimal performance and efficiency.

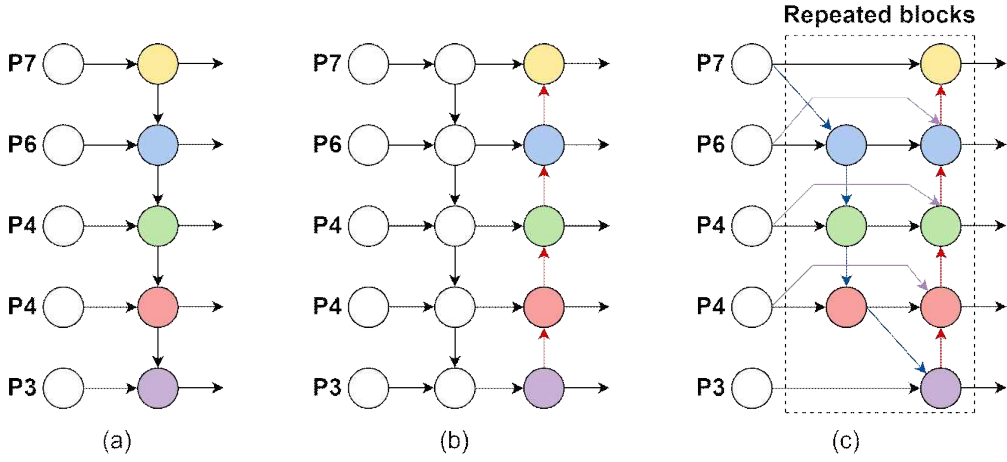


Figure 18. BiFPN module: (a) FPN, (b) PANet, and (c) BiFPN.

In summary, the C3CBAM enhances the detection of small objects through an attention mechanism, the C3-DCN improves robustness against deformed targets via deformable convolutions, and the BiFPN boosts overall detection accuracy by facilitating multiscale feature fusion. The critical improvements introduced by each module and the specific challenges they address are discussed below.

The C3CBAM module enhances the network sensitivity to small objects, significantly improving small-object feature extraction in complex backgrounds. This module addresses the challenge of small-object detection, particularly in low-resolution scenarios and

background interference, where small objects (e.g., helmets and gloves) are prone to misdetection or occlusion.

The C3-DCN integrates module convolutions to enhance target detection and robustness in recognizing workers' dynamic actions. This module mitigates challenges posed by variations in target position and spatial features due to worker movement, enabling the accurate detection of deformed objects (e.g., PPE components during actions, such as climbing a ladder or bending).

The BiFPN module optimizes multiscale feature integration through bidirectional feature fusion and a learnable weighting mechanism, improving its detection accuracy across object scales. This module addresses challenges posed by complex backgrounds and significant scale variations between objects, such as the simultaneous presence of small and large targets. This approach ensures reliable detection performance even in diverse and cluttered environments.

Combining these modules preserves computational economy while significantly improving detection accuracy, enabling this study to capture PPE features across diverse backgrounds and ensure accurate and efficient worker safety detection. With these optimizations, the PPE-YOLO framework employs the advantages of the C3CBAM, C3-DCN, and BiFPN modules to address the challenges associated with PPE detection in construction site environments. Through the effective integration of these three modules, the PPE-YOLO framework demonstrates superior capability in handling small-object detection, target deformation, and complex background interference. The C3CBAM enhances sensitivity to small objects. In addition, C3_DCN

strengthens robustness against deformed targets, and BiFPN significantly improves detection accuracy via multiscale feature fusion.

Table 9 presents the overall PPE-YOLO framework.

Table 9. PPE_YOLO model architecture.

Section	Layer	Operation	Parameters
Backbone	Conv (Layer 1)	Convolution	[64, 6, 2, 2]
	Conv (Layer 2)	Convolution	[128, 3, 2]
	C3CBAM (Layer 3)	C3CBAM	[128]
	Conv (Layer 4)	Convolution	[256, 3, 2]
	C3CBAM (Layer 5)	C3CBAM	[256]
	Conv (Layer 6)	Convolution	[512, 3, 2]
	C3_DCN (Layer 7)	C3_DCN	[512]
	Conv (Layer 8)	Convolution	[1024, 3, 2]
	C3 (Layer 9)	C3	[1024]
	SPPF (Layer 10)	SPPF	[1024, 5]
	Conv (Layer 11)	Convolution	[512, 1, 1]
	nn.Upsample (Layer 12)	Upsampling	[None, 2, "nearest"]
	Concat (Layer 13)	Concatenation	[1]
	C3 (Layer 14)	C3	[512, False]
	Conv (Layer 15)	Convolution	[256, 1, 1]
	nn.Upsample (Layer 16)	Upsampling	[None, 2, "nearest"]
Head	Concat (Layer 17)	Concatenation	[1]
	C3 (Layer 18)	C3	[256, False]
	Conv (Layer 19)	Convolution	[512, 3, 2]
	BiFPN_Add3 (Layer 20)	BiFPN	[256, 256]
	C3 (Layer 21)	C3	[512, False]
	Conv (Layer 22)	Convolution	[512, 3, 2]
	Concat (Layer 23)	Concatenation	[1]
	C3 (Layer 24)	C3	[1024, False]
	Detect (Layer 25)	Detection	[nc, anchors]

4.2.2 Loss function

The construction site environment is highly dynamic, with workers frequently changing postures while operating on ladders or elevated platforms. Their protective equipment, such as helmets, vests, and safety belts, often undergoes significant rotation. Traditional CIoU loss, lacking optimization for angle and distance factors, struggles to align predicted boxes with the ground truth accurately, leading to angular displacement and reduced reliability in PPE detection. Although the CIoU improves shape matching and accounts for aspect ratio differences, it neglects the effect of mismatched angles, which is crucial in detecting complex poses. In scenarios with significant rotation or irregular target shapes, this limitation hinders accurate localization, affects model convergence, and degrades detection performance (Du et al., 2021)¹⁴⁹).

To address this problem, we introduced the skew intersection over union (SIoU) loss function in target detection, replacing the CIoU. In addition to optimizing shape matching, the SIoU introduces an angle loss (angle cost) and a distance loss (distance cost), optimizing the spatial alignment of the predicted and ground-truth boxes. The design of SIoU accelerates the convergence of the distance between the centers of the two boxes and mitigates the negative influence of angle mismatches on the predicted box location. This approach optimizes the position and shape of the predicted box and considerably improves the accuracy of target localization. Furthermore, the SIoU employs an efficient formula design in loss calculation, reducing the inference time

and computational cost.

Incorporating SIoU enhances the capability of the model to capture the angular characteristics of targets, offering excellent detection of slight rotations of equipment, such as helmets or vests, in complex environments. Moreover, the shape optimization mechanism of the SIoU ensures a high degree of alignment between predicted bounding and ground-truth boxes, improving detection accuracy and enhancing model robustness. The computational steps of the SIoU loss function (Yi et al., 2023)¹⁵⁰⁾ are detailed below.

A. Angle Cost (A)

The angle cost quantifies the alignment between the predicted bounding box and the ground truth box, and is defined as follows (refer to Equation 1):

$$A = 1 - 2\sin^2\left(\arcsin\left(\frac{ch}{\sigma}\right) - \frac{\pi}{4}\right) \quad (1)$$

ch represents the height difference between the center points of the ground truth box and the predicted box.

σ denotes the Euclidean distance between the center points of the ground truth box and the predicted box, calculated as (refer to Equation 2):

$$\sigma = \sqrt{(b_{gtcx} - b_{cx})^2 + (b_{gtcy} - b_{cy})^2} \quad (2)$$

B. Distance Cost (Δ)

The distance cost quantifies the normalized Euclidean distance between the center points of the ground-truth and predicted boxes, defined as follows (refer to Equation 3 and 4):

$$\Delta = \sum_{t=x,y} (1 - e^{-\gamma \rho_t}) = 2 - e^{-\gamma \rho_x} - e^{-\gamma \rho_y} \quad (3)$$

Where

$$\begin{aligned} \rho_x &= \frac{b_{gtcx} - b_{cx}}{c_w} \\ \rho_y &= \frac{b_{gtcy} - b_{cy}}{c_h} \end{aligned} \quad (4)$$

b_{gtcx}, b_{gtcy} : Center coordinates of the ground truth box.

b_{cx}, b_{cy} : Center coordinates of the predicted box.

c_w, c_h : Width and height of the smallest enclosing rectangle covering both the ground truth box and the predicted box.

$\gamma = 2 - A$: A weight coefficient determined by the angle cost.

C. Shape Cost (Ω)

The shape cost quantifies the discrepancy in width and height between the ground truth box and the predicted box, defined as (refer to Equation 5 and 6):

$$\Omega = \sum_{t=w,h} (1 - e^{-w_t})^\theta = (1 - e^{-w_w})^\theta + (1 - e^{-w_h})^\theta \quad (5)$$

Where

$$\begin{aligned} w_w &= \frac{|w - w_{gt}|}{\max(w, w_{gt})} \\ w_h &= \frac{|h - h_{gt}|}{\max(h, h_{gt})} \end{aligned} \quad (6)$$

w, h : Width and height of the ground truth box.

w_{gt}, h_{gt} : Width and height of the ground truth box.

θ : A parameter controlling the attention to the shape cost, typically set to $\theta = 4$.

D. IoU Cost

The IoU cost (LIoU) measures the IoU between the predicted and ground-truth bounding boxes, defined as follows (refer to Equation 7 and 8):

$$L_{IoU} = 1 - IoU \quad (7)$$

Where

$$IoU = \frac{|b \cap b_{gt}|}{|b \cup b_{gt}|} \quad (8)$$

E. Final SIoU Loss Function

By integrating all the aforementioned components, the complete formulation of the SIoU loss function is defined as follows (refer to Equation 9):

$$L_{SIoU} = 1 - IoU + \Delta + \frac{\Omega}{2} \quad (9)$$

4.3 Action recognition model

At this stage, construction workers' actions are detected by improving YOLOv8. The YOLOv8 (Jocher et al., 2023)¹⁵¹⁾ model follows the YOLO series' single-stage detection framework, consisting of the backbone, neck, and head networks.

The backbone network primarily performs feature extraction, converting input images into multiscale, multilevel feature maps via a series of convolutional layers, pooling operations, and other transformations. These feature maps offer rich contextual information for the detection heads. The YOLOv8 model adopts the CSPNet as a

critical module in its backbone. The CSPNet enhances feature representation and reduces computational complexity by partially connecting feature maps across stages. This network splits the feature maps into two parts: one directly undergoes feature extraction, and the other passes through cross-stage connections before merging. This design improves feature expressiveness and ensures efficient feature extraction and processing.

The neck network, positioned between the backbone and detection heads, integrates and enhances features from the backbone. Critical components include the following:

- Cross-level feature fusion improves semantic feature expression by fusing features from various levels with contextual information. This approach improves accuracy and robustness, enhancing its ability to distinguish objects in images.
- The PANet aggregates features from multiple resolutions and semantic levels via a bottom-up pathway. The PANet improves feature representation and robustness in complex scenes by strengthening the understanding of contextual data and object relationships.

The head serves as the output layer and is responsible for predicting object classes, positions, and confidence scores. In YOLOv8, the head includes multiple detection heads, each tailored to process feature maps of various scales. Two major approaches are employed:

- Anchor-free detection directly predicts the center point, width, and height of objects, reducing computational complexity and increasing detection speed.
- Anchor-based detection adjusts predefined anchor boxes, making

it more suitable for detecting objects of varying shapes and sizes.

Figure 19 illustrates the overall network architecture.

The YOLOv8-based pose estimation method integrates fast and efficient object detection with direct human keypoint prediction, enabling end-to-end training in a single unified model. Unlike traditional object detection approaches, this method detects objects (i.e., human bodies) and accurately identifies keypoints, such as shoulders, elbows, and knees. A critical advantage of this approach is that it eliminates the post-processing steps commonly required by traditional bottom-up methods, such as aggregating detected keypoints into skeletal structures. Each detected bounding box is associated with a specific human pose, allowing keypoints to be naturally grouped in each detected instance. This streamlined process enhances efficiency while maintaining high accuracy in human pose estimation.

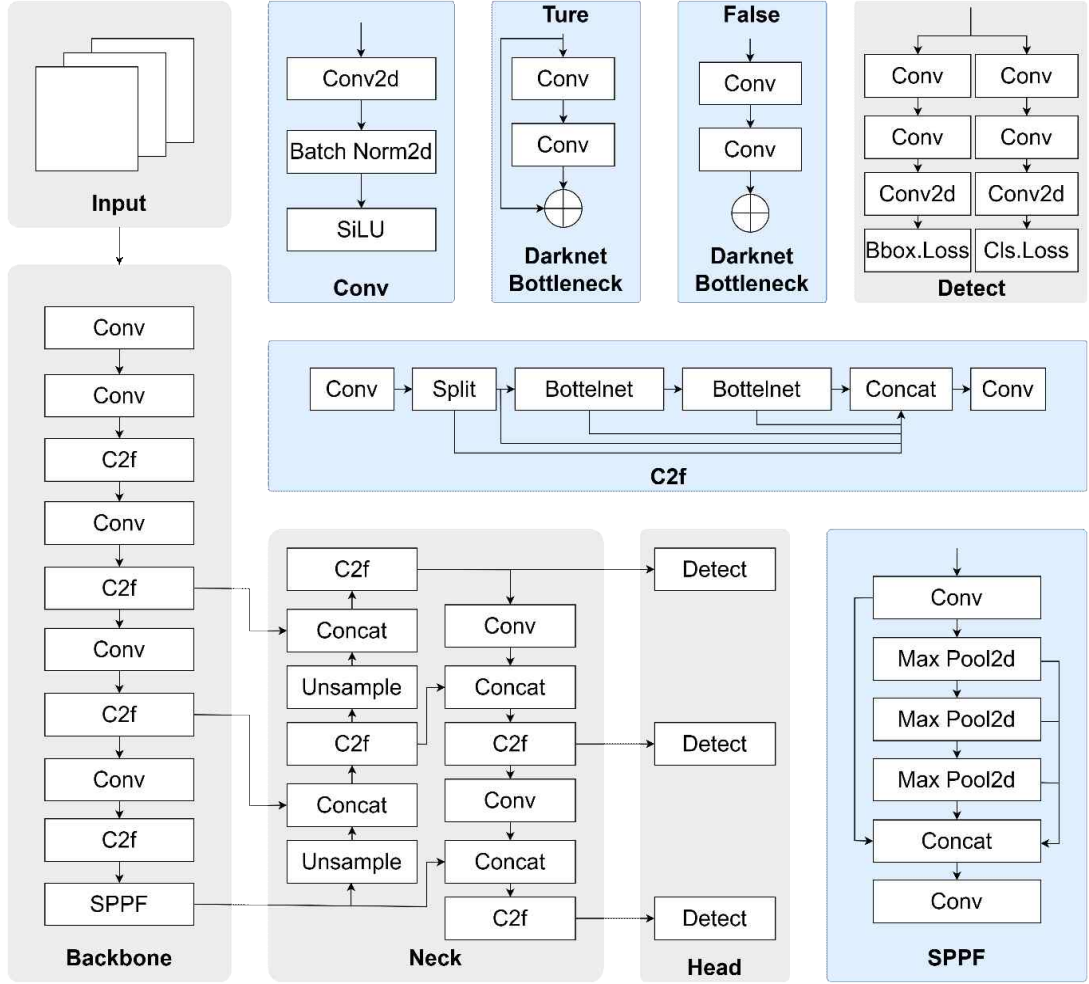


Figure 19. Network architecture of YOLOv8.

4.3.1 Model improvement

This study introduces the following modifications to the original YOLOv8 framework:

A. BottleneckCSP Module

The BottleneckCSP is a bottleneck layer model. This model consists of the following main components (Li et al., 2023)¹⁵²⁾.

The convolutional module, including the Conv2d layer, extracts

features from the input data. In the batch normalization (BatchNorm) layer, after the convolutional process, the feature maps are subjected to batch normalization, normalizing the activations throughout the mini-batch, enhancing training stability, and accelerating convergence. The leaky rectified linear units (ReLU) activation function introduces nonlinearity to the network, enabling it to model more complex relationships in the data while mitigating the problem of a “dying ReLU,” where neurons can become inactive.

The bottleneck layer includes 1×1 convolution. By condensing the number of channels in the feature map, this technique lowers memory consumption and computational complexity, facilitating more effective data processing. This layer also includes 3×3 convolution, performing intricate feature extraction, capturing detailed spatial information, preserving important features, and enabling the model to discern fine-grained differences in the input feature map.

The CSP structure includes the cross-stage connection. This structure enables partial connections across feature maps at various stages, enhances feature fusion, and improves the ability to capture multiscale information for robust performance in complex tasks.

Residual connection, including skip connections, add the input feature map directly to the output feature map after the convolutional operation. This layer facilitates training stability by mitigating the vanishing gradient problem and promoting better gradient flow during backpropagation.

In the convolutional module, the Conv2d and BatchNorm layers work together to extract and normalize features, contributing to enhanced

model stability during training. The LeakyReLU activation function introduces necessary nonlinearity, improving the representation power of the model by allowing it to learn more complex patterns. The bottleneck layer, with its 1×1 convolution, reduces the computational load, whereas the 3×3 convolution captures detailed features, permitting the model to detect subtle variations in keypoints. The CSP structure strengthens the ability to capture multiscale features via cross-stage fusion, boosting the accuracy of keypoint detection. The inclusion of residual connections contributes to model stability by addressing the vanishing gradient, which is common in deep neural networks. These elements enable the BottleneckCSP module to balance computational efficiency and detection accuracy in human keypoint detection tasks. Figure 20 depicts the module architecture.

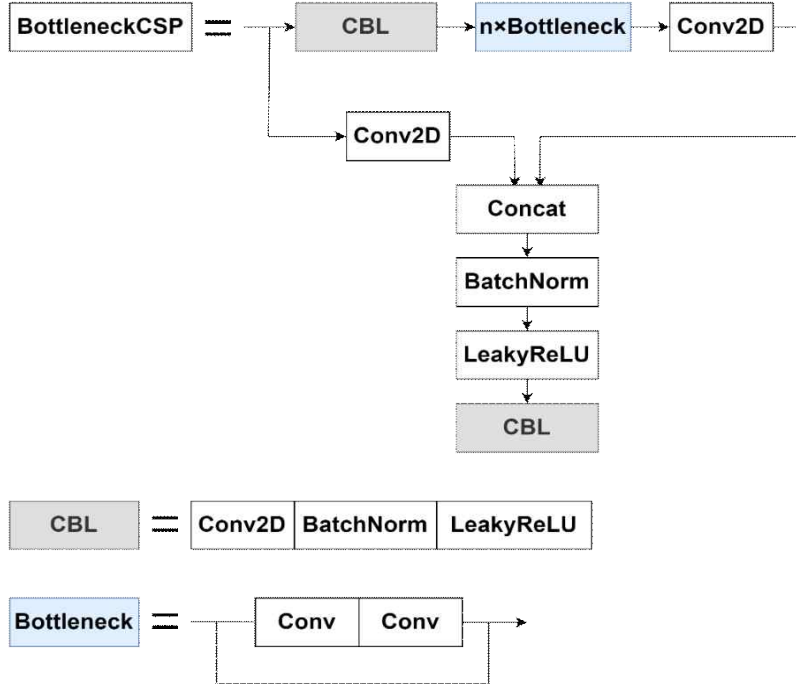


Figure 20. BottleneckCSP module.

B. GhostConv Module

The ghost module (Han et al., 2020)¹⁵³⁾ is designed to reduce computational cost and parameter count while preserving high performance by generating redundant feature maps. The GhostConv module contains two primary components: the primary convolution layer, which generates a limited number of primary feature maps via standard convolutional operations, and the ghost feature map generation layer. Additional redundant feature maps are produced via simple linear operations from these primary feature maps. Figure 21 depicts the GhostConv module structure.

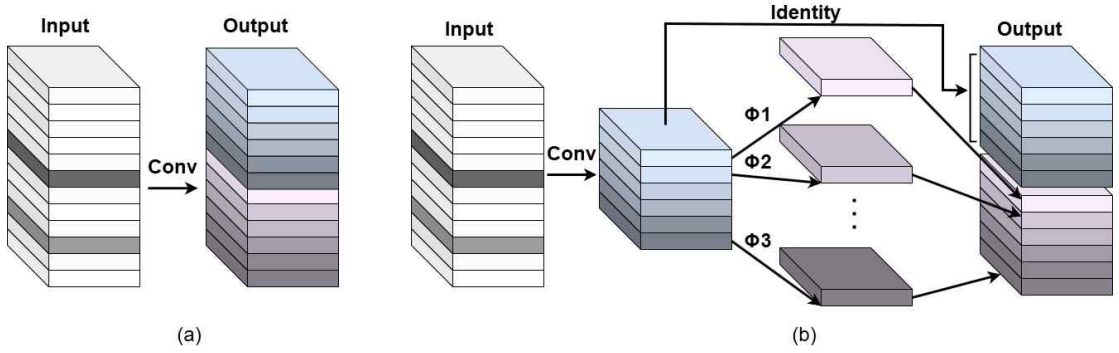


Figure 21. (a)Traditional convolution; (b)GhostConv module (Φ : Cheap operation).

In the conventional convolution that generates the feature map $\gamma \in R^{h' \times w' \times n}$, $X \in R^{c \times h \times w}$ represents the input data, where h and w denote the height and breadth of the feature map, respectively, and c indicates the number of channels in the input feature map. Furthermore, b indicates the bias term, $*$ represents the convolution operation, and $f \in R^{c \times k \times k \times n}$ denotes the convolution kernel. The calculation uses the following Equation 10:

$$\gamma = X^*f + b \quad (10)$$

Ghost convolution comprises two parts: several redundant feature maps generated by lightweight linear transformation layers and a few feature maps produced by traditional convolutional kernels, expressed as follows (refer to Equation 11):

$$\gamma' = X^*f' + b \quad (11)$$

where $\gamma' \in R^{h' \times w' \times m}$ represents a small number of original convolution feature maps in the output, and $f' \in R^{c \times k \times k \times m}$ denotes the convolution kernel. The output feature map (m) has fewer channels than a traditional convolution layer (n), i.e., $m < n$. Each feature map γ' undergoes a lightweight linear transformation operation, $\Phi_{i,j}$ (where $j=1,2,\dots,s$), to produce an s -dimensional ghost feature map. The calculation uses the following Equation 12.

$$Y_{i,j} = \Phi_{i,j}(Y_i') \quad (12)$$

Given that $m = \frac{n}{s}$ and the convolution kernel size for the linear operation is $d \times d$, compared to conventional convolution, ghost convolution can be approximated as s when $d \times d = k \times k$ and $s \ll c$. The computation ratio between the regular convolution and ghost convolution is as follows (refer to Equation 13):

$$R = \frac{n \times h' \times w' \times c \times k \times k}{\frac{n}{s} \times h' \times w' \times c \times k \times k + (s-1) \times \frac{n}{s} \times h' \times w' \times d \times d} \approx \frac{s \times c}{s + c - 1} \approx s \quad (13)$$

The equation indicates that the ghost convolution's computation is $\frac{1}{s}$ of that of the conventional convolution. Therefore, the ghost

convolution can significantly reduce the computation.

To quantitatively demonstrate the advantages of Ghost convolution in model compression and computational optimization, the following provides a detailed analysis of its parameter count and FLOPs, along with a comparison to traditional convolution.

1. Traditional Convolution Parameters (refer to Equation 14):

$$Params_{Conv} = c \times n \times k^2 \quad (14)$$

Where

Input channels: c ; Output channels: n ; Kernel size: $k \times k$.

2. Ghost-Conv Parameters (refer to Equation 15):

One part is generated through standard convolution to produce intrinsic feature maps, totaling $m = \frac{n}{s}$ maps.

The remaining $n - m = n \times \left(1 - \frac{1}{s}\right)$ maps are generated through linear transformation, not using the full convolution.

Assuming the linear transformation is a per-channel $d \times d$ convolution (typically $d=1$ or $d=3$, much smaller than k):

The total number of parameters for Ghost-Conv is:

$$Params_{Ghost} = c \times k^2 \times \frac{n}{s} + \frac{n(s-1)}{s} \times d^2 \quad (15)$$

3. Parameter Reduction Ratio (refer to Equation 16):

When $d \ll k$ and ignoring the second term:

$$\frac{Params_{Ghost}}{Params_{Conv}} = \frac{c \times k^2 \times \frac{n}{s} + \frac{n(s-1)}{s} \times d^2}{c \times n \times k^2} \approx \frac{1}{s} \quad (16)$$

4. Traditional Convolution FLOPs (refer to Equation 17):

Total FLOPs for n output channels:

$$FLOPs_{Conv} = c \times n \times k^2 \times h \times w \quad (17)$$

Where

Output size: $h \times w$; Output channels: n .

Each output channel requires $c \times k \times k$ multiply-add operations;

5. Ghost-Conv FLOPs (refer to Equation 18-20):

Standard convolution part: generates $\frac{n}{s}$ channels:

$$Params_{std} = \frac{n}{s} \times h \times w \times c \times k^2 \quad (18)$$

Linear transformation part (generating the remaining $n - \frac{n}{s} = \frac{n(s-1)}{s}$ channels):

$$Params_{lar} = \frac{n(s-1)}{s} \times h \times w \times d^2 \quad (19)$$

Total FLOPs for Ghost-Conv:

$$FLOPs_{Ghost} = \frac{n}{s} \times h \times w \times c \times k^2 + \frac{n(s-1)}{s} \times h \times w \times d^2 \quad (20)$$

6. FLOPs Reduction Ratio (refer to Equation 21):

When $d \ll k$ or $c \gg 1$:

$$\frac{FLOPs_{Ghost}}{FLOPs_{Conv}} = \frac{\frac{n}{s} \times h \times w \times c \times k^2 + \frac{n(s-1)}{s} \times h \times w \times d^2}{c \times n \times k^2 \times h \times w} \approx \frac{1}{s} \quad (21)$$

In summary, Ghost Convolution reduces both parameters and computation to approximately $\frac{1}{s}$ of those in traditional convolution, achieving higher efficiency while maintaining comparable performance.

Figure 22 illustrates the modified framework. The backbone

component integrates the GhostNet module, whereas the neck and backbone components incorporate the BottleneckCSP module, forming the new YOLOv8 model.

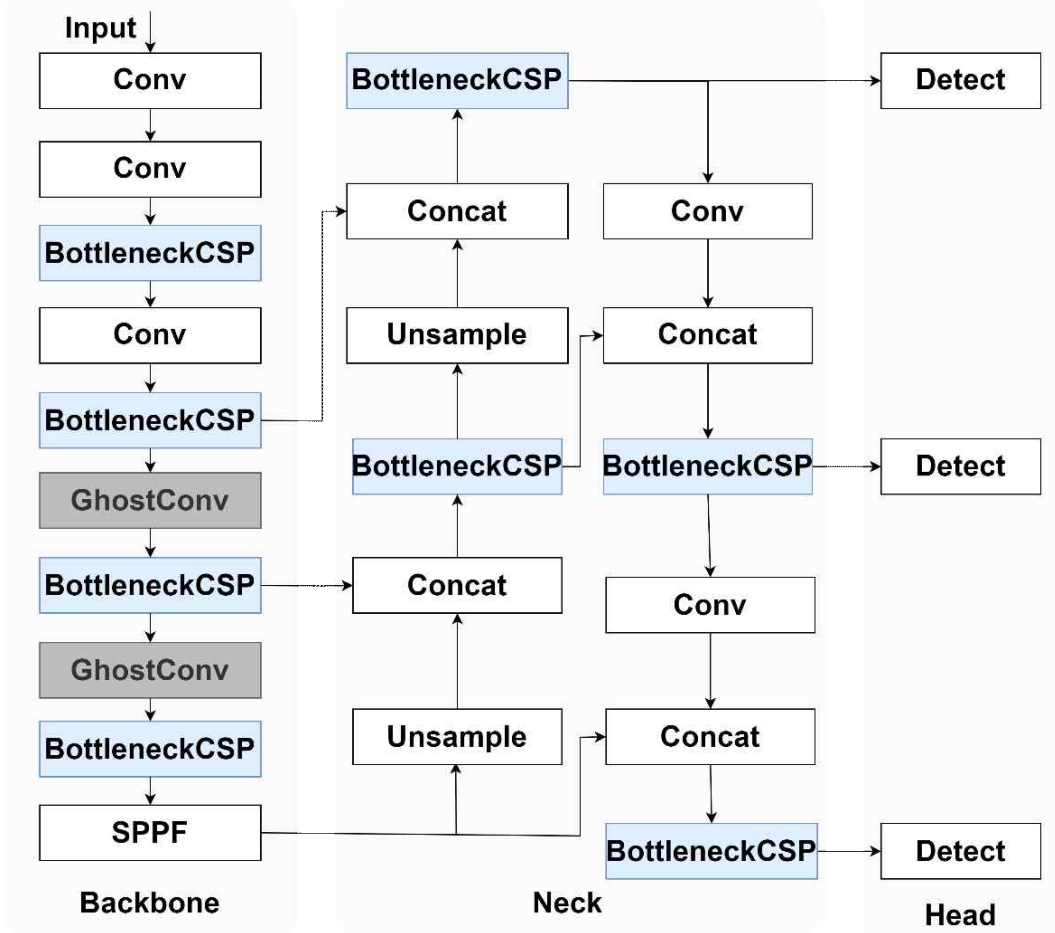


Figure 22. Customer_YOLOv8 network architecture.

4.4 Recognition of hazardous postures in construction workers

With the continuous enhancement of occupational health and safety management requirements on construction sites, efficiently and accurately identifying hazardous worker postures has become a critical research topic in the field of intelligent construction and safety management. To address the limitations of traditional assessment methods, which often suffer from high subjectivity and insufficient accuracy, this stage proposes an automated posture recognition and risk assessment approach that integrates deep learning with ergonomics. The objective is to enable intelligent recognition and quantitative analysis of hazardous postures during construction activities.

In the previous key point detection stage, this study employs an improved version of the YOLOv8 model as the framework for human pose estimation. Building upon the advanced object detection capabilities of the YOLO series, the model has been structurally optimized for human pose estimation tasks, significantly enhancing the precision and generalization performance of keypoint detection. This method is particularly well-suited for real-world construction scenarios characterized by complex backgrounds, uneven lighting, diverse human postures, and partial occlusions. Once the model is trained, it can automatically and accurately extract key body points—such as the head, shoulders, elbows, wrists, hips, knees, and ankles—from images or video frames, thereby providing essential data for risk evaluation.

Next, based on the extracted keypoint information, this study further develops a comprehensive posture assessment workflow. This workflow first performs joint angle calculations using the keypoints, and then uses the resulting angular features as input parameters for typical ergonomic risk assessment models such as REBA (Rapid Entire Body Assessment), RULA (Rapid Upper Limb Assessment), and AWOS (Assessment of Working Posture for Safety). These models ultimately output posture risk levels, enabling the quantitative analysis of poor postures among construction workers. This method not only achieves a full-process transformation from image data to risk assessment results, but also possesses good scalability, providing effective technical support for dynamic monitoring and health interventions on construction sites.

The following section provides a detailed explanation of the procedure for joint angle calculation.

4.4.1 Joint angle calculation

In this stage, the location of each skeleton point was projected onto a plane parallel to the imaging plane of the camera. This projection facilitated a simplified 2D body representation, which is advantageous for joint angle computation. Calculating the vectors between each limb and its adjacent limbs permits precisely determining angles between the limbs and analyzing the body posture. The angles between these vectors were computed using inverse trigonometric functions, with each vector defined by the joint positions and the neighboring keypoints. The detailed steps of this calculation process are provided

below.

A.Retrieve the three keypoint coordinates from the detection results

We retrieved the coordinates of three keypoints, which are the locations of three consecutive body joints. These are typically the joints connected by the limb of interest, and their coordinates are denoted by (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . These coordinates were derived from the keypoint detection results provided by the pose estimation model.

B.Calculate vector magnitudes (v_1 and v_2)

Next, the vectors between the consecutive joints were calculated to represent the direction and length of each limb. The first vector v_1 connects the first and second keypoints, calculated as $v_1 = (x_2 - x_1, y_2 - y_1)$. The second vector v_2 connects the second and third keypoints, calculated as $v_2 = (x_3 - x_2, y_3 - y_2)$. After obtaining the vectors, their magnitudes are calculated using the Euclidean distance formula. The magnitude of the first vector $|v_1|$ is $|v_1| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, and the second vector $|v_2|$ is $|v_2| = \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}$. These magnitudes represent the lengths of the limbs between the joints.

C.Compute angle (A_i)

Once the vectors and their magnitudes are calculated, the angle between the two vectors can be computed using the dot product formula. The dot product between two vectors v_1 and v_2 is given by (refer to Equation 14):

$$v_1 \cdot v_2 = (x_2 - x_1)(x_3 - x_2) + (y_2 - y_1)(y_3 - y_2) \quad (14)$$

The cosine of the angle A_i between the two vectors can then be

computed using the formula (refer to Equation 15):

$$\cos(A_i) = \frac{v_1 \cdot v_2}{|v_1| |v_2|} \quad (15)$$

Finally, the angle A_i is obtained by taking the inverse cosine (arccos) of the result, and converting it from radians to degrees using the formula (refer to Equation 16):

$$A_i = \arccos\left(\frac{v_1 \cdot v_2}{(|v_1| \times |v_2|)}\right) \times \left(\frac{180}{\pi}\right) \quad (16)$$

This step yields the joint angle in degrees, which is crucial for analyzing human posture and detecting potentially hazardous body positions. Algorithm 1 explains the steps for the implementation of the joint angle calculation.

Joint angle calculation algorithm

Input: keypoints

Output: angle

1. Get the keypoints of the detected object results
first_detection_keypoints = results[0].keypoints.xy
 2. Get the keypoint coordinates
keypoints = first_detection_keypoints[0]
 3. Define the indices of the keypoints to calculate the angle
index_1 = a
index_2 = b
index_3 = c
 4. Get the coordinates of the three keypoints
(x1, y1) = keypoints[index_1]
(x2, y2) = keypoints[index_2]
(x3, y3) = keypoints[index_3]
 5. Calculate vectors
Define function calculate_vector(p1, p2):
 return (p2[0] - p1[0], p2[1] - p1[1])
v1 = calculate_vector((x1, y1), (x2, y2))
v2 = calculate_vector((x1, y1), (x3, y3))
 6. Calculate the angle between the two vectors
Define function calculate_angle(v1, v2):
 - Calculate dot product: dot_product = v1[0] * v2[0] + v1[1] * v2[1]
 - Calculate vector lengths:
 v1_norm = sqrt(v1[0]^2 + v1[1]^2)
 v2_norm = sqrt(v2[0]^2 + v2[1]^2)
 - Calculate cos(theta): cos_theta = dot_product / (v1_norm * v2_norm)
 - Limit cos_theta to the range [-1, 1] to handle numerical errors
 - Calculate the angle in radians: angle_rad = acos(cos_theta)
 - Convert to degrees: angle_deg = degrees(angle_rad)
 - Return angle_degangle_deg = calculate_angle(v1, v2)
 7. Print the calculated result
Print "angle: " + angle_deg
-

Based on the joint angle calculations derived from the provided pseudocode, the subsequent step involves mapping these angles onto standardized ergonomic assessment scales, enabling the quantification of posture-related risks.

4.4.2 Scale selection and score calculation

The Ovako working posture analysis system (OWAS) is a method for evaluating worker postures, primarily designed to assess whether body posture during work adheres to ergonomic principles and whether it could lead to discomfort or injury. However, the application of OWAS in posture detection for construction workers is limited, primarily because this method is more suited to static work environments, whereas construction work often involves dynamic postures and complex working conditions. In contrast, REBA and rapid upper limb assessment (RULA) are better suited for evaluating postures in dynamic work environments. In construction site settings, where workers frequently change their posture, REBA and RULA are more capable of addressing this dynamism. Hence, this study adopts REBA and RULA for posture risk assessment (Kee, 2021)¹⁵⁴⁾.

A. Rapid upper limb assessment

In RULA (McAtamney & Corlett, 1993)¹⁵⁵⁾, the body is divided into two categories based on body parts. Group A includes arms, forearms, and wrists on both sides, whereas Group B comprises the neck, trunk, and legs. Each joint angle associated with a posture-related body part has a corresponding range of risk scores in the RULA table, resulting in distinct scores (A and B). These scores represent the

musculoskeletal system load imposed by the overall body position. Two additional values (C and D) are derived by incorporating scores for muscle use and force into the scores of A and B. Finally, using a coding system, these scores are aggregated to determine the overall risk level. The final score, ranging from 1 to 7, is determined based on an appropriate combination of individual scores. This final score corresponds to four action levels and four risk levels. Figure 23 illustrates the RULA scoring procedure.

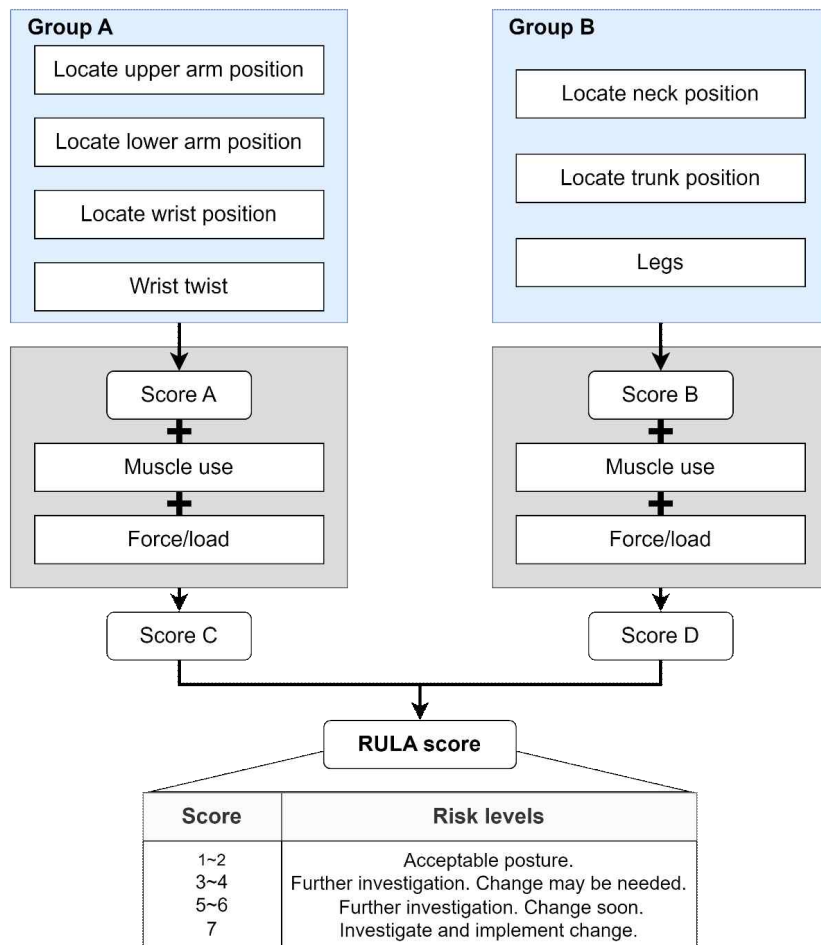


Figure 23. Rapid upper limb assessment.

B. Rapid entire body assessment

In REBA (Hignett & McAtamney, 2000)¹⁵⁶⁾, the body is segmented into two groups based on body parts. Group A consists of the trunk, neck, and legs, whereas Group B includes the upper arms, forearms, and wrists. Similar to RULA, each body part has corresponding joint angle ranges relevant for risk scoring. A coding system assigns a score to each combination of joint angles, reflecting the aggregate risk level for each group. Load-related risk scores are added to the aggregate score of Group A, whereas grip-related scores are incorporated into the aggregate score for Group B. The resulting scores are combined according to the coding system to obtain Score C. The overall risk level is determined by adding the activity scores to the total score. The final score, ranging from 1 to 12, corresponds to five action levels and five risk levels. Figure 24 depicts the REBA scoring procedure.

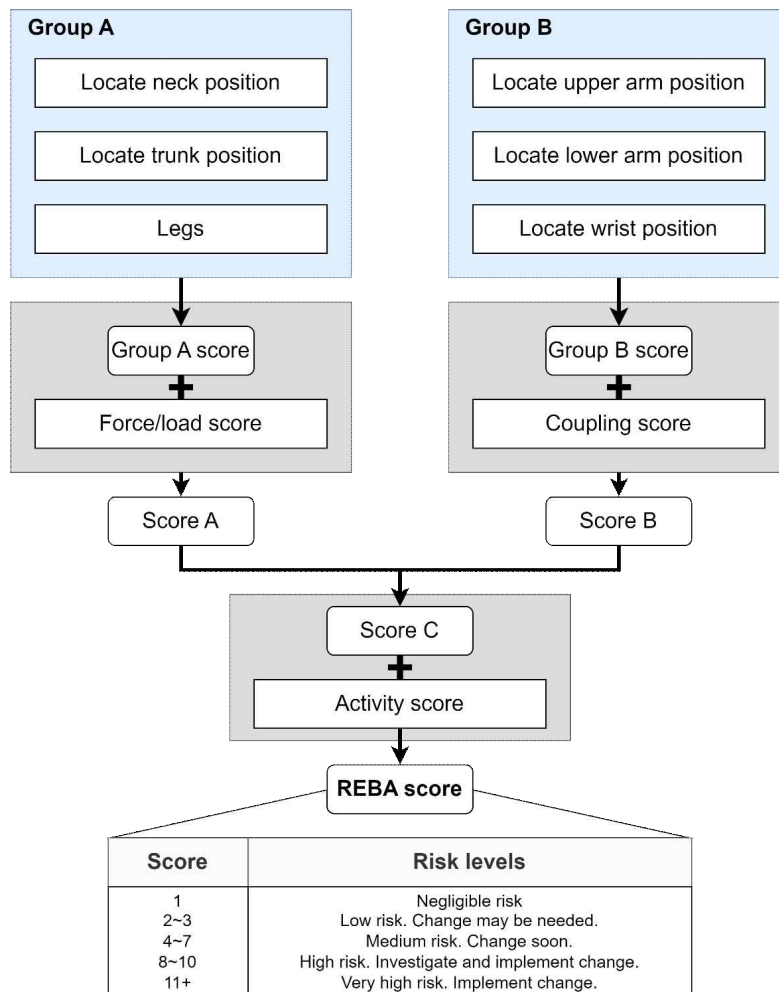


Figure 24. Rapid entire body assessment.

Chapter 5. Experiments and analysis

5.1 Experimental environment

The experimental environment comprises physical hardware and virtual software components, which work synergistically to facilitate efficient model learning, training, and optimization. The hardware environment includes high-performance computing resources, providing substantial computational power and data processing capabilities for handling complex deep learning tasks. Moreover, the virtual software environment enables the training and inference of deep learning models in isolated and efficient settings via optimized configurations and resource allocation, improving learning efficiency, performance stability, and optimization outcomes.

5.1.1 Hardware environment

The system operates on a 64-bit Windows 11 Professional (version 24H2) operating system, providing a stable and efficient runtime environment to support the high-performance demands of deep learning model training and inference. Windows 11 Professional is optimized for resource management and multitasking, enabling high-concurrency computational tasks while ensuring system efficiency in executing complex deep learning algorithms. Regarding the hardware, the system is powered by an Intel Core i7-14700F processor running at 2.10 GHz. This processor features a multicore, multithreaded architecture that delivers substantial parallel computing capabilities. By integrating multiple physical cores and threads, it handles concurrent tasks, enhancing performance in multitasking,

complex computations, and large-scale data processing. Given the computational intensity of deep learning tasks, particularly in matrix operations and large-scale data handling, the i7-14700F processor is well-suited to sustain large-scale model training, data preprocessing, and inference workloads. The system is equipped with 32 GB of high-performance memory, which plays a crucial role in large-scale data processing and high-concurrency computing. Sufficient memory capacity and bandwidth are essential for efficient deep learning model training because they prevent performance bottlenecks caused by memory constraints. The 32 GB memory configuration ensures smooth system operation under high workloads, minimizing delays from frequent data swapping and enhancing overall experimental efficiency and stability. Table 10 details the hardware specifications.

Table 10. Hardware environment.

Configuration Item	Detailed Description
Operating System	64-bit Windows 11 Professional (Version 24H2)
Processor	Intel(R) Core(TM) i7-14700F, Base Frequency: 2.10 GHz
Memory	32.0 GB

5.1.2 Virtual environment

This study aims to enhance the object detection and keypoint recognition performance by applying deep learning model training and inference in optimized hardware and software environments. Two separate Python-based virtual environments were configured to achieve this: one dedicated to object detection and the other to keypoint recognition.

For object detection tasks, the virtual environment is configured

with Python 3.12.4, PyTorch 2.4.1, and CUDA 11.8. This setup fully exploits modern GPU parallel computing capabilities, significantly accelerating the training process of the modified YOLOv5 model. By employing GPU-accelerated parallel processing, the model efficiently handles large datasets, substantially reducing the training time while ensuring stability in large-scale data processing. Furthermore, CUDA acceleration mitigates performance bottlenecks, particularly when processing high-resolution images or complex scenes, yielding a notable improvement in system efficiency.

For keypoint recognition tasks, the virtual environment is configured with Python 3.9.19, PyTorch 2.4.0, and CUDA 11.8, providing a stable runtime environment for training and inference of the improved YOLOv8 model. The YOLOv8 model demonstrates superior accuracy and enhanced processing capabilities in keypoint detection. The stability and efficiency of this configuration ensure the reliability of the model during data processing, further improving inference performance and enabling consistent and effective operation throughout the experiments. Table 11 presents detailed virtual environment configurations.

Table 11. Virtual Environment.

Item		Details
Python Version	Object Detection	3.12.4
	Keypoint Recognition	3.9.19
PyTorch Version	Object Detection	2.4.1
	Keypoint Recognition	2.4.0
CUDA Version	Object Detection	11.8
	Keypoint Recognition	11.8

5.2 Experimental analysis

5.2.1 Object detection model

This study develops a deep learning-based object detection model to assess whether workers are correctly wearing PPE to enhance safety and improve management efficiency on construction sites. The primary functions of the model include detecting and classifying PPE items and generating corresponding classification results based on actual wearing conditions.

5.2.1.1 Performance metrics

This study employs metrics and a confusion matrix to evaluate the performance of the proposed model regarding PPE recognition for construction workers (Figure 25). The confusion matrix categorizes predictions into true positives (TPs), FPs, true negatives (TNs), and FNs for a detailed classification assessment.

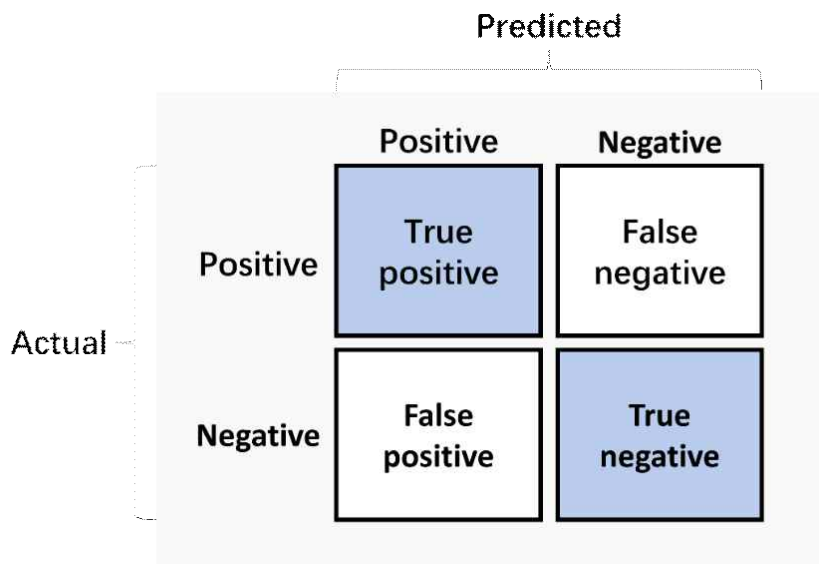


Figure 25. Confusion matrix.

For the worker PPE detection task, this study defines the basic classification results in the confusion matrix, serving as the foundation for calculating performance evaluation metrics (Figure 25). The four critical variables in the confusion matrix, TP, FP, FN, and TN, are defined as follows:

- The TP represents cases where the model correctly detects an image containing PPE and classifies it as PPE. It refers to instances where PPE is present in the image and is recognized by the model. This metric reflects the ability to identify PPE targets correctly and serves as a critical parameter for evaluating detection performance.

- The FP represents cases where the model incorrectly classifies an image without PPE as PPE. The FP refers to situations where the model falsely detects PPE. A high FP count indicates over-detection, which can cause unnecessary interventions or alarms.

- The FN refers to cases where PPE is present in the image but is not detected by the model, which classifies it as no PPE. This metric reflects the rate of missed detections. In PPE detection tasks, an FN can increase safety risks, making it essential to minimize FNs as much as possible.

- The TN represents cases where the model correctly detects an image without PPE and classifies it as no PPE. This metric reflects the ability to correctly identify non-PPE objects, indirectly indicating the robustness of the model.

In this experimental phase, beyond the basic statistics derived from the confusion matrix, four critical evaluation metrics were computed: precision, recall, the mAP at an IoU of 0.5 (mAP@50), and of the

range from 0.5 to 0.95 (mAP@50–95). These metrics offer a comprehensive assessment of the effectiveness of the model in PPE recognition, enabling a detailed analysis of its strengths and limitations.

Intersection over Union (IoU) is a metric used to evaluate the accuracy of predicted bounding boxes in object detection. It quantifies the overlap between a predicted bounding box and the ground truth bounding box, measuring how well the two regions align.

Given a predicted bounding box B_p and a ground truth bounding box B_{gt} , the IoU is defined as (refer to Equation 17):

$$IOU = \frac{\text{Area of intersection}}{\text{Area of } \cup} = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \quad (17)$$

where

$|B_p \cap B_{gt}|$ represents the area of intersection between the predicted and ground truth bounding boxes.

$|B_p \cup B_{gt}|$ represents the total area covered by both bounding boxes, ensuring no double counting of the intersection area.

The IoU value ranges from 0 to 1, where a higher value indicates better localization accuracy.

A. Precision

Precision represents the proportion of TP samples (i.e., correctly detected PPE) among all instances predicted as positive by the model. The TP quantifies the accuracy of the model’s positive predictions by measuring how many of the samples classified as PPE are actually PPE. A high precision score indicates that the model effectively identifies PPE targets while minimizing FPs. This capability is critical in real-world scenarios with substantial background noise or external

interference because it reduces the likelihood of misclassifying non-PPE objects as PPE.

B. Recall

Recall measures the ability to identify all positive instances, specifically the proportion of correctly detected PPE targets out of all actual PPE targets. In the context of worker PPE recognition, recall directly influences the rate of missed detections. A high recall score indicates that the model can detect the majority of PPE instances, minimizing FNs and improving the comprehensiveness of safety monitoring. In practical applications, a high recall ensures that as many PPE instances as possible are identified in complex working environments, reducing safety risks associated with undetected or improperly worn PPE.

C. mAP@50 (Mean Average Precision at IoU = 0.5)

The mAP@50 metric is a widely adopted evaluation metric in object detection, assessing model performance when the IoU threshold is set to 0.5. This metric quantifies the ability of the model to localize and classify objects accurately by averaging precision across all categories at this IoU level. In PPE detection, mAP@50 provides a crucial benchmark for evaluating baseline detection accuracy, ensuring that the model can reliably identify and localize PPE under standardized conditions.

D. mAP@50-95 (Mean Average Precision at IoU = 0.5:0.95)

The mAP@50-95 metric is a more stringent evaluation metric that considers multiple IoU thresholds (0.5 to 0.95) and calculates the average precision across these thresholds. Compared to mAP@50,

mAP@50-95 is more rigorous and provides a comprehensive assessment of the performance under varying overlapping requirements. This metric is valuable for evaluating the overall capability of the model because it has precision at higher IoU thresholds. In scenarios with significant object overlapping, mAP@50-95 evaluates whether the model maintains high detection performance. For PPE recognition tasks, mAP@50-95 assesses the adaptability and generalizability of the model across diverse conditions.

A thorough understanding of the effectiveness of the model in worker PPE recognition tasks can be obtained by comprehensively evaluating the performance of the model using precision, recall, mAP@50, and mAP@50-95. These metrics help analyze the ability of the model to improve PPE detection accuracy, minimize FPs and FNs, and balance precision and recall. In practical applications, an optimal balance is often necessary between precision and recall to ensure accurate PPE detection while minimizing missed detections. Based on these definitions, Table 12 summarizes the evaluation metrics in this study and their respective calculation formulas.

Table 12. Model performance evaluation formulas.

Metric	Formula	Description
Precision (P)	$Precision = \frac{TP}{TP + FP}$	–
Recall (R)	$Recall = \frac{TP}{TP + FN}$	–
mAP@50	$mAP@50 = \frac{1}{N} \sum_{i=1}^N AP_i$	N: number of categories (e.g., PPE classes). AP _i : average precision for class i, calculated when the intersection over union (IoU) is ≥ 0.5
mAP@50-95	$mAP@95-50 = \frac{1}{N} \sum_{i=1}^N \frac{1}{5} \sum_{j=1}^5 AP_{i,j}$	AP _{i,j} : average precision for the ith class at the jth IoU threshold, typically evaluated at multiple IoU values (i.e., 0.5, 0.55, 0.6, 0.65, 0.7)

5.2.1.2 Training and validation

A. Hyperparameters

In the construction worker PPE detection task, this study adopts a series of hyperparameter configurations (Table 13) for model training to ensure the accurate identification of PPE-wearing status and enhance the safety of construction sites. The aim is to optimize model performance and generalizability while ensuring training efficiency. The hyperparameter settings consider the diversity and complexity of tasks in the challenging construction site environment, focusing on improving model adaptability and robustness.

For the learning rate, the initial learning rate (l_{r0}) is set to 0.01, and the final learning rate (l_{rf}) is also 0.01. This setup ensures that the learning rate does not decrease too quickly during training, accelerating model convergence and reducing potential gradient

oscillations in the early stages. The combination of momentum (0.937) and weight decay (0.0005) helps stabilize the gradient descent and prevents overfitting, which is vital in complex construction environments where lighting and background conditions vary. A warm-up phase is implemented with warm-up epochs set to 3.0 to stabilize the training process. A lower momentum (warm-up_{momentum} = 0.8) and bias learning rate (warm-up_{bias_lr} = 0.1) are employed during this phase. This gradual increase in the learning rate helps alleviate early-stage instability and ensures that the model begins converging during a smooth optimization process.

The loss function design balances multitask trade-offs, particularly for the target detection task, by appropriately distributing loss weights for tasks. The settings for box loss weight (box = 0.05), classification loss weight (cls = 0.5), and object loss weight (obj = 1.0) balance the tasks of box regression, object classification, and target recognition, improving task performance. Adjustments to classification loss weight (cls_{pw} = 1.0) and object loss weight (obj_{pw} = 1.0) allow the model to focus more on critical targets in PPE detection, improving task-related learning efficiency. The IoU threshold (iou_t = 0.2) and anchor threshold (anchor_t = 4.0) are set to appropriate values to optimize the matching of anchor boxes to ground-truth boxes, ensuring high matching accuracy while avoiding the generation of redundant boxes and improving detection precision.

Concerning data augmentation, the model employs hue variation (hsv_h = 0.015), saturation variation (hsv_s = 0.7), and brightness variation (hsv_v = 0.4) to simulate scene changes under diverse lighting

conditions and address the complex environmental variations on construction sites, increasing the data diversity. These variations substantially enhance the robustness of the model to lighting changes, particularly in real construction sites where lighting is often an uncontrollable factor. Geometric transformations, including random translation (0.1), scaling (0.5), and horizontal flipping (fliplr = 0.5), strengthen the adaptability of the model to worker postures. Although rotation (0.0°) and shearing (0.0) were not enabled, mosaic augmentation (1.0) significantly increased the complexity of the training samples. This enhancement method better simulates multitarget scenarios and strengthens the ability of the model to detect multiple targets simultaneously on construction sites. Although MixUp (0.0) and copy-paste (0.0) were not used, which eliminates potential pseudo-label interference, the model learning process remained stable and accurate.

Regarding other configurations of the loss function, the focal loss gamma parameter ($f_{\text{gamma}} = 0.0$) is not enabled because the positive-to-negative sample ratio in the current task is relatively balanced, and there is no need to handle complex examples via focal loss. However, the focal loss alpha parameter ($f_{\text{alpha}} = 0.95$) is retained to adjust sample weights and enhance the focus on hard-to-detect targets during training.

The adaptability of the model to construction site scenarios is enhanced via this series of hyperparameter configurations, which fully consider the complexity of the construction site environment, diversity of worker postures, and variability of PPE-wearing status. Hence, its

robustness and generalizability in practical applications also significantly improved.

Table 13. Hyperparameters.

Category	Parameter Name	Value
Learning Rate	Initial (l_{r0})	0.01
	Final (l_{rf})	0.01
Optimizer	Momentum	0.937
	Weight decay	0.0005
Warm-up Phase	Epochs	3.0
	Momentum	0.8
	Bias LR	0.1
Loss Weights	Bbox (box)	0.05
	Classification loss weight (cls)	0.5
	Classification weight (cls _{pw})	1.0
	Object loss weight (obj)	1.0
	Object weight (obj _{pw})	1.0
	IoU (iou _t)	0.2
	Anchor (anchor _t)	4.0
Focal Loss	Alpha ($f_{l\alpha}$)	0.95
Data Augmentation	Hue (hsv _h)	0.0015
	Saturation (hsv _s)	0.7
	Brightness (hsv _v)	0.4
	Translation ratio (translate)	0.1
	Scaling ratio (scale)	0.5
	Flip (fliplr)	0.5
Mixing Techniques	Mosaic augmentation (mosaic)	1.0

B. Output curve analysis

This section presents an analysis of the curves. The curves were generated after the training and validation in this phase.

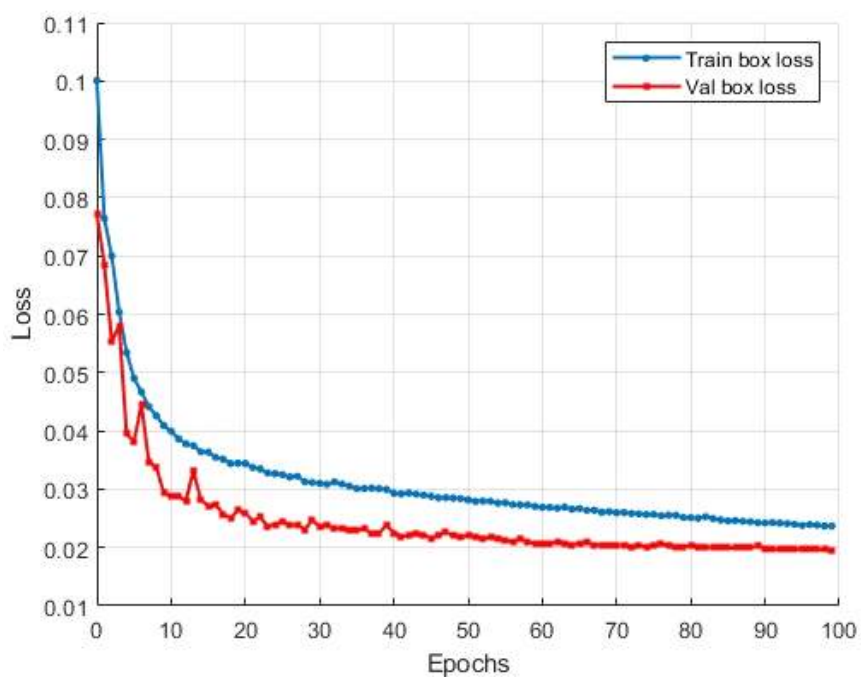
Box loss curve (Figure 26(a)). The box loss on the training set starts at 0.11 and decreases rapidly, particularly between 0 and 20

epochs, indicating that the model quickly learns effective features and optimizes the bounding boxes during the early stages of training. The stabilization of training loss from 20 to 100 epochs suggests the model is nearing convergence. By epoch 100, the box loss dropped to 0.025, reflecting good fitting during training. The box loss on the validation set also decreased rapidly between epochs 0 and 30, suggesting effective fitting on the validation set and the capacity to learn critical features for PPE detection in construction workers. However, fluctuations in the box loss of the validation set between epochs 0 and 30 indicate potential overfitting or instability when handling some validation samples. These fluctuations may be attributed to the small validation dataset, insufficient sample diversity, or increased difficulty detecting specific PPE categories. In construction worker scenarios, workers may wear protective gear (e.g., helmets, gloves, goggles) in complex environments, which can vary significantly in appearance due to changes in angle, lighting, or occlusions. Moreover, worker posture variability, obstructions, and gear deformation may contribute to detection instability. The fluctuations in validation box loss could also stem from an unrepresentative validation dataset, as the sample size may be small and insufficiently diverse in terms of environmental changes, equipment types, and worker postures. Thus, the model may not perform consistently across all scenarios, yielding fluctuating errors. Notably, by epoch 100, the box loss on the validation set tends to reach 0.02, indicating further improvement in the performance of the model on the validation set in the later stages of training. This reduction in validation loss suggests that, despite early fluctuations,

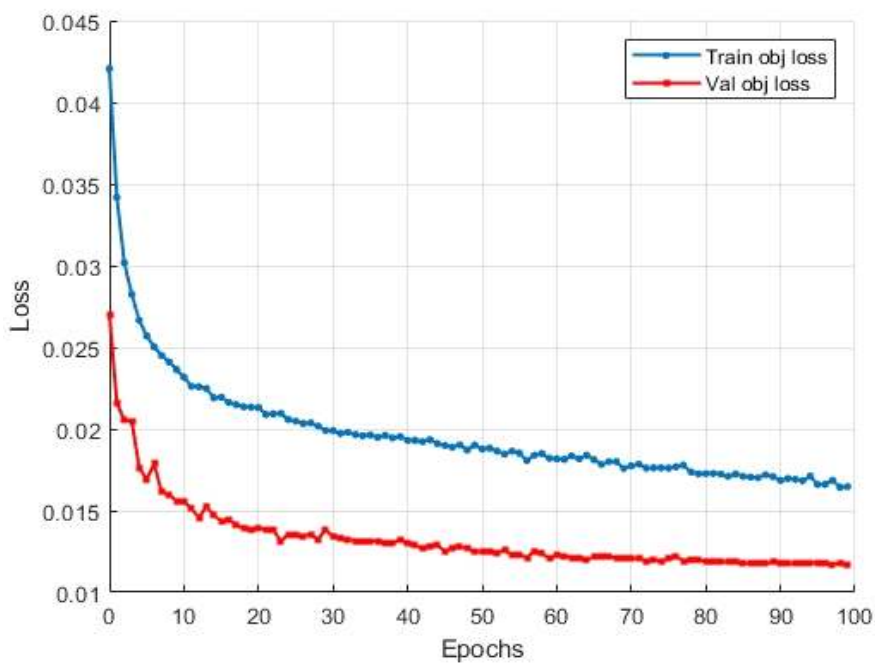
the model gradually learned more accurate features, particularly in handling complex environments and variable postures, enhancing robustness. This trend may indicate that, after sufficient training cycles, the model better adapts to the validation set samples, demonstrating stronger generalizability. Moreover, the lower validation loss compared with the training loss supports the absence of significant overfitting during training, with the model performing well across diverse scenarios in the validation set. This phenomenon suggests that the model can handle diverse types of protective equipment, environmental variations, and worker movements in the construction worker PPE detection task, improving detection performance in real-world scenarios.

Object loss and classification loss curve analysis (Figure 26(b,c)). In object detection tasks, object loss (obj loss) and classification loss (cls loss) are crucial loss functions that focus on object existence and classification accuracy, respectively. On the training set, obj loss starts at 0.045 at epoch 0, decreases rapidly, and stabilizes after approximately 25 epochs, reaching 0.016 at epoch 100. This outcome indicates that the model quickly learns effective object localization features, optimizing the existence of the bounding box early in the training process. As training progresses, the loss stabilizes, demonstrating the ability of the model to recognize and accurately locate PPE items. The cls loss follows a similar pattern. Initially starting at 0.045, it decreases rapidly after about 10 epochs and approaches 0 by epoch 100, signifying the efficient learning of the model in the classification task. By the end of training, the near-zero

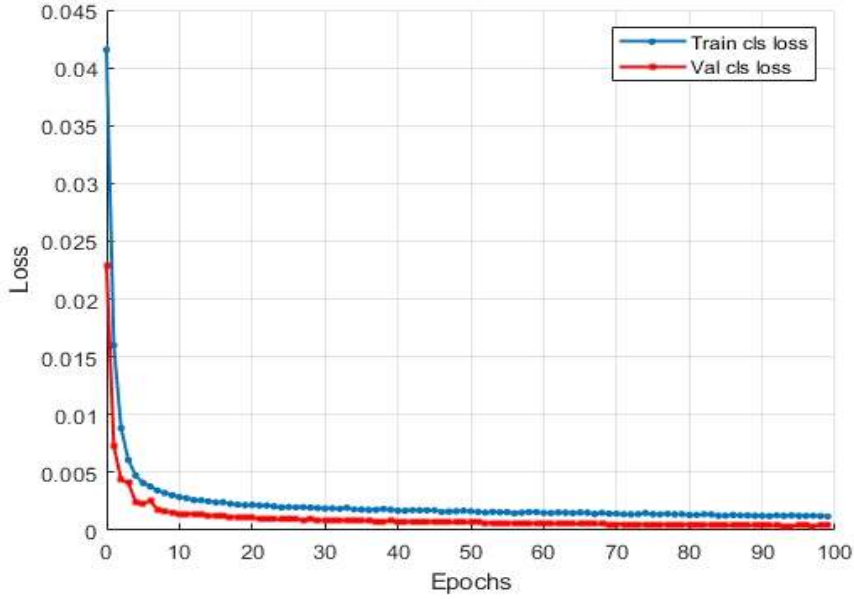
cls loss indicates that the model is highly effective in identifying PPE types, improving classification accuracy significantly. The optimization of obj loss and cls loss occurs concurrently, enhancing the overall performance in object detection and classification tasks. In comparison to the training set, the obj loss on the validation set starts at 0.027 at epoch 0, stabilizes around epoch 20, and drops to 0.012 by epoch 100. The obj loss for the validation set consistently remains lower than that of the training set, with a smoother decline, highlighting that the model effectively recognizes PPE in unseen data. Similarly, the cls loss on the validation set starts at 0.024, decreases steadily, and stabilizes near 0 by epoch 100. The cls loss curve of the validation set is lower than that of the training set, suggesting higher classification accuracy and better generalization. The training and validation loss curves indicate that the optimization of object existence and classification tasks happens in tandem, each task complementing the other. The low loss values on the validation set demonstrate the strong generalizability of the model, ensuring it can adapt to diverse data distributions while maintaining high accuracy. This finding suggests that the model performs well on unseen data and can detect and classify PPE effectively in diverse scenarios.



(a) Box loss



(b) Object (Obj) loss



(c) Classification (Cls) loss

Figure 26. Model loss curve: (a) Box loss; (b) Object (Obj) loss; (c) Classification (Cls) loss.

Precision curve (Figure 27). At epoch 0, the precision starts at 0.15, indicating that the model has a high FP rate in the early stages, incorrectly identifying many nonexistent objects as PPE. However, as training progresses, precision rises rapidly, revealing that the model reduces false detections by optimizing bounding-box regression and classification accuracy. By epoch 10, precision reaches 0.95, demonstrating a significant improvement in correctly identifying PPE worn by construction workers. As training continues, the precision curve flattens, and by epoch 100, it stabilizes around 0.98, indicating that the model maintains a low FP rate and achieves high accuracy and robustness in object localization and classification. This steady increase in precision reflects successful convergence, ensuring reliable PPE detection even in complex environments and diverse data

scenarios. The final precision value suggests that the model generalizes well and is highly effective in reducing misclassifications in real-world applications.

Recall (Figure 27). At epoch 0, the recall starts at 0.2, indicating a high FN rate, meaning the model fails to detect many actual PPE instances. However, as training progresses, recall increases rapidly, demonstrating that the model gradually improves its detection capability by recognizing more PPE equipment and minimizing missed detections. By epoch 10, recall reaches nearly 0.9, aligning closely with the precision, suggesting a significant reduction in FNs. As training continues, recall stabilizes at 0.97 by epoch 100, indicating that the model successfully detects almost all actual PPE instances. Minor fluctuations (≤ 0.03) between epochs 0 and 10 reflect slight instability, likely caused by data diversity, training adjustments (e.g., learning rate changes), or data challenges (e.g., occlusions or varying PPE appearances). Despite this, the model demonstrates strong generalization and robust recall performance in PPE detection tasks.

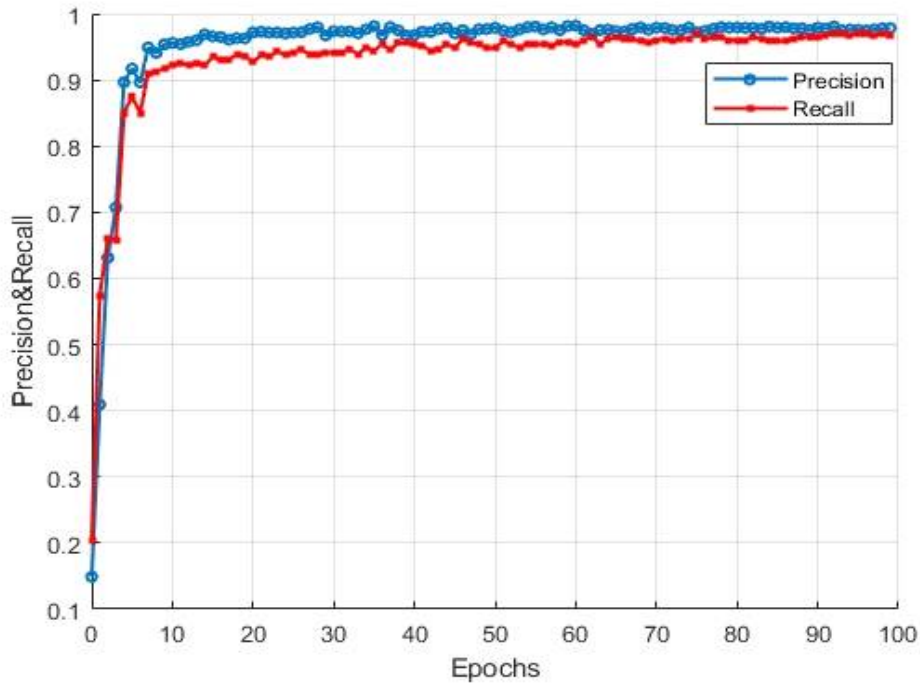


Figure 27. Model precision and recall curve.

The mAP curve analysis (Figure 28). In this model, the mAP curve includes two indicators, mAP@50 and mAP@50-90, to measure the detection accuracy at an IoU threshold of 0.5 and across multiple thresholds from 0.5 to 0.95. The mAP@50 curve reflects the detection performance under relatively low localization precision requirements (IoU = 0.5). At epoch 0, the mAP@50 value is approximately 0.1, indicating that the model initially struggles to detect and localize construction worker PPE accurately. However, as training progresses, the mAP@50 value increases rapidly, reaching approximately 0.99 by epoch 100. This sharp improvement suggests that the model quickly learns the distinguishing features of PPE and achieves high localization accuracy at lower IoU thresholds. The results demonstrate that the model recognizes and localizes various PPE types, including helmets,

safety vests, and gloves, even under variations in pose and viewpoint.

The mAP@50-90 curve represents performance under stricter localization precision requirements ($\text{IoU} = 0.5$ to 0.95). Compared to mAP@50 , the mAP@50-90 values are lower, with fluctuations observed between epochs 0 and 15, with a fluctuation range of less than 0.1. These variations suggest instability in the optimization process when the model must perform high-precision object localization. Several factors may contribute to this fluctuation in the construction worker PPE detection task. First, PPE occlusion and object overlapping with other elements (e.g., tools or structural components) may hinder accurate boundary detection. Second, appearance variations among PPE types (e.g., helmets vs. safety vests) increase the complexity of fine-grained object localization. Third, the dynamic construction environment and diverse worker postures introduce additional challenges to high-precision detection at higher IoU thresholds. Despite these challenges, the model maintains an mAP@50-90 value above 0.75, indicating strong overall performance in high-precision object detection. Given that PPE in real-world construction sites is often partially occluded, improperly worn, or relatively small, the model demonstrates robustness and adaptability by maintaining high detection accuracy in such complex scenarios.

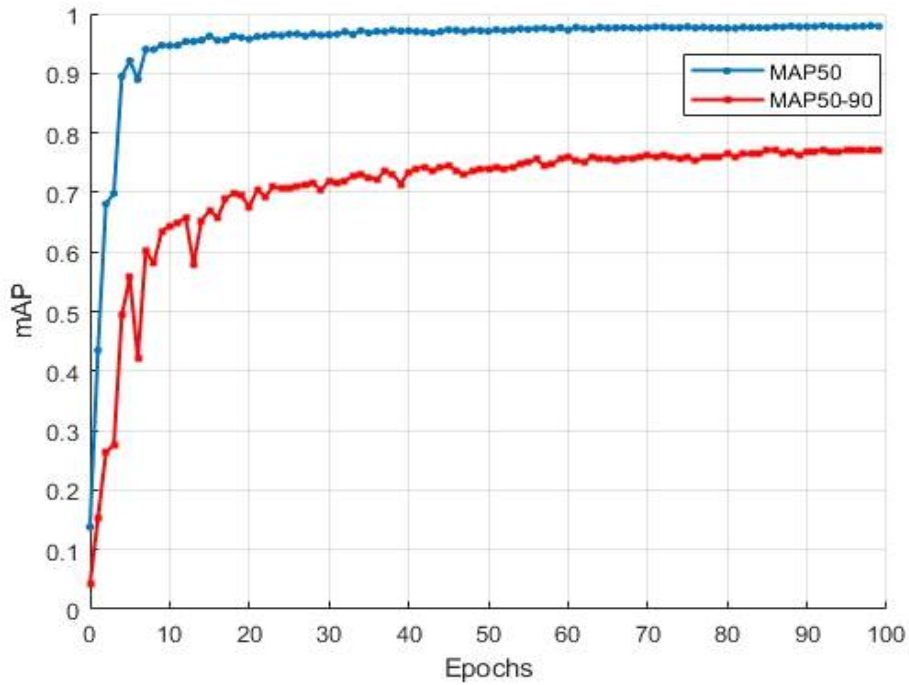


Figure 28. Mean average precision (mAP) curve.

The model performs well across multiple evaluation metrics. The loss curve displays stable box loss and low object and classification loss, with the validation loss lower than the training loss, indicating effective object localization, high PPE recognition accuracy, good generalization, and no severe overfitting. The precision curve reflects a significant reduction in FPs, stabilizing at a high precision level, ensuring reliable identification of PPE types (e.g., helmets and gloves), and enabling effective detection in complex construction site environments. Similarly, the recall curve indicates a substantial decrease in FNs, with a stable recall curve across data distributions, demonstrating strong adaptability to real-world construction scenarios. Finally, the mAP curve highlights exceptional detection performance, with mAP@50 reaching 0.99, confirming the superior PPE detection

capability and low FP rate of the model. In contrast, mAP@50-90 exceeds 0.75, indicating its effectiveness in high-precision localization, robust recognition across PPE types, and adaptability to occlusion and complex environments.

5.2.1.3 Visualization

Figure 29 indicates that the visualized PPE detection results across three typical construction scenes display substantial performance improvements of the optimized object detection framework in addressing complex tasks on actual construction sites. The figure highlights the potential of the framework for adaptability and scalability in handling multiobject detection.

In the ladder work scenario, the optimized framework significantly improved the detection accuracy of safety gloves from 0.8 to 0.9 and helmets from 0.4 to 0.8, while the detection performance for safety shoes remained stable. This notable enhancement in performance underscores the superior feature extraction capabilities of the optimized framework, particularly in handling small or low-resolution targets. The framework is more adept at accurately identifying and recognizing targets that may be challenging to detect due to their small size, partial occlusion, or placement against complex backgrounds. This improvement emphasizes advancements in fine-detail processing and enhances the adaptability and robustness of the framework in complex scenes.

In the scaffold work scenario, the optimized framework significantly improved the detection accuracy of gloves (from 0.6 to 0.7) and

increased the accuracy of safety shoe detection from 0.8 to 0.9. This result indicates that the optimized framework achieves more stable and efficient performance for medium-sized targets. The framework supports the detection of safety ropes and safety vests, with the detection accuracy for safety vests reaching 0.8. This expanded capability reflects a substantial improvement in the adaptability of the framework to multiclass, multiobject detection tasks. The ability to accommodate new object categories is attributed to enhancements in the feature extraction module and network architecture, meeting the practical demands of complex and diverse scenarios.

In the elevated platform work scenario, the optimized framework exhibited marginal improvements in the detection accuracy of gloves and boots. In contrast, the detection performance for other targets (e.g., helmets, safety ropes, and safety vests) remained consistent with that of the original framework. The detection accuracy for safety ropes and safety vests remained at the highest level of 1. This outcome suggests that the optimized framework demonstrates strong stability in detecting high-confidence targets, maintaining consistent and reliable performance even in dynamic and complex environments.

In summary, the optimized object detection framework exhibits exceptional capabilities in detecting small targets and a positive overall performance improvement trend. The framework also demonstrates strong adaptability to new target detection tasks across work scenarios. Notably, the framework offers significant advantages, especially in scenarios characterized by complex details and diverse target categories.

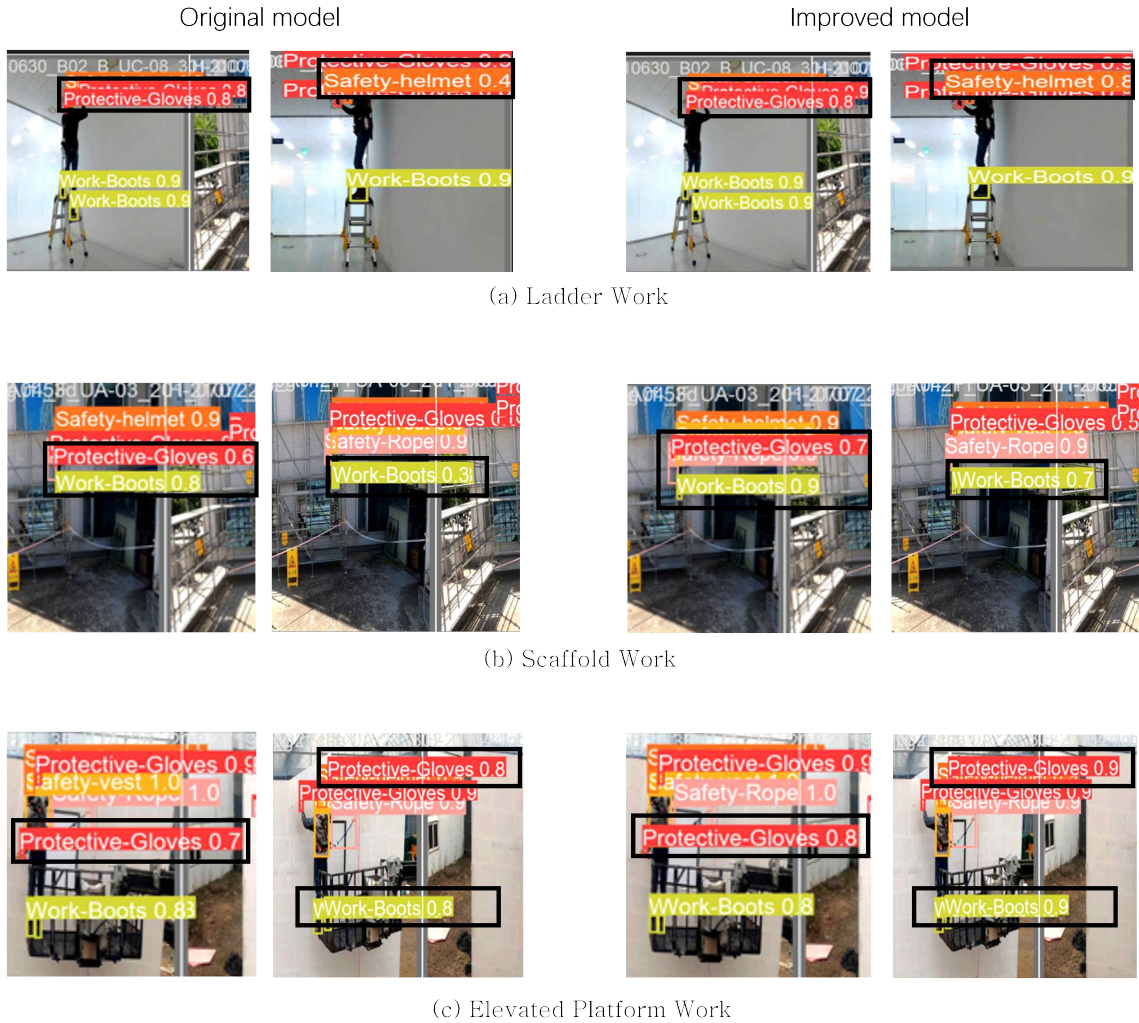


Figure 29. Visualization Results: (a) Ladder Work, (b) Scaffold Work, (c) Elevated Platform Work.

5.2.1.4 Model optimization analysis

The confusion matrix, offering insight into predictions across categories, is crucial for assessing classification performance. The x- and y-axes of the matrix represent six categories: gloves, safety ropes, safety helmets, safety vests, safety shoes, and background. Figure 30 illustrates the prediction accuracy of the optimized object

detection model compared with the original YOLOv5 for each category.

1. Confusion matrix of the initial model

The confusion matrix of the initial model (Figure 30(a)) reveals several distinct misclassification trends. Notably, frequent misclassification occurs between the background and target categories. For instance, the value at $(6,1) = 0.08$ indicates that the background was misclassified as gloves, suggesting difficulty in differentiating between the background and small objects (e.g., gloves). Similarly, the values at $(5,6) = 0.14$ and $(3,6) = 0.09$ highlight misclassifications between the background and safety shoes or safety helmets. These errors are likely due to the minimal feature differences between the background and target classes or the small size, partial occlusion, or presence of complex backgrounds, making accurate distinction challenging for the model.

Moreover, the value at $(1,6) = 0.73$ demonstrates a significant misclassification between the background and gloves, likely attributable to the tendency of gloves, as small objects, to become occluded or overlap with the background in complex construction environments, hindering their recognition. Misclassifications between safety shoes and other categories, such as safety ropes and gloves, also occurred. For example, $(6,5) = 0.01$ indicates that safety shoes were misclassified as background, whereas $(3,6) = 0.09$ and $(2,6) = 0.02$ reflect misclassifications of safety helmets and safety ropes as background. These errors underscore the challenges in distinguishing between the background and targets, especially when targets are small or partially occluded.

Further, minor misclassifications occurred between safety ropes, safety helmets, and gloves (e.g., $(1,2) = 0.01$ and $(1,3) = 0.03$). These misclassifications may result from visual similarities between these categories, particularly in complex backgrounds or when multiple targets exist in the same area. For instance, safety helmets and gloves may share similar shapes, especially when worn, with the edges of the gloves overlapping the helmet outline. Similarly, safety ropes can be confused with other protective equipment when only partially visible or occluded.

Overall, the confusion matrix of the initial model highlights several limitations related to misclassifications between targets and background interference, especially when dealing with small or occluded objects. These problems contribute to a reduction in the overall detection accuracy of the model.

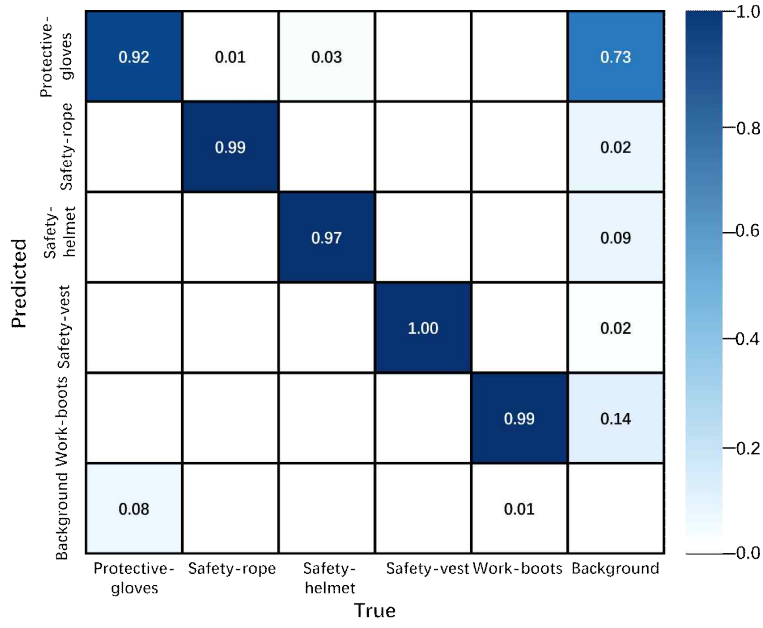
2. Confusion matrix of the optimized model

The confusion matrix of the optimized model demonstrates substantial improvements in performance, particularly in terms of reducing misclassifications and enhancing overall accuracy (Figure 30(b)). For instance, $(1,2) = 0$ and $(1,3) = 0$ indicate that the optimized model has eliminated the misclassifications between gloves and safety ropes and between gloves and safety helmets. This enhancement suggests that the ability to distinguish subtle differences between these target categories has been significantly improved, particularly in cases where the shapes, sizes, or textures of the targets are similar. The optimized model demonstrates a strong capability to handle similar targets within complex scenarios.

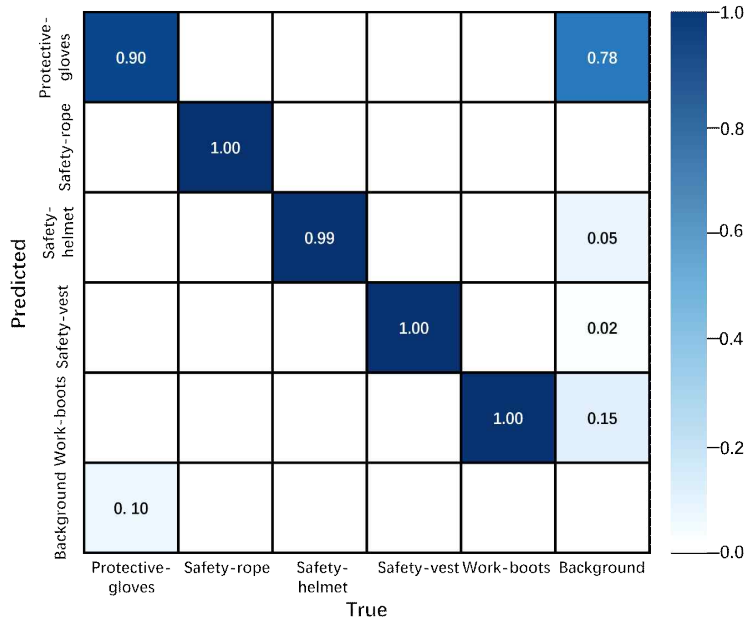
Moreover, $(3,6) = 0.05$ highlights a notable improvement in the accuracy of safety helmet detection. The optimized model reduces the likelihood of safety helmets being misclassified as the background. This result is significant in complex environments where safety helmets, as medium-sized objects, may overlap with the background color and texture, leading to misclassification in the original model. The optimized framework excels at recognizing this target category, demonstrating its superiority in managing complex backgrounds and small targets.

Further, $(6,5) = 0$ signifies the excellent performance of the optimized model in detecting safety shoes. In the original model, safety shoes were occasionally misclassified as the background, but the optimized model eliminated this error. This improvement reflects a significant improvement in detecting small targets, such as safety shoes, which are challenging to identify due to their size or partial occlusion.

Tables 14 and 15 present the critical performance indicators generated by the model following training, including the precision, recall, and mAP, among others.



(a) YOLOv5



(b) PPE_YOLO

Figure 30. Confusion matrix of the (a)YOLOv5 and (b)PPE_YOLO models.

Table 14. Performance metrics of the YOLOv5 model.

Class	Images	Instances	P	R	mAP50	mAP50-95
all	200	1160	0.970	0.971	0.977	0.753
Protective-Gloves	200	302	0.891	0.865	0.913	0.509
Safety-Rope	200	139	0.993	1.000	0.995	0.870
Safety-helmet	200	200	0.995	0.996	0.995	0.800
Safety-vest	200	138	0.982	1.000	0.988	0.867
Work-Boots	200	381	0.989	0.992	0.995	0.720

The original YOLOv5 model demonstrates excellent overall performance, with precision reaching 0.97, recall at 0.971, and mAP@0.5 as high as 0.977, indicating strong accuracy in object detection tasks. However, the mAP@0.5-0.95 value is 0.753, reflecting a slight decline in performance under stricter IoU thresholds, particularly in scenarios involving small targets and complex backgrounds. For large targets, such as safety ropes (mAP@0.5: 0.995, mAP@0.5-0.95: 0.87) and safety helmets (mAP@0.5: 0.995, mAP@0.5-0.95: 0.8), detection performance is robust, illustrating the strong ability of the model to identify prominent targets. However, for smaller targets, such as gloves, precision and recall are 0.891 and 0.865, respectively, whereas mAP@0.5-0.95 is only 0.509, indicating the need for further refinement in distinguishing fine-grained target categories. Some categories face misdetections and missed detections in complex backgrounds, such as gloves being misclassified as background and safety shoes exhibiting an mAP@0.5-0.95 value of 0.72, revealing the limitations of distinguishing small targets.

Table 15. Performance metrics of the YOLOv11 model.

Class	Images	Instances	P	R	mAP50	mAP50-95
all	200	1160	0.967	0.957	0.962	0.758
Protective-Gloves	200	302	0.876	0.791	0.856	0.509
Safety-Rope	200	139	0.992	1.000	0.995	0.883
Safety-helmet	200	200	1.000	1.000	0.995	0.811
Safety-vest	200	138	0.981	1.000	0.969	0.869
Work-Boots	200	381	0.987	0.992	0.995	0.722

The YOLOv11 model demonstrates balanced and competitive performance compared to YOLOv5. Its overall precision and recall are 0.967 and 0.957, respectively, with an mAP@0.5 of 0.962 and an mAP@0.5-0.95 of 0.758. These results indicate a slight reduction in precision but improved robustness under stricter IoU thresholds relative to YOLOv5. For large targets such as safety ropes (mAP@0.5: 0.995; mAP@0.5-0.95: 0.883) and safety helmets (mAP@0.5: 0.995; mAP@0.5-0.95: 0.811), the model maintains excellent detection accuracy and consistency, reflecting strong localization and classification capabilities. However, small object detection remains challenging. For the glove category, precision and recall are 0.876 and 0.791, respectively, and the mAP@0.5-0.95 remains at 0.509, showing no improvement over YOLOv5 and highlighting persistent difficulties under complex backgrounds. Despite these limitations, YOLOv11 exhibits slight gains in certain PPE categories. Notably, the mAP@0.5-0.95 for safety vests increased from 0.867 to 0.869, and for safety boots from 0.720 to 0.722, indicating incremental improvements in the recognition of these PPE items.

Table 16. Performance metrics of the PPE_YOLO model.

Class	Images	Instances	P	R	mAP50	mAP50-95
all	200	1160	0.980	0.970	0.980	0.772
Protective-Gloves	200	302	0.922	0.858	0.925	0.554
Safety-Rope	200	139	1.000	1.000	0.995	0.889
Safety-helmet	200	200	1.000	1.000	0.995	0.821
Safety-vest	200	138	0.986	1.000	0.989	0.869
Work-Boots	200	381	0.992	0.992	0.995	0.726

The PPE_YOLO model displays significant improvement in overall performance, with precision increasing from 0.97 to 0.98, indicating a further reduction in misdetection rates. Moreover, mAP@0.5 rises to 0.98, and mAP@0.5-0.95 increases to 0.772, demonstrating the enhanced detection of the optimized model across IoU thresholds. In terms of category performance, the detection accuracy for gloves improved from 0.891 to 0.922, and mAP@0.5-0.95 increased from 0.509 to 0.554, significantly reducing misdetections for small targets. Large targets, such as safety ropes (mAP@0.5-0.95: 0.889) and safety helmets (mAP@0.5-0.95: 0.821), demonstrated more stable performance, with recall reaching 1. Safety shoe detection slightly improved in mAP@0.5-0.95 to 0.726, optimizing the ability of the model to handle complex backgrounds and subtle differences. The model demonstrated a notable enhancement in consistency for precision and recall across all categories, indicating that it reduces misdetections and missed detections and improves adaptability to small targets and complex backgrounds.

In summary, the PPE-YOLO model outperforms the original model at global and category levels, presenting significant improvement under

stricter IoU thresholds (mAP@0.5–0.95). This finding indicates that the optimized model made notable advancements in fine-grained target classification, complex scene resolutions, and small target detection. The original model achieved high performance in detecting large targets; however, it faced challenges in feature extraction for small targets (e.g., gloves) and background differentiation, which became the primary bottleneck affecting the overall performance. The optimized model addresses these limitations by enhancing feature extraction capabilities and refining the model architecture, considerably reducing misdetections and missed detections. This model also improves the handling of confusion between gloves and the background and the subtle category differences in complex scenes.

5.2.2 Keypoint detection model

Keypoint detection is pivotal for analyzing worker posture and recognizing actions because it allows the precise identification of critical body parts (e.g., the head, shoulders, elbows, and knees). This identification allows for a detailed understanding of the worker's dynamic behavior in work scenarios. In the complex environment of construction sites, keypoint detection encounters multiple challenges, including occlusion, lighting variations, irregular backgrounds, and accurate recognition of diverse worker actions. This study proposes a deep learning-based keypoint detection algorithm to overcome these obstacles and achieve high-precision keypoint localization.

5.2.2.1 Performance metrics

This study employs multiple metrics, including object keypoint

similarity (OKS), precision, recall, and mAP, to evaluate performance comprehensively. Precision is the ratio of TPs to predicted positives, and recall is the ratio of TPs to actual positives. These metrics reflect the detection effectiveness of the model. In addition, mAP is the area under the precision-recall curve, providing an evaluation across varying detection thresholds. Table 17 presents the relevant formulas for model performance evaluation to explain the calculation of these evaluation metrics.

Table 17. Model performance evaluation metrics.

Metric	Formula	Description
Precision (P)	$P = \frac{TP}{TP + FP}$	–
Recall (R)	$R = \frac{TP}{TP + FN}$	–
Mean Average Precision (mAP)	$mAP = \frac{1}{n} \sum_{i=1}^n AP_i$	where n represents the total number of keypoint categories (e.g., shoulders, elbows, knees), and AP_i denotes the Average Precision (AP) for the i -th keypoint category.

To assess the performance of the model in human keypoint detection tasks, we employ the Object Keypoint Similarity (OKS) as the evaluation metric. OKS is specifically designed for keypoint detection and quantifies the similarity between predicted keypoints and ground truth annotations. Analogous to the Intersection over Union (IoU) metric in object detection, OKS evaluates model accuracy by computing the Euclidean distance between corresponding true and predicted keypoints. A higher OKS score indicates that the detected keypoints are more closely aligned with the ground truth, reflecting improved detection accuracy.

The OKS calculation considers not only the spatial distance between detected and ground truth keypoints but also incorporates additional factors to enhance evaluation precision. First, OKS introduces a visibility flag (v_i) to exclude occluded or invisible keypoints, ensuring a fair assessment by considering only visible keypoints in the computation. This approach effectively mitigates errors caused by occlusion and other visual obstructions. Furthermore, OKS accounts for the area occupied by the detected human body in the image (s), normalizing the results to maintain consistency across varying scenarios. The OKS formulation is given as follows (refer to Equation 18):

$$OKS = \frac{\sum_i \exp\left(-\frac{d_i^2}{2s^2k_i^2}\right) \delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \quad (18)$$

where:

i : Index of the annotated keypoint.

d_i : Euclidean separation between the i -th ground truth keypoint and the detected keypoint.

s : Area occupied by the detected person in the image.

k_i : Decay constant used to control keypoint category i .

v_i : Visibility flag for the i -th ground truth keypoint, where 0 denotes no label, 1 denotes an obscured label, and 2 denotes a visible label.

$\delta(v_i > 0)$: Indicator function that only determines the OKS value for keypoints that are visible in the ground truth annotations. It equals 1 if the i -th keypoint is visible, and 0 otherwise.

TP: Correctly detected and localized human keypoints accurately

identified, such as successfully recognizing and locating body joints, with the predicted positions closely matching the ground truth.

FP: Incorrectly detected human keypoints when the model falsely identifies a keypoint that does not exist in the image or misclassifies a nonexistent body part (e.g., mistakenly detecting a shoulder keypoint where none is present).

FN: Keypoints that should have been detected but were missed, referring to instances where the model fails to identify an actual keypoint in the image or does not recognize a visible keypoint (e.g., failing to detect a clearly visible knee position).

mAP: The mAP is a fundamental evaluation metric for human keypoint detection, assessing a model's ability to accurately localize keypoints across multiple anatomical landmarks (e.g., shoulders, elbows, knees). The mAP is computed by determining the Average Precision (AP) for each keypoint category and averaging these values across all keypoints. AP is derived from the area under the Precision-Recall (PR) curve, which quantifies the trade-off between precision and recall at varying confidence thresholds.

mAP@0.5: Computes AP at $\text{OKS} \geq 0.5$, where a keypoint prediction is considered correct if its OKS exceeds 0.5. The final mAP is obtained by averaging AP across all keypoints.

mAP@0.5:0.95: Evaluates model robustness by computing AP at multiple OKS thresholds (0.5, 0.55, 0.6, ..., 0.95), reflecting performance across a range of localization tolerances. The final mAP is derived by averaging AP values across all keypoints and OKS thresholds.

Higher AP values indicate superior detection performance for individual keypoints, demonstrating the model’s capability to accurately localize keypoints while minimizing false positives and false negatives. Consequently, mAP serves as a comprehensive measure of overall model effectiveness, with higher values signifying improved consistency and accuracy in keypoint predictions across different categories and spatial scales.

In summary, this study provides a comprehensive and systematic evaluation of the performance of the model in human keypoint detection tasks using OKS, precision, recall, AP, and mAP. These metrics assess the detection accuracy and robustness of the model across scenarios.

5.2.2.2 Training and validation

In the construction worker keypoint detection task, a series of hyperparameter configurations (Table 18) were employed during the model training phase to achieve efficient and accurate posture and keypoint detection. These configurations were selected to balance training efficiency, model performance, and generalizability.

Table 18. Hyperparameters.

Category	Parameter Name	Value
Training Settings	Epochs (epochs)	90
	Batch Size (batch)	16
	Input Image Size (imgsz)	640
Learning Rate	Initial (l_{r0})	0.01
	Final (l_{rf})	0.01
	Momentum (momentum)	0.937
	Weight Decay (weight_decay)	0.0005
Warm-up Settings	Epochs (warmup_epochs)	3.0
	Momentum (warmup_momentum)	0.8
	Bias LR (warmup_bias_lr)	0.1
Loss Function	Box Loss Weight (box)	7.5
	Class Loss Weight (cls)	0.5
	Depthwise Convolution Loss (dfl)	1.5
	Pose Loss Weight (pose)	12.0
	Label Smoothing (label_smoothing)	0.0
	Key Object Weight (kobj)	1.0
Data Augmentation	Hue (hsv_h)	0.015
	Saturation (hsv_s)	0.7
	Brightness (hsv_v)	0.4
	Translation (translate)	0.1
	Scaling (scale)	0.5
	Horizontal Flip (fliplr)	0.5
	Mosaic Augmentation (mosaic)	1.0
	Auto Augmentation (auto_augment)	randaugment
	Random Erasing (erasing)	0.4
	Crop Fraction (crop_fraction)	1.0
Evaluation Settings	IoU Threshold (iou)	0.7
	Maximum Detections (max_{det})	300

First, the number of training epochs was set to 90, with a batch size of 16 and an input image size of 640, ensuring stable training while accommodating large-scale construction site data. The number of

worker threads was set to eight to enhance computational efficiency during training. A pretrained model was used (`pretrained = true`), and the optimizer was set to `auto` to facilitate rapid convergence.

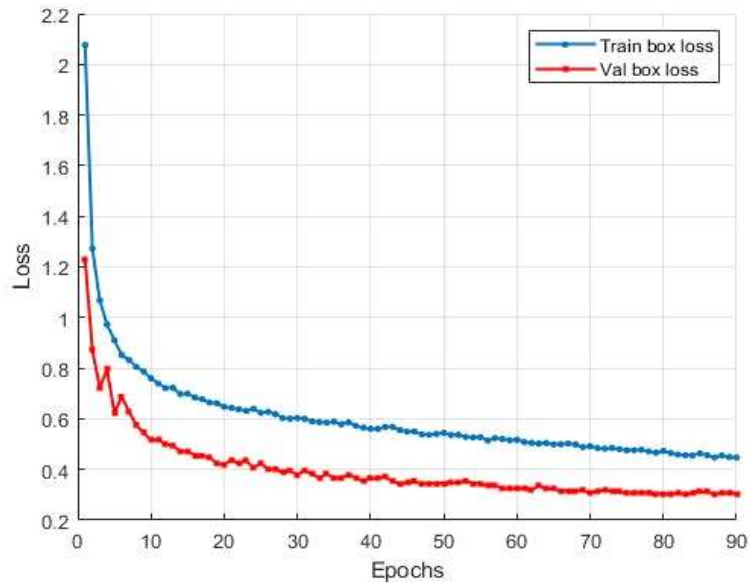
Regarding learning rate-related hyperparameters, the initial learning rate (l_{r0}) and final learning rate (l_{rf}) were set to 0.01. The optimizer momentum was configured to 0.937, whereas the weight decay was set to 0.0005. These configurations help mitigate overfitting while ensuring stable gradient updates during training. A warm-up phase was introduced to enhance training stability, with warm-up epochs set to 3.0. During this phase, the warm-up momentum (0.8) and bias learning rate (`warmup_bias_lr=0.1`) were adjusted to reduce instability at the initial training stage.

The loss function configuration was designed to balance multiple tasks in the object detection framework. The box loss weight (`box = 7.5`), classification loss weight (`cls = 0.5`), and depth-separable convolutional loss weight (`dfc = 1.5`) collectively contributed to improving the detection precision. The pose loss weight (12.0) strongly emphasized posture information in keypoint detection, ensuring accurate worker posture recognition. Label smoothing (0.0) and critical object weight (`kobj = 1.0`) were applied to enhance model robustness, preventing overfitting.

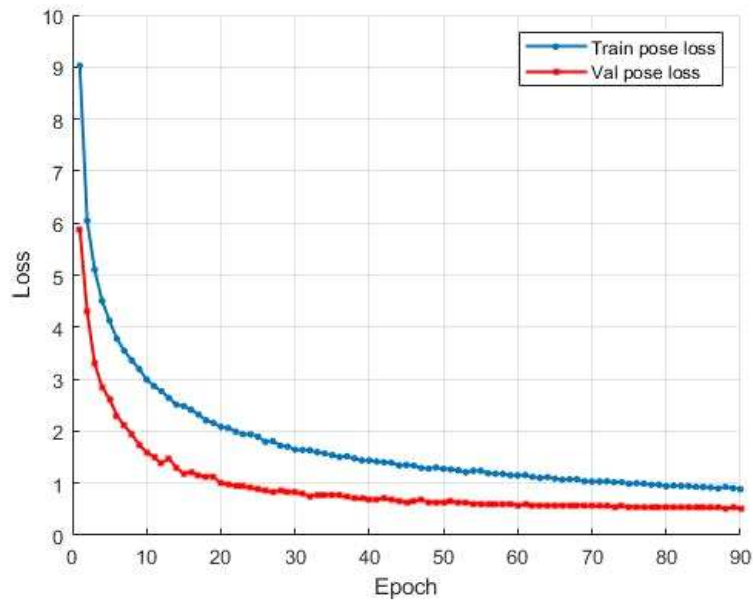
For data augmentation, techniques were implemented to improve the generalizability of the model. Color-based augmentations, including hue ($hsv_h = 0.015$), saturation ($hsv_s = 0.7$), and brightness adjustments ($hsv_v = 0.4$), enhanced the adaptability of the model to diverse lighting conditions. Geometric transformations, such as translation (`translate =`

0.1), scaling (scale = 0.5), and horizontal flipping (fliplr = 0.5), improved the robustness of the model to worker posture variations. Mosaic augmentation (mosaic = 1.0) was employed to increase the diversity of the training data, enhancing model performance in complex construction site environments.

Although MixUp (0.0) and copy-paste (0.0) were not enabled, their omission helped avoid the interference of pseudo-labels, ensuring stable training. The model also employed automatic data augmentation (auto_augment = randaugment) and such techniques as random erasing (0.4) and cropping (crop_fraction = 1.0) to enhance the diversity of training samples. During the evaluation, an IoU threshold (0.7) was applied to maintain high detection precision, whereas the maximum detection count ($\max_{\text{det}} = 300$) allowed the model to handle the complexity of construction site scenes effectively.



(a) Box loss



(b) Pose loss

Figure 31. Model loss curve: (a) Box loss; (b) Pose loss.

Figure 31 presents the training results for the worker keypoint recognition task, illustrating the changes and convergence of the performance over 90 training epochs. Figure 31(a) presents the loss curve for bounding-box predictions (box loss), depicting the error trend in localizing the bounding boxes around keypoints. As training progresses, the model progressively improves its ability to identify the regions corresponding to worker keypoints accurately. Figure 31(b) presents the loss curve for posture estimation (posture loss), reflecting the convergence process of keypoint recognition and localization errors. This finding demonstrates the ability of the model to optimize posture detection continuously.

As the number of training epochs increases, the model improves its feature extraction and keypoint prediction capabilities, as indicated by the gradual decline in loss values, which eventually stabilize. By the 85th epoch, all loss curves reach their lowest values and display a stable convergence trend. This finding suggests that the model has learned the underlying features of the training data, with minimal performance improvement observed in subsequent epochs. The convergence of the loss curves indicates the stability and robustness of the model in keypoint detection tasks.

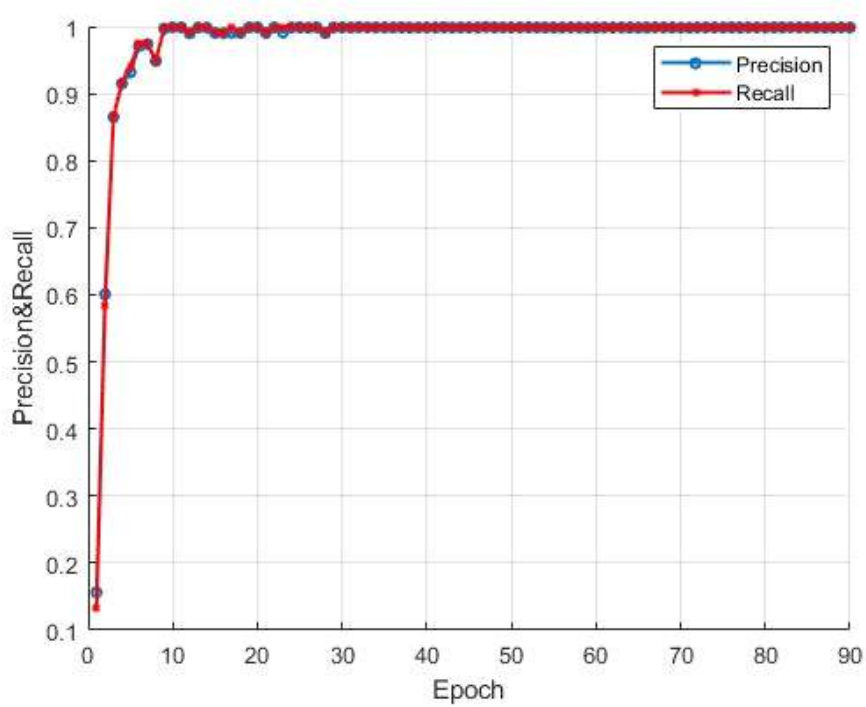


Figure 32. Model precision and recall curve.

Figure 32 illustrates the variations in precision and recall values across training epochs for the worker keypoint recognition task. In the initial training phase, the predictive accuracy is relatively low because it has yet to capture the critical features of worker postures fully. This phenomenon is common in early-stage model training, particularly in complex construction site environments characterized by diverse worker poses and occlusions. Hence, precision and recall remain low in the first few epochs, with the model struggling to detect worker keypoints effectively.

As training progresses, the model incrementally learns patterns related to worker postures and movements. The model gradually refines its understanding of the keypoint locations and posture angles

via backpropagation and parameter updates, leading to noticeable improvements in precision and recall. By epoch 10, a significant performance gain is observed, with precision and recall rising rapidly. This trend suggests that the model has captured the underlying structure of worker movements, resulting in enhanced keypoint prediction accuracy.

In later stages, precision and recall stabilize, demonstrating that the model has balanced accuracy and generalizability, reducing overfitting to specific postures or actions. Furthermore, the maximization of the area under the curve (AUC) highlights the robust performance across training epochs. The increasing AUC values suggest that the model maintains an effective trade-off between precision and recall, ensuring high detection accuracy while capturing a broad range of worker postures.

Moreover, the maximization of the AUC highlights the robust performance throughout the training process. The increase in AUC indicates that the model balances precision and recall at each training stage, ensuring high accuracy while covering more true targets. This result is vital for the worker keypoint recognition task, where the construction site environment is complex, and worker postures and movements vary substantially. Ensuring high accuracy and comprehensive coverage of worker postures is crucial for improving construction site safety and work efficiency.

Furthermore, the peak AUC value represents the optimization of model performance across thresholds, fully demonstrating its robustness and flexibility in posture detection. In the following training

stages, the model continues to learn efficiently, gradually reducing FNs and FPs, indicating that it identifies worker keypoints and maintains efficient detection capabilities when handling various postures.

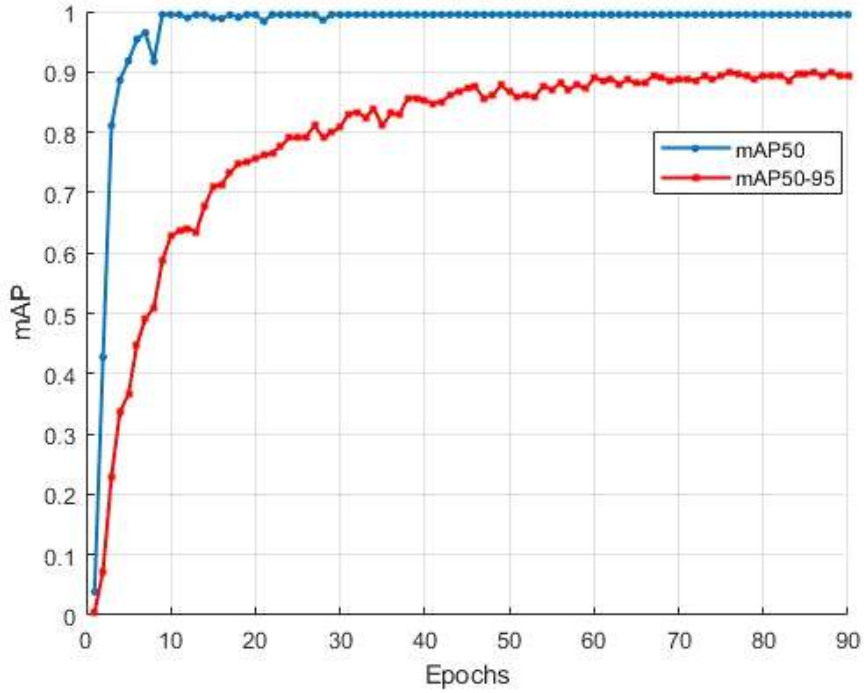


Figure 33. Model mean average precision (mAP) curve.

In keypoint detection tasks, two important metrics commonly applied to evaluate model performance are $\text{mAP}@50$ and $\text{mAP}@50-95$. First, $\text{mAP}@50$ represents the accuracy when the OKS exceeds 0.5, reflecting the performance under a higher precision threshold. This metric is valuable because it indicates whether the model can accurately detect worker keypoints with high precision, ensuring its ability to meet stringent requirements in practical applications. In contrast, $\text{mAP}@50-95$ evaluates the average detection accuracy of keypoints across multiple thresholds, from 0.5 to 0.95 in increments of 0.05. This metric provides a more thorough assessment of the

performance under various tolerance levels, highlighting its adaptability and robustness in real-world scenarios.

The curves in Figure 33 demonstrate a consistent improvement in mAP@50 and mAP@50-95 during the first 60 training epochs. During this phase, the model progressively learned to detect worker keypoints with increasing accuracy via continuous parameter optimization and learning strategy refinement. In the earlier stages, as the number of training epochs increased, the feature extraction capabilities of the model were enhanced, enabling it to identify critical patterns in complex worker postures and movements, improving detection accuracy. Notably, the mAP@50 curve reveals a rapid increase, approaching a high value near 0.99, indicating that the model effectively identifies worker keypoints under strict precision requirements and demonstrates robust accuracy. The mAP@50-95 curve exhibits a similar upward trend, although slightly lower than that of mAP@50, reaching approximately 0.90. This result suggests that, although the model maintains high detection accuracy under higher OKS thresholds, its performance varies across thresholds, reflecting potential instability at looser thresholds, such as when slight keypoint deviations or partial occlusion occur, slightly decreasing detection accuracy. Nevertheless, the mAP@50-95 value of around 0.90 still demonstrates the overall stability and balanced performance of the model.

Following the 60th training epoch, performance began to stabilize, and the growth rates of mAP@50 and mAP@50-95 stabilized. This plateau typically indicates that the model is nearing its optimal

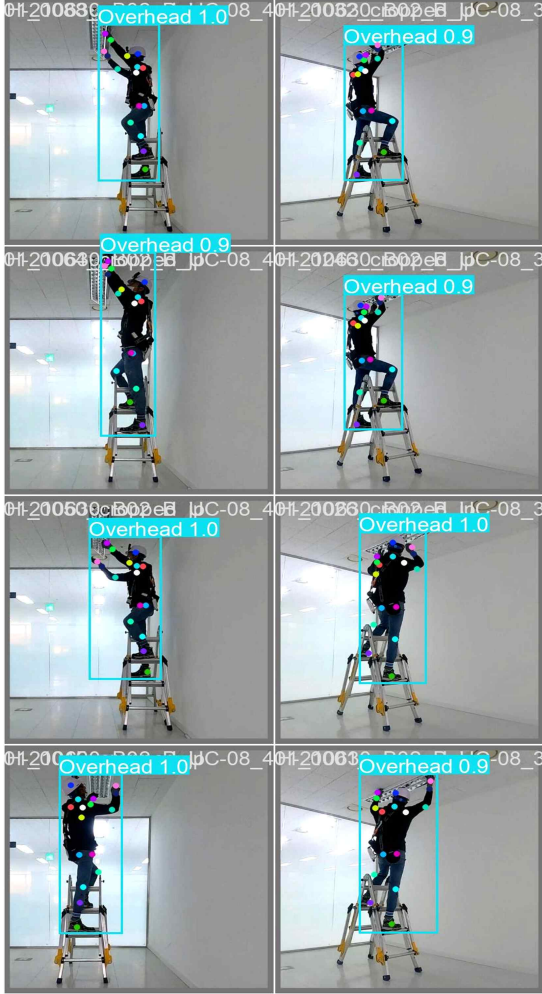
performance, with further training likely to yield diminishing returns. The stability observed during this phase underscores the robustness and generalizability of the model, suggesting that it can maintain high recognition accuracy across diverse worker postures and movements while adapting to the dynamic conditions of construction site environments.

In conclusion, the box and posture losses gradually decreased and stabilized after the 85th epoch, indicating that the model achieved high accuracy and convergence in keypoint localization and posture estimation. The mAP@50 value approached 0.99, demonstrating the ability to detect worker keypoints efficiently under strict precision requirements, whereas mAP@50-95 remained around 0.90, reflecting the stability and reliability of the model across thresholds. Precision was relatively low in the early training stages but steadily improved and eventually stabilized, revealing the effectiveness of the model in reducing FPs while maintaining high accuracy in keypoint recognition. Initially low, the recall rapidly increased as the model continued to learn and stabilize, ensuring the model covered truer targets and enhanced the worker posture detection.

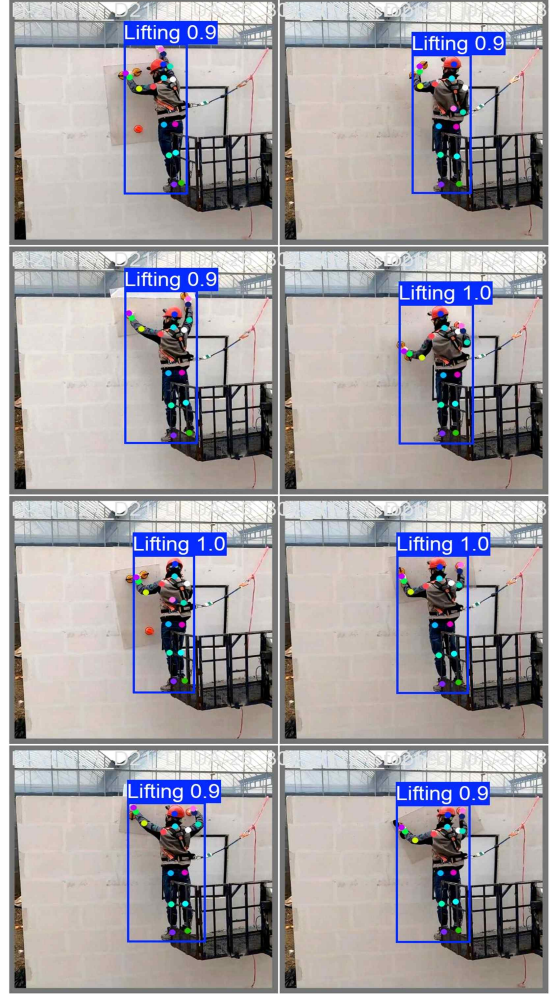
5.2.2.3 Visualization

Figure 34 visualizes the results of construction worker posture detection and human keypoint estimation based on the improved YOLOv8 algorithm. The left section illustrates workers performing overhead tasks on ladders, whereas the right section depicts workers engaged in lifting tasks using aerial work platforms. Each detected

worker is enclosed in a blue bounding box, with the corresponding task category and confidence score (e.g., 1.0 or 0.9) displayed at the top of the box. Human keypoints, including joints, arms, and legs, are marked with colored dots, demonstrating the ability of the model to recognize body parts accurately.



(a) Overhead



(b) Lifting

Figure 34. Results visualization: (a) Overhead; (b) Lifting.

From the visualization, this method achieves precise detection and localization of critical body joints across work scenarios, covering the upper limbs, lower limbs, and torso. Such high-precision keypoint detection enhances the accuracy of task classification and provides a reliable data foundation for joint angle calculations and potential posture anomaly identification. In construction environments, accurate joint localization plays a significant role in evaluating the compliance and safety of worker operations.

5.2.2.4 Ablation study

This study systematically improved the YOLOv8 model, significantly enhancing critical performance metrics, including the parameter count, inference speed, computational efficiency, and accuracy (Table 19).

Table 19. Ablation test results.

Model	Params /M	GFLOPS /g	Speed /ms	Model Layers
Yolov8	3.27	9.2	0.6	250
Yolov8+ BottleneckCSP + GhostConv	3.03	8.8	0.4	290
Model	Box		Pose	
	mAP0.5 /%	mAP0.5-0.95 /%	mAP0.5 /%	mAP0.5-0.95 /%
Yolov8	99.5	96.3	99.5	88.0
Yolov8+ BottleneckCSP + GhostConv	99.5	96.9	99.5	90.9

This paper proposes an improved YOLOv8 object detection model that preserves high detection accuracy while significantly reducing

model size and computational overhead. Experimental results show that the model achieves a 7.3% reduction in parameter count, effectively lowering dependence on storage and computational resources. This is particularly beneficial for deployment on memory- and computation-constrained terminal devices such as mobile platforms and edge computing nodes. The reduction in parameters enhances execution efficiency during both training and inference, while also reducing runtime memory usage and energy consumption. Furthermore, the model achieves a 33.33% increase in inference speed and a 4.35% reduction in floating-point operations, which accelerates inference and further minimizes energy usage. These optimization strategies greatly improve the model's adaptability for deployment in energy-sensitive environments. Despite its lightweight architecture, the model consistently achieves higher mAP scores on multi-class detection tasks, demonstrating the effectiveness of the proposed optimization strategy in balancing accuracy and efficiency.

Construction sites—characterized by high dynamism and uncertainty—place stringent demands on the real-time performance, robustness, and deployment flexibility of detection models. Traditional deep learning models, reliant on high-performance hardware, often fall short under low-power, low-bandwidth, and long-duration operation constraints typical of real-world construction scenarios. In contrast, the improved YOLOv8 model demonstrates excellent potential for edge-side deployment due to its high precision, low computational load, and efficient runtime performance. In edge computing scenarios such as embedded cameras, smart safety helmets, and mobile

inspection platforms, the model's lightweight structure enhances environmental adaptability and system integration, improving real-time responsiveness and maintainability in intelligent safety monitoring systems.

From a generalization and scalability perspective, the proposed optimization strategy is not tailored to specific tasks or scenarios. Instead, it adopts a structurally generalized design that systematically enhances inference efficiency and transferability. As a result, the model is broadly applicable across diverse construction activities, including high-altitude work, material handling, and scaffold operations. Even in the absence of large volumes of scene-specific training data, the model maintains strong recognition accuracy and robustness, showcasing its cross-task generalization capabilities.

Overall, the proposed improvements to YOLOv8 strike an effective balance between accuracy and efficiency, offering a scalable and energy-efficient solution for intelligent monitoring and worker behavior analysis in resource-constrained and dynamically changing environments such as construction sites.

5.3 Joint angle analysis and risk assessment

Based on the improved YOLOv8 model, this research accurately detected critical points of construction workers during their tasks. Using the detected critical point coordinates, joint angles were calculated using the vector dot product method, providing foundational support for the quantitative analysis of work postures.

This section focuses on the results of joint angle calculations for

two typical tasks—lifting and overhead work—to further explore posture characteristics and potential risks in working scenarios.

5.3.1 Calculation of joint angles

This study reveals potentially hazardous postures associated with work postures via the systematic calculation and visualization of joint angles.

In lifting tasks, excessive flexion of the knee and trunk joints may cause muscle fatigue or structural damage to the lower back, particularly when improper lifting techniques are employed. In overhead tasks, fluctuations in the angles of the shoulder and elbow joints may impose excessive stress on the shoulder muscles and joints, increasing the risk of strain and injury. For each detected keypoint set, this study employs the vector dot product method to compute the body joint angles, facilitating a quantitative analysis of worker postures.

A. Joint angle calculation process

Keypoint pairing: Based on the human anatomical structure and biomechanical characteristics of body movement, the detected keypoints are logically grouped into triplets for angle calculation. For example, for upper limb movement, the shoulder, elbow, and wrist are selected to compute the elbow joint angle, and for lower limb movement, the hip, knee, and ankle serve as the basis for knee joint angle calculation.

Angle calculation: The vector dot product method is employed to compute the angle for each triplet. This calculation method is derived

from a geometric formula that determines angles based on three points, as detailed in Section 4.4 (refer to Equation 19):

$$\theta = \arccos\left(\frac{\vec{v_1} \cdot \vec{v_2}}{\|\vec{v_1}\| \|\vec{v_2}\|}\right) \quad (19)$$

where v_1 and v_2 are vectors derived from keypoint triplets, $\|v\|$ denotes the magnitude of a vector, and $v_1 \cdot v_2$ represents the dot product of the two vectors.

Batch calculation: This study automates the calculation process through programming to enhance efficiency and ensure consistency. This calculation facilitating the extraction of keypoints and the computation of joint angles across all frames in the task images.

B. Results analysis and visualization

This study visualizes and analyzes the joint angle calculation results for two typical work scenarios—lifting and overhead tasks. This process captures the dynamic variations in joint angles during tasks and enables the quantitative identification of potentially hazardous postures.

1. Lifting task: In the lifting task, this study focuses on the knee, trunk, and neck joints for calculation and analysis. During the process of lifting or lowering heavy objects, the angles of the knee and trunk undergo significant variations. The specific analysis includes the following:

- Neck joint: Variations in the neck angle indicate potential instability in head posture during lifting, potentially increasing the load on the cervical spine and elevating the risk of musculoskeletal strain.
- Knee joint: Substantial changes in the knee angle suggest that the

lower limbs bear considerable dynamic loads during lifting, potentially exerting excessive stress on the soft tissue and ligaments of the knee joint.

- Trunk joint: Excessive trunk joint angle deviations imply that improper lifting techniques may contribute to muscle fatigue or increase the risk of lower back injuries.

2. Overhead task: In the overhead task, this study calculates and analyzes the shoulder, elbow, and wrist joints. As the arms are raised and held overhead, the shoulder angle undergoes significant changes, indicating increased pressure on the upper limb joints. The specific analysis includes the following:

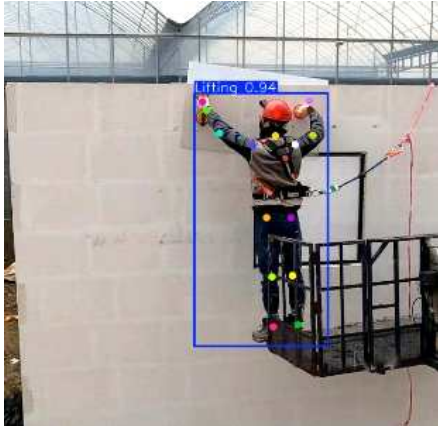
- Wrist joint: Variations in the wrist angle reveal the potential overextension of the hand during overhead tasks, possibly elevating the risk of wrist strain or injury.

- Shoulder joint: A continuous increase in shoulder angle suggests that the shoulder muscles and joints endure a considerable load when the arms are elevated. Prolonged maintenance of this posture may cause muscle tension, fatigue, or strain.

- Elbow joint: Fluctuations in the elbow angle reflect the stability of the arms throughout task stages. Abnormal variations in elbow angle may indicate posture misalignment, potentially increasing the risk of joint stress or inefficient task execution.

Table 20 lists the results of joint angle calculations and their corresponding visualizations.

Table 20. Joint angle calculation and visualization.

Worker task types	Joint name	Joint degree	Visualization
 Lifting	Neck joint	3.52°	
	knee joint	174.81°	
	Trunk joint	8.35°	
 Overhead	Wrist joint	175.37°	
	Shoulder joint	33.62°	
	Elbow joint	139.78°	

In this study, an improved YOLOv8 model was utilized to accurately recognize keypoints of construction workers, enabling precise calculation of joint angles associated with two typical working postures. For the lifting posture, the angles of the neck, knee, and trunk were analyzed, while for the overhead posture, the wrist, shoulder, and elbow joints were examined. Additionally, visualizations of the angle variations were generated to intuitively support the analysis of posture characteristics.

Excessive flexion of the knee and trunk during lifting tasks can lead to prolonged tension in the back muscles, potentially causing muscle strain or structural injuries. Improper lifting techniques—particularly those involving excessive bending of the trunk or knees—can significantly increase spinal load. Maintaining such postures over time is likely to cause chronic lower back pain, acute injuries, or even degenerative spinal conditions. During overhead tasks, workers are required to keep their arms raised for extended periods, leading to frequent changes in the angles of the shoulder and elbow joints. This results in sustained high loads on the shoulder muscles and joints, significantly increasing the risk of muscle fatigue and joint degeneration, which may eventually develop into occupational musculoskeletal disorders such as tendinitis or frozen shoulder. Through the systematic calculation and visualization of key joint angles presented in this study, high-risk postures can be more intuitively identified, providing a scientific foundation for posture analysis, optimization, and the development of effective intervention strategies.

However, relying solely on joint angle analysis is insufficient to

comprehensively assess posture-related risk levels. Therefore, it is necessary to incorporate standardized ergonomic assessment tools such as RULA (Rapid Upper Limb Assessment), REBA (Rapid Entire Body Assessment), and OWAS (Ovako Working Posture Analysis System). Integrating these methods enables a more scientific and systematic quantification and classification of posture-related risks. This, in turn, provides practical guidance for on-site managers to implement targeted posture improvement measures, thereby enhancing occupational safety and reducing the incidence of musculoskeletal disorders.

5.3.2 Assessment of worker risk levels

Based on the posture risk assessment decision tree developed in this study (Figure 35), researchers can select appropriate evaluation methods according to the characteristics of each task after completing keypoint detection and joint angle calculation. The corresponding angle data are then input into the selected assessment model to systematically identify, analyze, and categorize posture-related risks. In selecting the appropriate evaluation method, this study adheres to the standards recognized by labor research institutes. The specific steps are as follows:

1. Input Data:

Joint angle data: obtained from the preceding stage of keypoint detection and joint angle calculation analysis.

Force-related data: which can be collected using appropriate sensors (e.g., force plates, wearable pressure sensors, or EMG

devices).

2. Activity Duration Determination (≥ 30 Seconds):

If the activity duration is less than 30 seconds, the task is classified as a non-activity, and alternative methods (e.g., OWAS) are recommended for assessment.

If the activity lasts 30 seconds or longer, the process proceeds to the next decision point.

3. Involvement of Major Upper Limb Joints (Head/Shoulder/Elbow):

If these joints are not involved, the task is again categorized as a non-activity and labeled as N/A (Not Applicable/Unevaluable).

If these joints are involved, the process advances to the next step.

4. Involvement of the Knee Joint:

If knee activity is present, the task is classified as a full-body activity, and the appropriate assessment method is REBA (Rapid Entire Body Assessment).

If knee involvement is absent, the task is categorized as an upper body activity, and RULA (Rapid Upper Limb Assessment) is the recommended method.

In addition, when neither RULA nor REBA is suitable—particularly in cases involving static or low-movement tasks—OWAS (Ovako Working Posture Analysis System) can be applied. Tasks that cannot be categorized under any of the available evaluation methods are designated as “unevaluable”, indicating the need for alternative or supplementary assessment approaches.

The applicable evaluation methods are described in detail below:

- **RULA:** RULA is suitable for evaluating full-body activities,

particularly tasks involving repetitive or non-repetitive movements of both the upper and lower limbs. It focuses on analyzing the postures, force, and muscle use of the neck, trunk, shoulders, upper arms, forearms, and wrists. RULA is commonly used for short-term observations and is ideal for dynamic work environments such as office tasks, assembly lines, and caregiving. Its main advantages include simplicity, quick application, and the ability to promptly identify high-risk tasks on-site. However, it may be less effective for continuous assessments of highly dynamic jobs.

- **REBA:** REBA is primarily designed for tasks that emphasize upper limb activity, particularly those requiring frequent bending, stretching, twisting, or other joint movements. It provides a comprehensive evaluation of the entire body—including the neck, trunk, legs, and upper limbs—while also considering load handling, movement frequency, and applied force. REBA is well-suited for complex work scenarios in sectors such as healthcare, construction, and food processing. By integrating multiple risk factors, it offers a more adaptable and robust tool for assessing moderately complex tasks.

- **OWAS:** When neither RULA nor REBA is applicable, OWAS is recommended for analyzing static or low-movement tasks, especially those involving minimal body motion sustained over extended periods. This method classifies body postures and combines them with load factors to determine risk levels. OWAS is suitable for scenarios such as machinery monitoring, prolonged standing, or low-frequency manual handling, commonly found in industrial and manufacturing settings. Although effective for evaluating static work, OWAS has limited

sensitivity to short-duration or high-intensity dynamic activities.

- **Unevaluable cases:** In certain situations where the characteristics of a task—such as its brevity, complexity, or high degree of freedom—do not meet the criteria of any existing evaluation method, the data are categorized as "unevaluable."

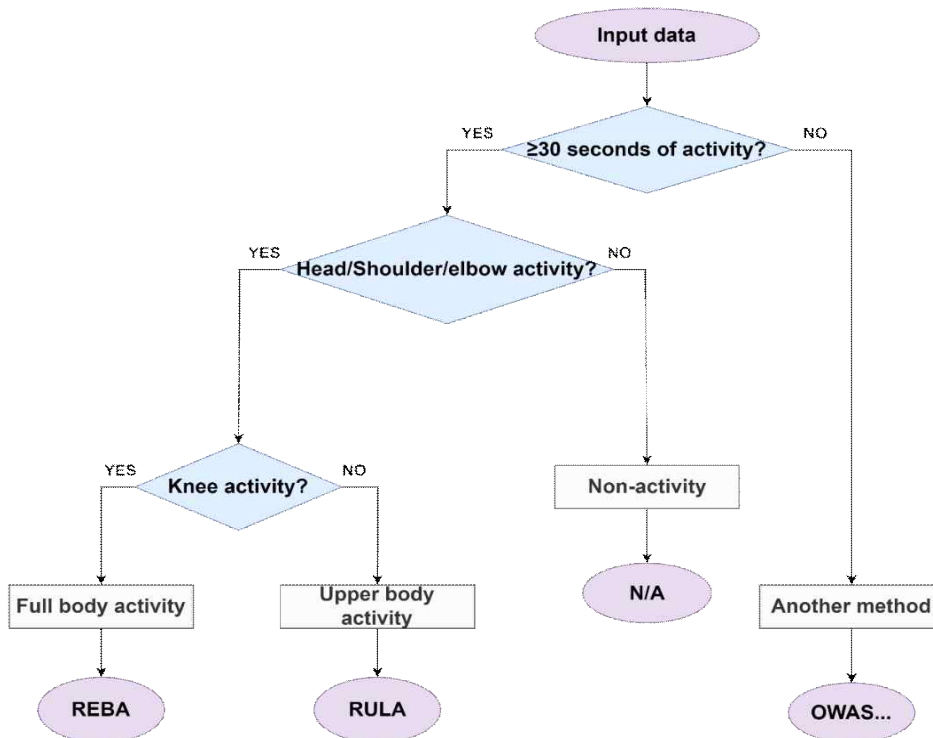


Figure 35. Risk level assessment process.

Once the evaluation method is established, the joint angle data obtained in earlier stages can be fed into the ergonomic assessment module for scoring. Based on these results, representative high-risk postures in construction tasks can be identified, enabling the implementation of targeted intervention strategies. This systematic approach to posture recognition and risk evaluation offers practical

value in multiple dimensions.

At the individual level, the system enables real-time monitoring and feedback on high-risk postures, prompting workers to adjust their body positions during tasks. Such proactive adjustments help mitigate the risk of work-related musculoskeletal disorders, including lower back strain, neck and shoulder fatigue, and joint degeneration. Long-term use of the system is expected to enhance workers' physical comfort and reduce physiological strain.

At the management level, the quantitative data generated by the system supports evidence-based decision-making on construction sites. Managers can utilize posture risk ratings to optimize task allocation and workflow arrangements, implement more rational shift schedules, and conduct targeted ergonomic training—thereby improving overall operational efficiency and site safety.

Additionally, system data can inform decisions on equipment placement and workspace layout, such as tool positioning, workstation height, and spatial organization, to reduce the prevalence of high-risk postures and enhance the sustainability and comfort of manual operations.

In summary, the proposed ergonomic evaluation method—based on joint angle recognition and risk scoring—offers practical support for individual health monitoring, task process optimization, and ergonomic workspace design. With continued operation and data accumulation, the system holds significant potential for long-term health trend analysis, facilitating personalized interventions and macro-level safety management within the construction industry.

Chapter 6. Conclusion and future work

6.1 Conclusion

This study addresses critical challenges in construction site safety detection, including dataset limitations, the reliance on a single approach for posture estimation, the lack of diversity in risk assessment methods, and the narrow focus on worker PPE detection. An optimized deep learning detection framework is proposed to overcome these limitations. By integrating C3CBAM, C3-DCN, and BiFPN modules, along with the SLoU loss function, the performance and robustness of the improved PPE detection model based on YOLOv5 were significantly enhanced for construction worker PPE detection tasks. Furthermore, improvements to the YOLOv8 model facilitated efficient keypoint detection, joint angle calculation, and risk assessment, providing strong technical support for construction site monitoring and worker safety management. The primary contributions of this research are summarized below.

A.Generalization capability of the dataset and improvement in PPE detection

This study utilizes a real-world construction site dataset provided by the AI-Hub platform, with a strong emphasis on authenticity and diversity in dataset selection. The dataset covers various typical construction scenarios, including ladder work, scaffolding work, and aerial platform operations. Compared to small-scale datasets collected in controlled laboratory environments, the AI-Hub dataset more accurately reflects the complexity of real construction sites, enhancing

both environmental and object diversity. Specifically, the AI-Hub dataset was collected from diverse construction sites under various challenging conditions, such as occlusion, non-standard lighting, complex backgrounds, and different types of site facilities. This greatly improves the generalization capability of the model by providing training data that closely resembles real-world environments.

The optimized object detection model demonstrated significant performance improvements on this dataset, particularly for the PPE detection task. The model achieved mAP@50 and mAP@50-95 scores of 0.98 and 0.772, respectively. These results indicate high-precision multi-task detection, with notable advantages in detecting small objects such as gloves, helmets, and safety shoes. Furthermore, confusion matrix analysis revealed a substantial reduction in misclassification, especially in distinguishing small PPE objects from the background. Even under adverse conditions such as occlusion and variable lighting, the model maintained high detection accuracy, demonstrating strong applicability in complex construction environments. These experimental results validate the reliability of the optimized PPE detection model for practical applications, particularly in providing accurate safety monitoring in real-world construction sites.

B. Innovation in human keypoint estimation and risk level assessment

Human keypoint detection and risk assessment represent another innovative aspect of this study. By optimizing the YOLOv8 model with efficient feature extraction modules, attention mechanisms, and a lightweight network structure, the computational complexity of the model was significantly reduced, resulting in a substantial improvement

in inference speed. In the keypoint detection task, the optimized YOLOv8 model achieved outstanding performance in recognizing construction worker postures, with mAP@50 and mAP@50-95 scores reaching 99.5% and 90.9%, respectively. The model effectively identified keypoints associated with worker movements even in complex construction environments such as ladder work, scaffolding work, and aerial platform operations, enabling more precise posture analysis.

This study also introduces a novel risk assessment method based on joint angle variations. By analyzing deviations in joint angles, this method quantitatively evaluates workers' postures and identifies potentially hazardous positions, such as excessive bending or twisting, which are often associated with a high risk of occupational injuries. Unlike traditional single-class classification methods, this approach offers multiple evaluation criteria and can flexibly adapt to different work scenarios and worker needs. For instance, in ladder operations, changes in joint angles can indicate poor climbing posture, highlighting potential safety risks. Based on such analysis, safety managers can provide targeted training to workers, thereby reducing the occurrence of unsafe behaviors.

Ablation studies further confirmed the robustness and accuracy of the optimized model in complex construction site environments. The model maintained stable detection performance across different working conditions and postural variations, and was capable of providing real-time risk assessments. This is of great significance for practical deployment in construction sites.

C. Practical applications

This study demonstrates significant application value in the field of construction site safety management, particularly through high-precision PPE detection and human keypoint-based posture recognition. The proposed method provides effective data support and technical means for safety evaluation and management on construction sites.

Firstly, the developed PPE detection model is trained on real-world construction site data collected from the AI-Hub platform. It effectively handles challenging conditions such as uneven lighting, complex backgrounds, and object occlusion, accurately identifying whether workers are equipped with essential safety gear including helmets, gloves, and safety shoes. The method can be deployed as a periodic offline analysis tool on the backend of construction monitoring systems, enabling scheduled processing of collected video footage. This allows for timely detection of PPE violations, assisting safety personnel in conducting more precise inspections and enhancing the efficiency and targeting of safety supervision.

Secondly, in terms of human keypoint detection and posture assessment, the optimized YOLOv8 model integrates efficient feature extraction modules and attention mechanisms, significantly improving the accuracy of keypoint recognition. The method analyzes key worker movements during construction tasks, identifying hazardous postures—such as excessive bending or joint twisting—by monitoring joint angle variations. Compared to traditional point-based monitoring approaches, this posture-based risk assessment method offers more dynamic and

detailed insights, supporting incident tracing, safety data archiving, and behavioral optimization. Additionally, safety managers can periodically analyze site surveillance data using the model to assess PPE compliance and posture-related risk distributions, thereby identifying high-risk operations or frequent violations and adjusting safety supervision priorities or training programs accordingly.

Furthermore, the model is designed with a balance between accuracy and computational efficiency. By incorporating a lightweight network structure and reducing computational complexity, the system exhibits high deployment flexibility and operational efficiency. It can function effectively in resource-constrained environments without relying on high-performance servers or cloud-based infrastructure. The model supports inference on edge devices such as embedded GPUs or industrial camera processing units. This feature is particularly critical in the construction industry, where many projects are conducted in remote areas with limited infrastructure. The lightweight design ensures suitability for low-power, mobile scenarios, significantly expanding the feasibility of on-site deployment.

Meanwhile, the method's multi-task detection capabilities provide valuable, data-driven insights for managerial decision-making. By analyzing PPE compliance and posture risk frequency across different worker roles, tasks, and time periods, managers can identify high-risk activities and prevalent unsafe behaviors. This information can then be used to optimize construction workflows, reallocate resources, and improve the cost-effectiveness of safety investments. Additionally, the method's visual feedback functionality supports worker training and

behavioral correction by illustrating real-world unsafe actions along with quantitative risk explanations, thereby raising workers' safety awareness and self-protection capabilities.

In summary, the proposed PPE detection and posture assessment method achieves high accuracy, strong adaptability, and flexible deployment capability, offering an effective and practical intelligent solution for construction site safety management.

6.2 Future work

Future research will focus on the following directions.

First, to enhance detection accuracy and model generalizability, future efforts should further expand the diversity and scale of the dataset. This includes incorporating a broader range of construction scenarios, work environments, and types of worker activities, as well as accounting for variations in equipment, ergonomic conditions, and regional work practices. Such expansions will ensure the model performs effectively and reliably across diverse real-world settings. By constructing richer and more representative datasets, the robustness and adaptability of the model in complex and dynamic environments can be significantly improved.

Second, current posture detection models primarily concentrate on single-worker behavior analysis; however, actual construction sites typically involve multiple workers operating simultaneously. Therefore, the development of multi-person posture detection methods—particularly those capable of analyzing worker interactions and collaborative activities in complex environments—will be a critical

research direction. Advancements in this area will not only improve monitoring accuracy in multi-worker scenarios but also provide more comprehensive decision-making support for safety management in dynamic and high-density work environments, thereby addressing broader real-world application demands.

Finally, lighting variability remains a significant challenge affecting detection performance. Construction environments often experience rapid changes in lighting, including strong shadows, glare, and reflections, which can severely compromise image quality and model accuracy. As such, future research should prioritize enhancing model robustness and stability under extreme lighting conditions. This includes exploring advanced image enhancement techniques and adaptive model architectures, as well as developing targeted optimization strategies to address specific environmental challenges. These improvements will help ensure consistent performance across various construction site scenarios.

Through these efforts, future research aims to support more efficient, intelligent, and comprehensive safety monitoring and management on construction sites, contributing to the broader goal of intelligent transformation in construction safety practices.

References

1. Zhang, S., Loosemore, M., Sunindijo, R. Y., Galvin, S., Wu, J., & Zhang, S. (2022). Assessing safety risk management performance in Chinese subway construction projects: A multistakeholder perspective. *Journal of Management in Engineering*, 38(4), 05022009.
2. Berglund, L., Johansson, M., Nygren, M., Samuelson, B., Stenberg, M., & Johansson, J. (2021). Occupational accidents in Swedish construction trades. *International Journal of Occupational Safety and Ergonomics*.
3. Guo, S., Zhao, Y., Luoren, Y., Liang, K., & Tang, B. (2022). Knowledge discovery of correlations between unsafe behaviors within construction accidents. *Engineering, Construction and Architectural Management*, 29(4), 1797–1816.
4. Islam, M. R., Nazifa, T. H., Priyanka, H. K., Ahmed, A., & Shahid, S. (2019, July). Identifying factors and mitigation measures of safety practices for sustainable building construction. In *IOP Conference Series: Earth and Environmental Science* (Vol. 294, No. 1, p. 012041). IOP Publishing.
5. Li, L., & Li, Y. (2022). The spatial relationship between CO2 emissions and economic growth in the construction industry: based on the Tapio decoupling model and STIRPAT model. *Sustainability*, 15(1), 528.
6. Zhang, J., Zhang, W., Xu, P., & Chen, N. (2019). Applicability of accident analysis methods to Chinese construction accidents. *Journal of safety research*, 68, 187–196.
7. Lu, Y., Yin, L., Deng, Y., Wu, G., & Li, C. (2023). Using cased based reasoning for automated safety risk management in construction industry.

Safety science, 163, 106113.

8. Xia, X., Xiang, P., Khanmohammadi, S., Gao, T., & Arashpour, M. (2024). Predicting safety accident costs in construction projects using ensemble data-driven models. *Journal of Construction Engineering and Management*, 150(7), 04024054.
9. Yoon, Y. G., Ahn, C. R., Yum, S. G., & Oh, T. K. (2024). Establishment of safety management measures for major construction workers through the association rule mining analysis of the data on construction accidents in Korea. *Buildings*, 14(4), 998.
10. Statistics, U. B. O. L. (2020). Census of fatal occupational injuries (CFOI).
11. Statistics, B. O. L. (2016). Census of fatal occupational injuries. Washington, DC: Bureau of Labor Statistics.
12. Hu, K., Rahmandad, H., Smith-Jackson, T., & Winchester, W. (2011). Factors influencing the risk of falls in the construction industry: a review of the evidence. *Construction Management and Economics*, 29(4), 397–416.
13. OSHA (Occupational Safety and Health Administration). (2015). OSHA commonly used statistics.
14. Khosravi, Y., Asilian-Mahabadi, H., Hajizadeh, E., Hassanzadeh-Rangi, N., Bastani, H., & Behzadan, A. H. (2014). Factors influencing unsafe behaviors and accidents on construction sites: A review. *International journal of occupational safety and ergonomics*, 20(1), 111–125.
15. HSE, H. (2019). Construction statistics in Great Britain.
16. Zid, C., Kasim, N., & Soomro, A. R. (2020). Effective project management approach to attain project success, based on

- cost-time-quality. *International Journal of Project Organisation and Management*, 12(2), 149–163.
17. Wu, H., Zhong, B., Li, H., Love, P., Pan, X., & Zhao, N. (2021). Combining computer vision with semantic reasoning for on-site safety management in construction. *Journal of Building Engineering*, 42, 103036.
 18. Moohialdin, A., Lamari, F., Marc, M., & Trigunarsyah, B. (2021). A Real-Time Computer Vision System for Workers' PPE and Posture Detection in Actual Construction Site Environment. In *EASEC16: Proceedings of The 16th East Asian-Pacific Conference on Structural Engineering and Construction*, 2019 (pp. 2169–2181). Springer Singapore.
 19. Fang, W., Ding, L., Love, P. E., Luo, H., Li, H., Peña-Mora, F., ... & Zhou, C. (2020). Computer vision applications in construction safety assurance. *Automation in Construction*, 110, 103013.
 20. Balakreshnan, B., Richards, G., Nanda, G., Mao, H., Athinarayanan, R., & Zaccaria, J. (2020). PPE compliance detection using artificial intelligence in learning factories. *Procedia Manufacturing*, 45, 277–282.
 21. Lee, J., & Lee, S. (2023). Construction site safety management: a computer vision and deep learning approach. *Sensors*, 23(2), 944.
 22. Luo, H., Wang, M., Wong, P. K. Y., & Cheng, J. C. (2020). Full body pose estimation of construction equipment using computer vision and deep learning techniques. *Automation in construction*, 110, 103016.
 23. Kolar, Z., Chen, H., & Luo, X. (2018). Transfer learning and deep convolutional neural networks for safety guardrail detection in 2D images. *Automation in Construction*, 89, 58–70.
 24. Fang, Q., Li, H., Luo, X., Ding, L., Luo, H., Rose, T. M., & An, W.

- (2018). Detecting non-hardhat-use by a deep learning method from far-field surveillance videos. *Automation in construction*, 85, 1-9.
25. Wu, J., Cai, N., Chen, W., Wang, H., & Wang, G. (2019). Automatic detection of hardhats worn by construction personnel: A deep learning approach and benchmark dataset. *Automation in Construction*, 106, 102894.
26. Nath, N. D., Behzadan, A. H., & Paal, S. G. (2020). Deep learning for site safety: Real-time detection of personal protective equipment. *Automation in construction*, 112, 103085.
27. Jeelani, I., Asadi, K., Ramshankar, H., Han, K., & Albert, A. (2021). Real-time vision-based worker localization & hazard detection for construction. *Automation in Construction*, 121, 103448.
28. Antwi-Afari, M. F., Qarout, Y., Herzallah, R., Anwer, S., Umer, W., Zhang, Y., & Manu, P. (2022). Deep learning-based networks for automated recognition and classification of awkward working postures in construction using wearable insole sensor data. *Automation in construction*, 136, 104181.
29. Kelm, A., Laußat, L., Meins-Becker, A., Platz, D., Khazaei, M. J., Costin, A. M., ... & Teizer, J. (2013). Mobile passive Radio Frequency Identification (RFID) portal for automated and rapid control of Personal Protective Equipment (PPE) on construction sites. *Automation in construction*, 36, 38-52.
30. Barro-Torres, S., Fernández-Caramés, T. M., Pérez-Iglesias, H. J., & Escudero, C. J. (2012). Real-time personal protective equipment monitoring system. *Computer Communications*, 36(1), 42-50.
31. Naticchia, B., Vaccarini, M., & Carbonari, A. (2013). A monitoring

- system for real-time interference control on large construction sites. *Automation in Construction*, 29, 148–160.
32. Zhang, H., Yan, X., Li, H., Jin, R., & Fu, H. (2019). Real-time alarming, monitoring, and locating for non-hard-hat use in construction. *Journal of construction engineering and management*, 145(3), 04019006.
 33. Kim, J. H., Jo, B. W., Jo, J. H., Lee, Y. S., & Kim, D. K. (2021). Autonomous detection system for non-hard-hat use at construction sites using sensor technology. *Sustainability*, 13(3), 1102.
 34. Kim, S. H., Wang, C., Min, S. D., & Lee, S. H. (2018). Safety helmet wearing management system for construction workers using three-axis accelerometer sensor. *Applied Sciences*, 8(12), 2400.
 35. Márquez-Sánchez, S., Campero-Jurado, I., Herrera-Santos, J., Rodríguez, S., & Corchado, J. M. (2021). Intelligent platform based on smart PPE for safety in workplaces. *Sensors*, 21(14), 4652.
 36. Nath, N. D., Behzadan, A. H., & Paal, S. G. (2020). Deep learning for site safety: Real-time detection of personal protective equipment. *Automation in construction*, 112, 103085.
 37. Choo, H., Lee, B., Kim, H., & Choi, B. (2023). Automated detection of construction work at heights and deployment of safety hooks using IMU with a barometer. *Automation in Construction*, 147, 104714.
 38. Pisu, A., Elia, N., Pompianu, L., Barchi, F., Acquaviva, A., & Carta, S. (2024). Enhancing workplace safety: A flexible approach for personal protective equipment monitoring. *Expert Systems with Applications*, 238, 122285.
 39. Seo, J., Han, S., Lee, S., & Kim, H. (2015). Computer vision techniques for construction safety and health monitoring. *Advanced Engineering*

- Informatics, 29(2), 239–251.
40. Du, S., Shehata, M., & Badawy, W. (2011, March). Hard hat detection in video sequences based on face features, motion and color information. In 2011 3rd international conference on computer research and development (Vol. 4, pp. 25–29). iee.
 41. Shrestha, K., Shrestha, P. P., Bajracharya, D., & Yfantis, E. A. (2015). Hard-hat detection for construction safety visualization. *Journal of Construction Engineering*, 2015(1), 721380.
 42. Park, M. W., Elsafty, N., & Zhu, Z. (2015). Hardhat-wearing detection for enhancing on-site safety of construction workers. *Journal of Construction Engineering and Management*, 141(9), 04015024.
 43. Park, M. W., & Brilakis, I. (2012). Construction worker detection in video frames for initializing vision trackers. *Automation in Construction*, 28, 15–25.
 44. Fang, W., Ding, L., Luo, H., & Love, P. E. (2018). Falls from heights: A computer vision-based approach for safety harness detection. *Automation in construction*, 91, 53–61.
 45. Al-Azani, S., Luqman, H., Alfarraj, M., Sidig, A. A. I., Khan, A. H., & Al-Hammed, D. (2024). Real-Time Monitoring of Personal Protective Equipment Compliance in Surveillance Cameras. *IEEE Access*.
 46. Riaz, M., He, J., Xie, K., Alsagri, H. S., Moqurrab, S. A., Alhakbani, H. A. A., & Obidallah, W. J. (2023). Enhancing Workplace Safety: PPE_Swin—A Robust Swin Transformer Approach for Automated Personal Protective Equipment Detection. *Electronics*, 12(22), 4675.
 47. Velasco, M. A., & Marasigan, R. I. (2024, July). Enhancing Personal Protective Equipment Compliance and Usage Accuracy on Construction

- Sites through YOLOv7 Detection. In 2024 IEEE 7th International Conference on Big Data and Artificial Intelligence (BDAI) (pp. 205–212). IEEE.
48. Gil, D., & Lee, G. (2024). Zero-shot monitoring of construction workers' personal protective equipment based on image captioning. *Automation in Construction*, 164, 105470.
49. Kumar, S., Poyyamozi, M., Murugesan, B., Rajamanickam, N., Alroobaea, R., & Nureldeen, W. (2024). Investigation of Unsafe Construction Site Conditions Using Deep Learning Algorithms Using Unmanned Aerial Vehicles. *Sensors*, 24(20), 6737.
50. Nguyen, N. T., Tran, Q., Dao, C. H., Nguyen, D. A., & Tran, D. H. (2024). Automatic Detection of Personal Protective Equipment in Construction Sites Using Metaheuristic Optimized YOLOv5. *Arabian Journal for Science and Engineering*, 1–19.
51. Zhang, M., Shi, R., & Yang, Z. (2020). A critical review of vision-based occupational health and safety monitoring of construction site workers. *Safety science*, 126, 104658.
52. HAN, SangUk; LEE, SangHyun. A vision-based motion capture and recognition framework for behavior-based safety management. *Automation in Construction*, 2013, 35: 131–141.
53. Han, S., Lee, S., & Peña-Mora, F. (2014). Comparative study of motion features for similarity-based modeling and classification of unsafe actions in construction. *Journal of Computing in Civil Engineering*, 28(5), A4014005.
54. Han, S., Lee, S., & Peña-Mora, F. (2012). Vision-based motion detection for safety behavior analysis in construction. In *Construction Research*

Congress 2012: Construction Challenges in a Flat World (pp. 1032-1041).

55. Han, S., & Lee, S. (2013). A vision-based motion capture and recognition framework for behavior-based safety management. *Automation in Construction*, 35, 131-141.
56. Han, S., Lee, S., & Peña-Mora, F. (2014). Comparative study of motion features for similarity-based modeling and classification of unsafe actions in construction. *Journal of Computing in Civil Engineering*, 28(5), A4014005.
57. Han, S., Lee, S., & Peña-Mora, F. (2013). Vision-based detection of unsafe actions of a construction worker: Case study of ladder climbing. *Journal of Computing in Civil Engineering*, 27(6), 635-644.
58. Guo, H., Yu, Y., Ding, Q., & Skitmore, M. (2018). Image-and-skeleton-based parameterized approach to real-time identification of construction workers' unsafe behaviors. *Journal of Construction Engineering and Management*, 144(6), 04018042.
59. Fang, W., Ding, L., Love, P. E., Luo, H., Li, H., Peña-Mora, F., ... & Zhou, C. (2020). Computer vision applications in construction safety assurance. *Automation in Construction*, 110, 103013.
60. DING, Lieyun, et al. A deep hybrid learning model to detect unsafe behavior: Integrating convolution neural networks and long short-term memory. *Automation in construction*, 2018, 86: 118-124.
61. Luo, X., Li, H., Yang, X., Yu, Y., & Cao, D. (2019). Capturing and understanding workers' activities in far-field surveillance videos with deep action recognition and Bayesian nonparametric learning. *Computer-Aided Civil and Infrastructure Engineering*, 34(4), 333-351.

62. Zhang, H., Yan, X., & Li, H. (2018). Ergonomic posture recognition using 3D view-invariant features from single ordinary camera. *Automation in Construction*, 94, 1-10.
63. Chu, W., Han, S., Luo, X., & Zhu, Z. (2020). Monocular vision-based framework for biomechanical analysis or ergonomic posture assessment in modular construction. *Journal of Computing in Civil Engineering*, 34(4), 04020018.
64. Chu, W., Han, S., Luo, X., & Zhu, Z. (2020). Monocular vision-based framework for biomechanical analysis or ergonomic posture assessment in modular construction. *Journal of Computing in Civil Engineering*, 34(4), 04020018.
65. Kong, L., Li, H., Yu, Y., Luo, H., Skitmore, M., & Antwi-Afari, M. F. (2018). Quantifying the physical intensity of construction workers, a mechanical energy approach. *Advanced Engineering Informatics*, 38, 404-419.
66. Yu, Y., Yang, X., Li, H., Luo, X., Guo, H., & Fang, Q. (2019). Joint-level vision-based ergonomic assessment tool for construction workers. *Journal of construction engineering and management*, 145(5), 04019025.
67. Yang, M., Wu, C., Guo, Y., Jiang, R., Zhou, F., Zhang, J., & Yang, Z. (2023). Transformer-based deep learning model and video dataset for unsafe action identification in construction projects. *Automation in Construction*, 146, 104703.
68. Girshick, R., Donahue, J., Darrell, T., & Malik, J. (2015). Region-based convolutional networks for accurate object detection and segmentation. *IEEE transactions on pattern analysis and machine intelligence*, 38(1), 142-158.

69. He, K., Zhang, X., Ren, S., & Sun, J. (2015). Spatial pyramid pooling in deep convolutional networks for visual recognition. *IEEE transactions on pattern analysis and machine intelligence*, 37(9), 1904–1916.
70. Ren, S., He, K., Girshick, R., & Sun, J. (2016). Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, 39(6), 1137–1149.
71. He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 2961–2969).
72. Lin, T. Y., Dollár, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2117–2125).
73. Redmon, J. (2016). You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.
74. Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C. Y., & Berg, A. C. (2016). Ssd: Single shot multibox detector. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14* (pp. 21–37). Springer International Publishing.
75. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).
76. Fang, Q., Li, H., Luo, X., Ding, L., Luo, H., Rose, T. M., & An, W.

- (2018). Detecting non-hardhat-use by a deep learning method from far-field surveillance videos. *Automation in construction*, 85, 1-9.
77. Fang, Q., Li, H., Luo, X., Ding, L., Luo, H., & Li, C. (2018). Computer vision aided inspection on falling prevention measures for steeplejacks in an aerial environment. *Automation in Construction*, 93, 148-164.
78. Roberts, D., & Golparvar-Fard, M. (2019). End-to-end vision-based detection, tracking and activity analysis of earthmoving equipment filmed at ground level. *Automation in Construction*, 105, 102811.
79. Fang, W., Zhong, B., Zhao, N., Love, P. E., Luo, H., Xue, J., & Xu, S. (2019). A deep learning-based approach for mitigating falls from height with computer vision: Convolutional neural network. *Advanced Engineering Informatics*, 39, 170-177.
80. Nath, N. D., Behzadan, A. H., & Paal, S. G. (2020). Deep learning for site safety: Real-time detection of personal protective equipment. *Automation in construction*, 112, 103085.
81. Guo, Y., Xu, Y., & Li, S. (2020). Dense construction vehicle detection based on orientation-aware feature fusion convolutional neural network. *Automation in Construction*, 112, 103124.
82. Sun, Z., Ke, Q., Rahmani, H., Bennamoun, M., Wang, G., & Liu, J. (2022). Human action recognition from various data modalities: A review. *IEEE transactions on pattern analysis and machine intelligence*, 45(3), 3200-3225.
83. Girdhar, R., Ramanan, D., Gupta, A., Sivic, J., & Russell, B. (2017). Actionvlad: Learning spatio-temporal aggregation for action classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 971-980).

84. Diba, A., Sharma, V., & Van Gool, L. (2017). Deep temporal linear encoding networks. In Proceedings of the IEEE conference on Computer Vision and Pattern Recognition (pp. 2329-2338).
85. Feichtenhofer, C., Pinz, A., & Wildes, R. P. (2017). Spatiotemporal multiplier networks for video action recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 4768-4777).
86. Zong, M., Wang, R., Chen, X., Chen, Z., & Gong, Y. (2021). Motion saliency based multi-stream multiplier ResNets for action recognition. *Image and Vision Computing*, 107, 104108.
87. Wang, P., Wang, S., Gao, Z., Hou, Y., & Li, W. (2017). Structured images for RGB-D action recognition. In Proceedings of the IEEE international conference on computer vision workshops (pp. 1005-1014).
88. Xiao, Y., Chen, J., Wang, Y., Cao, Z., Zhou, J. T., & Bai, X. (2019). Action recognition for depth video using multi-view dynamic images. *Information Sciences*, 480, 287-304.
89. Zhang, B., Wang, L., Wang, Z., Qiao, Y., & Wang, H. (2016). Real-time action recognition with enhanced motion vector CNN. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 2718-2726).
90. Piergiovanni, A. J., & Ryoo, M. S. (2019). Representation flow for action recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 9945-9953).
91. Donahue, J., Anne Hendricks, L., Guadarrama, S., Rohrbach, M., Venugopalan, S., Saenko, K., & Darrell, T. (2015). Long-term recurrent convolutional networks for visual recognition and description. In

- Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 2625–2634).
92. Majd, M., & Safabakhsh, R. (2020). Correlational convolutional LSTM for human action recognition. *Neurocomputing*, 396, 224–229.
 93. He, J. Y., Wu, X., Cheng, Z. Q., Yuan, Z., & Jiang, Y. G. (2021). DB-LSTM: Densely-connected Bi-directional LSTM for human action recognition. *Neurocomputing*, 444, 319–331.
 94. Zhao, H., & Jin, X. (2020, June). Human action recognition based on improved fusion attention CNN and RNN. In 2020 5th international conference on computational intelligence and applications (ICCIA) (pp. 108–112). IEEE.
 95. Sudhakaran, S., Escalera, S., & Lanz, O. (2019). Lsta: Long short-term attention for egocentric action recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 9954–9963).
 96. Wu, Z., Xiong, C., Ma, C. Y., Socher, R., & Davis, L. S. (2019). Adaframe: Adaptive frame selection for fast video recognition. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 1278–1287).
 97. Liu, Z., Li, Z., Wang, R., Zong, M., & Ji, W. (2020). Spatiotemporal saliency-based multi-stream networks with attention-aware LSTM for action recognition. *Neural Computing and Applications*, 32(18), 14593–14602.
 98. Kim, P. S., Lee, D. G., & Lee, S. W. (2018). Discriminative context learning with gated recurrent unit for group activity recognition. *Pattern Recognition*, 76, 149–161.

99. Ji, S., Xu, W., Yang, M., & Yu, K. (2012). 3D convolutional neural networks for human action recognition. *IEEE transactions on pattern analysis and machine intelligence*, 35(1), 221–231.
100. Li, J., Liu, X., Zhang, M., & Wang, D. (2020). Spatio-temporal deformable 3d convnets with attention for action recognition. *Pattern Recognition*, 98, 107037.
101. Yang, C., Xu, Y., Shi, J., Dai, B., & Zhou, B. (2020). Temporal pyramid network for action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 591–600).
102. Varol, G., Laptev, I., Schmid, C., & Zisserman, A. (2021). Synthetic humans for action recognition from unseen viewpoints. *International Journal of Computer Vision*, 129(7), 2264–2287.
103. Fayyaz, M., Bahrami, E., Diba, A., Noroozi, M., Adeli, E., Van Gool, L., & Gall, J. (2021). 3D CNN with adaptive temporal feature resolutions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 4731–4740).
104. Li, K., Li, X., Wang, Y., Wang, J., & Qiao, Y. (2021). Ct-net: Channel tensorization network for video classification. *arXiv preprint arXiv:2106.01603*.
105. Liang, D., & Thomaz, E. (2019). Audio-based activities of daily living (adl) recognition with large-scale acoustic embeddings from online videos. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 3(1), 1–18.
106. Xiao, F., Lee, Y. J., Grauman, K., Malik, J., & Feichtenhofer, C. (2020). Audiovisual slowfast networks for video recognition. *arXiv preprint arXiv:2001.08740*.

107. Khan, A. M., Lee, Y. K., Lee, S. Y., & Kim, T. S. (2010, May). Human activity recognition via an accelerometer-enabled-smartphone using kernel discriminant analysis. In 2010 5th international conference on future information technology (pp. 1-6). IEEE.
108. Ignatov, A. (2018). Real-time human activity recognition from accelerometer data using convolutional neural networks. *Applied Soft Computing*, 62, 915-922.
109. Shah, S. A., & Fioranelli, F. (2019, September). Human activity recognition: Preliminary results for dataset portability using FMCW radar. In 2019 international radar conference (RADAR) (pp. 1-4). IEEE.
110. Zhou, H., He, Y., Huang, H., & Jing, X. (2020). Two-stream convolutional neural network for action recognition in radar. In *Signal and Information Processing, Networking and Computers: Proceedings of the 6th International Conference on Signal and Information Processing, Networking and Computers (ICSINC)* (pp. 19-26). Springer Singapore.
111. Chen, Z., Zhang, L., Jiang, C., Cao, Z., & Cui, W. (2018). WiFi CSI based passive human activity recognition using attention based BLSTM. *IEEE Transactions on Mobile Computing*, 18(11), 2714-2724.
112. Li, Z., & Li, D. (2022). Action recognition of construction workers under occlusion. *Journal of Building Engineering*, 45, 103352.
113. Cheng, M. Y., Khitam, A. F., & Tanto, H. H. (2023). Construction worker productivity evaluation using action recognition for foreign labor training and education: A case study of Taiwan. *Automation in Construction*, 150, 104809.
114. Guo, H., Zhang, Z., Yu, R., Sun, Y., & Li, H. (2023). Action recognition based on 3D skeleton and LSTM for the monitoring of construction

- workers' safety harness usage. *Journal of Construction Engineering and Management*, 149(4), 04023015.
115. Tian, Y., Lin, S., Xu, H., & Chen, G. (2024). A Spatial-Temporal Multi-Feature Network (STMF-Net) for Skeleton-Based Construction Worker Action Recognition. *Sensors*, 24(23), 7455.
 116. Genc, E., Yildirim, M. E., & Salman, Y. B. (2024). Human activity recognition with fine-tuned CNN-LSTM. *Journal of Electrical Engineering*, 75(1), 8-13.
 117. Duan, P., Goh, Y. M., & Zhou, J. (2023). Personalized stability monitoring based on body postures of construction workers working at heights. *Safety science*, 162, 106104.
 118. Terven, J., Córdova-Esparza, D. M., & Romero-González, J. A. (2023). A comprehensive review of yolo architectures in computer vision: From yolov1 to yolov8 and yolo-nas. *Machine Learning and Knowledge Extraction*, 5(4), 1680-1716.
 119. Maas, A. L., Hannun, A. Y., & Ng, A. Y. (2013, June). Rectifier nonlinearities improve neural network acoustic models. In *Proc. icml* (Vol. 30, No. 1, p. 3).
 120. Redmon, J., & Farhadi, A. (2017). YOLO9000: better, faster, stronger. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7263-7271).
 121. Redmon, J. (2018). Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*.
 122. Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020). Yolov4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*.

123. Jocher, G. (2023). YOLOv5 by Ultralytics. 2020. URL <https://github.com/ultralytics/yolov5>.
124. Li, C., Li, L., Jiang, H., Weng, K., Geng, Y., Li, L., ... & Wei, X. (2022). YOLOv6: A single-stage object detection framework for industrial applications. arXiv preprint arXiv:2209.02976.
125. Wang, C. Y., Bochkovskiy, A., & Liao, H. Y. M. (2023). YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 7464–7475).
126. Jocher, G., Chaurasia, A., & Qiu, J. (2023). YOLO by Ultralytics.
127. Terven, J., Córdova-Esparza, D. M., & Romero-González, J. A. (2023). A comprehensive review of yolo architectures in computer vision: From yolov1 to yolov8 and yolo-nas. Machine Learning and Knowledge Extraction, 5(4), 1680–1716.
128. <https://docs.ultralytics.com/models/yolov10/#architecture>
129. <https://docs.ultralytics.com/models/yolo11/>
130. Park, M., Bak, J., & Park, S. (2023). Small and overlapping worker detection at construction sites. Automation in Construction, 151, 104856.
131. Li, P., Wu, F., Xue, S., & Guo, L. (2023). Study on the Interaction Behaviors Identification of Construction Workers Based on ST-GCN and YOLO. Sensors, 23(14), 6318.
132. Wang, X., Niu, D., Luo, P., Zhu, C., Ding, L., & Huang, K. (2020, November). A safety helmet and protective clothing detection method based on improved-yolo v 3. In 2020 Chinese automation congress (CAC) (pp. 5437–5441). IEEE.

133. Kim, D., & Xiong, S. (2025). Enhancing Worker Safety: Real-Time Automated Detection of Personal Protective Equipment to Prevent Falls from Heights at Construction Sites Using Improved YOLOv8 and Edge Devices. *Journal of Construction Engineering and Management*, 151(1), 04024187.
134. Mahmud, S. S., Islam, M. A., Ritu, K. J., Hasan, M., Kobayashi, Y., & Mohibullah, M. (2023, December). Safety Helmet Detection of Workers in Construction Site using YOLOv8. In *2023 26th International Conference on Computer and Information Technology (ICCIT)* (pp. 1-6). IEEE.
135. Gu, Y., Wang, Y., Shi, L., Li, N., Zhuang, L., & Xu, S. (2021). Automatic detection of safety helmet wearing based on head region location. *IET Image Processing*, 15(11), 2441-2453.
136. Wu, F., Jin, G., Gao, M., Zhiwei, H. E., & Yang, Y. (2019, May). Helmet detection based on improved YOLO V3 deep model. In *2019 IEEE 16th International conference on networking, sensing and control (ICNSC)* (pp. 363-368). IEEE.
137. Liu, Y., Zhang, J., Shi, L., Huang, M., Lin, L., Zhu, L., ... & Zhang, C. (2024). Detection method of the seat belt for workers at height based on UAV image and YOLO algorithm. *Array*, 22, 100340.
138. Song, H. (2022). Multi-scale safety helmet detection based on RSSE-YOLOv3. *Sensors*, 22(16), 6061.
139. Ngoc-Thoan, N., Bui, D. Q. T., Tran, C. N., & Tran, D. H. (2024). Improved detection network model based on YOLOv5 for warning safety in construction sites. *International Journal of Construction Management*, 24(9), 1007-1017.

140. Wang, H., Hu, Z., Guo, Y., Yang, Z., Zhou, F., & Xu, P. (2020). A real-time safety helmet wearing detection approach based on CSYOLOv3. *Applied Sciences*, 10(19), 6732.
141. Li, H., Wu, D., Zhang, W., & Xiao, C. (2024). YOLO-PL: Helmet wearing detection algorithm based on improved YOLOv4. *Digital Signal Processing*, 144, 104283.
142. Zhang, C., & Shen, J. (2024, June). YOLO-DS: Application of One-Stage Instance Segmentation in the Dark at Construction Sites. In *2024 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8). IEEE.
143. Apeināns, I. (2024, June). Optimal Size of Agricultural Dataset for YOLOv8 Training. In *ENVIRONMENT. TECHNOLOGIES. RESOURCES. Proceedings of the International Scientific and Practical Conference* (Vol. 2, pp. 38-42).
144. Rahimi, M., Mostafavi, M., & Arabameri, A. (2024, March). Automatic Detection of Brain Tumor on MRI Images Using a YOLO-Based Algorithm. In *2024 13th Iranian/3rd International Machine Vision and Image Processing Conference (MVIP)* (pp. 1-5). IEEE.
145. Jocher, G., Stoken, A., Borovec, J., Chaurasia, A., Changyu, L., Hogan, A., ... & Ingham, F. (2021). ultralytics/yolov5: v5. 0-YOLOv5-P6 1280 models, AWS, Supervise. ly and YouTube integrations. Zenodo.
146. Xin, H., & Song, J. (2024). YOLOv5-ACCOF Steel Surface Defect Detection Algorithm. *IEEE Access*.
147. Peng, J., Zhou, W., Han, Y., Li, M., & Liu, W. (2023). Deep learning-based autonomous real-time digital meter reading recognition method for natural scenes. *Measurement*, 222, 113615.

148. Tan, M., Pang, R., & Le, Q. V. (2020). Efficientdet: Scalable and efficient object detection. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 10781–10790).
149. Du, S., Zhang, B., Zhang, P., & Xiang, P. (2021, July). An improved bounding box regression loss function based on CIOU loss for multi-scale object detection. In 2021 IEEE 2nd International Conference on Pattern Recognition and Machine Learning (PRML) (pp. 92–98). IEEE.
150. Yi, W., Ma, S., & Li, R. (2023). Insulator and defect detection model based on improved YOLO-S. IEEE Access.
151. Jocher, G., Chaurasia, A., & Qiu, J. (2023). YOLO by Ultralytics.
152. Li, L., Wang, Z., & Zhang, T. (2023). Gbh-yolov5: Ghost convolution with bottleneckcsp and tiny target prediction head incorporating yolov5 for pv panel defect detection. Electronics, 12(3), 561.
153. Han, K., Wang, Y., Tian, Q., Guo, J., Xu, C., & Xu, C. (2020). Ghostnet: More features from cheap operations. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 1580–1589).
154. Kee, D. (2021). Comparison of OWAS, RULA and REBA for assessing potential work-related musculoskeletal disorders. International Journal of Industrial Ergonomics, 83, 103140.
155. McAtamney, L., & Corlett, E. N. (1993). RULA: a survey method for the investigation of work-related upper limb disorders. Applied ergonomics, 24(2), 91–99.
156. Hignett, S., & McAtamney, L. (2000). Rapid entire body assessment (REBA). Applied ergonomics, 31(2), 201–205.

딥러닝 기반 고소 작업자 안전 분석을 위한 개인 보호 장비 탐지 및 작업 자세 인식 모델 개발

미 이 룡

경북대학교 대학원 건설환경에너지공학부 건축공학전공
(지도교수 이동은)

(초 록)

건설 산업의 급속한 발전에 따라 건설 현장의 안전 문제는 점차 중요한 사회적 이슈로 부각되고 있으며, 이는 전 세계적으로 주목받는 과제 중 하나이다. 통계에 따르면, 건설 산업의 사고율 및 사망률은 타 산업에 비해 현저히 높은 수준이며, 특히 고소 작업 중 발생하는 사고가 전체 사고에서 큰 비중을 차지하고 있다. 각국에서는 이러한 문제에 대응하기 위해 다양한 안전 관리 방안을 도입하고 있으나, 건설 환경의 복잡성과 작업자 행동의 예측 어려움으로 인해 안전 문제는 여전히 지속되고 있는 실정이다. 최근 컴퓨터 비전 기술은 건설 현장 작업자 안전 분야에서 주목을 받고 있으며, 특히 딥 러닝 기반의 객체 탐지 및 행동 인식 기술은 건설 현장 안전 모니터링을 향상시키는 혁신적인 솔루션으로 떠오르고 있다. 연구자들은 컴퓨터 비전, 의미 추론, TensorFlow 등의 기술과 데이터 융합 기법을 결합하여 PPE 착용 상태의 지능적인 인식과 작업자 자세 분석을 실현하고, 이를 통해 작업자의 위험도를 평가하고 있다. 머신 비전 및 전이 학습의 적용은 PPE 착용 여부와 규정 준수 여부를 판단할 수 있게 하였다. 특히 딥 러닝 기술 중 CNN 모델은 안전모 착용 인식 및 위험 행동 탐지에서 큰 진전을 이루었다. 또한, 웨어러블 장치와 인공지능을 결합한 실시간 추적 시스템이 개발 중에 있으며, 이는 안전 모니터링과 작업 행동 분석 역량을 강화하고 부적절한 작업 자세를 인식하여 작업자의 안전과 건강을 향상시키는 것을 목표로 하고 있다. 그러나 현재의 연구는 몇 가지 도전에 직면해 있으며, 그 중에는 데이터셋 규모의 부족, 센서 의존도 과다, 평가 방법의 단일성, 그리고 개별 작업에 대한 과도한 집중 등이 있다.

본 연구는 고소 작업자의 안전 상태를 효과적으로 감지하기 위해 딥러닝 기반 모델을 개선하였다. AI-HUB 데이터셋을 활용하였으며, YOLO 모델을 기반 모델을 중심으

로 개인 보호 장비(PPE) 탐지와 골격 키포인트 인식을 통합한 다중 작업 탐지 방식을 적용하였다. 모델의 정확도와 신뢰성을 향상시키기 위해 특성 추출 방식의 최적화, 주의 메커니즘 도입, 실험 설계의 경량화를 수행하였다. 그 결과, 제안된 모델은 높은 탐지 정확도를 유지하면서도 연산 효율성을 확보하였으며, 제한된 자원 환경에서도 안정적으로 작동할 수 있음을 확인하였다. 실제 건설 현장에서 모델은 조명 변화, 복잡한 배경, 물체 가림 등 다양한 어려운 조건에서도 작업자의 PPE 착용 여부(예: 안전모, 장갑, 안전화)를 정확하게 판별할 수 있었다. 또한, 골격 키포인트 기반 자세 분석을 통해 과도한 굽힘, 관절 비틀림 등의 비정상 작업 자세를 인식하고, 동작 위험도를 평가함으로써 부적절한 자세를 조기에 식별할 수 있다. 아울러, 본 모델은 다양한 작업 역할과 시간대에 따른 PPE 착용 상태 및 자세 위험도를 분석하여 안전 관리자에게 신뢰성 있는 데이터를 제공하며, 반복적인 규정 위반 및 고위험 행동의 패턴을 파악하고 안전 교육 및 감독 방안을 효과적으로 수립할 수 있도록 지원한다. 시각 피드백 기능을 통해 작업자는 자신의 위험 행동을 인지하고 즉시 교정할 수 있어, 자가 보호 능력 향상에도 기여한다. 본 연구는 건설 현장의 안전 모니터링 수준을 향상시키고, 지능형 건설 안전 관리 체계 구축에 실질적인 기여를 할 수 있을 것으로 기대된다.

Appendix A: Dataset Overview

The dataset used in this study is publicly available and was obtained from the following source:

Name: Real-time Video Data of High-Altitude Work Sites (고소작업 현장 실시간 영상 데이터)

Source: AI-Hub (Korea)

Access Link:

<https://www.aihub.or.kr/aihubdata/data/view.do?currMenu=115&topMenu=100&dataSetSn=507>

Metadata Summary:

Item	Description
Data Domain	Disaster safety environment
Data Type	Image
Data Format	JPG
Label Format	JSON
Label Types	Bounding boxes, keypoints
Data Source	Videos generated from typical high-place work and hazardous situation scenarios
Use Case	Risk state assessment in construction and high-place work environments
Dataset Volume	330 hours of multi-angle video; 687,895 annotated frames (recorded in 2021)

Description:

This dataset was developed to support intelligent safety management at high-altitude construction sites. It includes annotated video data for detecting personal protective equipment (PPE), verifying the use or

omission of safety hooks, identifying overload conditions, recognizing unsafe behaviors such as leaning on guardrails, and flagging safety risks. The videos were captured using a combination of CCTV, LTE-based helmet-mounted cameras, and action cameras, ensuring full visual coverage of the work areas without blind spots.

The dataset consists of high-definition video recordings from multiple angles, totaling 330 hours of footage collected from 19 different construction sites. Its primary goal is to aid AI training for the early detection of hazardous behaviors and environmental risks in elevated work environments. Through video-based AI analysis, this dataset contributes to accident prevention and supports the identification of root causes of safety incidents.